

# **Exploring Excess Return Generation through European Fixed Income Arbitrages**

Alessio Ottaviani

EDHEC Business School

Supervisor: Prof. Riccardo Rebonato

**Abstract:**

# Contents

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| <b>Introduction</b>                                                   | <b>5</b>  |
| <b>I Data, Strategy Construction, and Performance</b>                 | <b>9</b>  |
| A Data and Strategy Construction . . . . .                            | 10        |
| B Index Construction and Weighting Schemes . . . . .                  | 13        |
| C Performance Analysis . . . . .                                      | 15        |
| <b>II Benchmark Factor Models</b>                                     | <b>17</b> |
| A Benchmark Specifications . . . . .                                  | 17        |
| B Results and Discussion . . . . .                                    | 20        |
| <b>III Factor Selection and Spanning Regressions</b>                  | <b>23</b> |
| A Candidate Factor Universe . . . . .                                 | 24        |
| B Principal Component Analysis . . . . .                              | 25        |
| C Best-Subset Selection . . . . .                                     | 26        |
| D Conditional Alpha . . . . .                                         | 29        |
| E Alpha Across Layers . . . . .                                       | 30        |
| <b>IV Cross-Strategy Interdependencies</b>                            | <b>30</b> |
| A Mispricing Construction and Stress Regimes . . . . .                | 31        |
| B P1: Cross-Market Co-Movement . . . . .                              | 32        |
| C P2: Stress Amplification . . . . .                                  | 33        |
| D P3: Lead–Lag Transmission . . . . .                                 | 35        |
| E P4: Persistence and Funding-Dependence . . . . .                    | 36        |
| F P5: The Common Factor . . . . .                                     | 37        |
| G Discussion: Why These Arbitrages Persist . . . . .                  | 38        |
| <b>V Conclusion</b>                                                   | <b>41</b> |
| <b>A Appendix</b>                                                     | <b>49</b> |
| A.1 BTP Italia: Indexation Coefficient and Deflation Floors . . . . . | 49        |
| A.2 Trade Construction Details . . . . .                              | 50        |
| A.2.1 Entry/Exit Rules and Thresholds . . . . .                       | 50        |
| A.2.2 Trade-Level Statistics . . . . .                                | 50        |
| A.2.3 Transaction Costs . . . . .                                     | 51        |
| A.3 Signal-Weighted vs Equal-Weighted Comparison . . . . .            | 52        |
| A.4 Volatility-Managed Portfolios . . . . .                           | 53        |
| A.5 Benchmark Regressions with US Factors . . . . .                   | 54        |
| A.6 Subperiod and Rolling Window Analysis . . . . .                   | 56        |
| A.6.1 Duarte et al. (2007) . . . . .                                  | 56        |
| A.6.2 Brooks et al. (2020) — Active Fixed Income . . . . .            | 56        |
| A.6.3 Fung & Hsieh (2004) . . . . .                                   | 57        |
| A.6.4 Out-of-Sample Alpha: Rolling-Window Robustness . . . . .        | 58        |

|      |                                                                         |    |
|------|-------------------------------------------------------------------------|----|
| A.7  | VIF Diagnostics . . . . .                                               | 59 |
| A.8  | Candidate Factor Universe . . . . .                                     | 61 |
|      | A.8.1 Complete Factor List with Sources . . . . .                       | 61 |
| A.9  | PCA Robustness . . . . .                                                | 66 |
|      | A.9.1 Robustness across the Number of Components . . . . .              | 66 |
|      | A.9.2 Subperiod and Regime Analysis . . . . .                           | 66 |
|      | A.9.3 Conditional Alpha: Christopherson–Ferson–Glassman (PCA) . . . . . | 68 |
| A.10 | Factor-Selection Robustness . . . . .                                   | 69 |
|      | A.10.1 Sensitivity to the Subset Cardinality . . . . .                  | 69 |
|      | A.10.2 Factor Selection Across Methods . . . . .                        | 69 |
|      | A.10.3 Conditional Alpha: Detailed Results (Best-Subset) . . . . .      | 70 |
|      | A.10.4 Rolling Alpha . . . . .                                          | 71 |
| A.11 | Cross-Strategy Interdependencies: Additional Tests . . . . .            | 72 |
|      | A.11.1 DCC-GARCH . . . . .                                              | 72 |
|      | A.11.2 Granger Causality: Lag Robustness . . . . .                      | 73 |

|               |           |
|---------------|-----------|
| <b>Tables</b> | <b>75</b> |
|---------------|-----------|

## List of Tables

|         |                                                                                                          |    |
|---------|----------------------------------------------------------------------------------------------------------|----|
| A.I     | Entry and Exit Thresholds by Strategy . . . . .                                                          | 50 |
| A.II    | Trade-Level Statistics by Strategy . . . . .                                                             | 51 |
| A.III   | Transaction Fee Schedule (bps per unit DV01 or duration) . . . . .                                       | 52 |
| A.IV    | Equal-Weighted Strategy Returns . . . . .                                                                | 52 |
| A.V     | Manipulation-Proof Performance Measure by Risk Aversion (Equal-Weighted)                                 | 53 |
| A.VI    | Moreira–Muir Volatility-Managed Portfolio Performance . . . . .                                          | 53 |
| A.VII   | Duarte Factor Model Regressions (US Factors) . . . . .                                                   | 54 |
| A.VIII  | Fung & Hsieh Factor Model Regressions (US Factors) . . . . .                                             | 55 |
| A.IX    | Active Fixed Income Factor Model Regressions (US Factors) . . . . .                                      | 55 |
| A.X     | Sub-Period and Regime Analysis: Duarte et al. (2007) . . . . .                                           | 56 |
| A.XI    | Sub-Period and Regime Analysis: Brooks et al. (2020) . . . . .                                           | 57 |
| A.XII   | Sub-Period and Regime Analysis: Fung & Hsieh (2004) . . . . .                                            | 58 |
| A.XIII  | Out-of-Sample Alpha: Rolling-Window Robustness . . . . .                                                 | 59 |
| A.XIV   | Variance Inflation Factors — Benchmark Factor Models, EUR Factors . . . .                                | 60 |
| A.XV    | Candidate Risk Factors . . . . .                                                                         | 61 |
| A.XVI   | PCA Spanning Robustness across the Number of Components . . . . .                                        | 66 |
| A.XVII  | PCA Subperiod Analysis (Contemporaneous) . . . . .                                                       | 66 |
| A.XVIII | Conditional Alpha (PCA, Contemporaneous): Christopherson, Ferson, and<br>Glassman (1998) Model . . . . . | 68 |
| A.XIX   | Best-Subset Support as the Cardinality $k$ Varies . . . . .                                              | 69 |
| A.XX    | Selected Factors Across Selection Methods . . . . .                                                      | 70 |
| A.XXI   | Conditional Alpha: Christopherson, Ferson, and Glassman (1998) Model . . .                               | 71 |
| A.XXII  | Granger Causality: Robustness to Lag Length . . . . .                                                    | 74 |
| I       | Summary Statistics for Fixed Income Arbitrage Strategies . . . . .                                       | 76 |
| II      | Manipulation-Proof Performance Measure by Risk Aversion . . . . .                                        | 77 |
| III     | Duarte Factor Model Regressions . . . . .                                                                | 78 |
| IV      | Fung & Hsieh Factor Model Regressions . . . . .                                                          | 79 |
| V       | Active Fixed Income Factor Model Regressions . . . . .                                                   | 80 |
| VI      | Alpha Estimates Across Benchmark Factor Models . . . . .                                                 | 81 |
| VII     | PCA Spanning Regressions . . . . .                                                                       | 82 |
| VIII    | Top Factor Loadings by Principal Component . . . . .                                                     | 82 |
| IX      | Out-of-Sample Alpha: PCA and Best-Subset . . . . .                                                       | 83 |
| X       | Post-Selection OLS on the Best-Subset Factors . . . . .                                                  | 84 |
| XI      | Alpha by Stress Regime: PCA and Best-Subset . . . . .                                                    | 85 |
| XII     | Alpha Across Empirical Layers: Factor Benchmarks, PCA, and Best-Subset .                                 | 86 |
| XIII    | Alpha Across Factor-Selection Methods . . . . .                                                          | 87 |
| XIV     | Unconditional and Regime-Dependent Correlations . . . . .                                                | 88 |
| XV      | Cross-Strategy Spanning Regressions ( $\Delta m$ ) . . . . .                                             | 89 |
| XVI     | Stress Amplification: Interaction Regressions . . . . .                                                  | 90 |
| XVII    | AR(1) Persistence of Mispricing Levels . . . . .                                                         | 91 |
| XVIII   | Common Factor (PC1) and Funding Stress . . . . .                                                         | 92 |

## List of Figures

|     |                                                 |    |
|-----|-------------------------------------------------|----|
| A.1 | Rolling Alpha with Regime Shading (Best-Subset) | 72 |
| A.2 | Dynamic Conditional Correlations (DCC-GARCH)    | 73 |
| 1   | The TIPS–Treasury Basis, 2005–2025              | 93 |
| 2   | Market-Level Basis Time Series                  | 94 |
| 3   | Cumulative Return Indices and Drawdowns         | 95 |
| 4   | Rolling Alpha Across Benchmark Factor Models    | 96 |
| 5   | PCA Scree Plot                                  | 97 |
| 6   | Pairwise Correlations by Stress Regime          | 98 |
| 7   | Generalized Impulse Response Functions          | 99 |

## Introduction

Duarte et al. (2007) pose the question that motivates our analysis: do fixed-income arbitrage strategies earn genuine alpha, or are their returns simply compensation for systematic risk—“picking up nickels in front of a steamroller”? Using a ten-factor spanning regression, they find that the more complex strategies produce significant alpha that survives equity and bond market risk controls. We take this question to a new setting—European fixed-income markets—and to three strategies that have not previously been studied jointly as systematic arbitrages, namely the BTP Italia inflation-linked basis, the European corporate CDS–bond basis, and the iTraxx index skew.

A second set of questions then arises. Does this alpha withstand a far more demanding test of risk—and which risks does each strategy ultimately bear? Is the alpha that survives roughly constant, or time- and state-dependent? And, perhaps above all, why does it persist? The presidential address of Duffie (2010) provides a framework: when intermediary balance-sheet capacity is the binding constraint on arbitrage, mispricings across different markets should co-move, the co-movement should intensify in stress, and the transmission should follow a liquidity pecking order from more liquid to less liquid markets. Brunnermeier and Pedersen (2009) provide the mechanism—shocks to funding liquidity and market liquidity reinforce each other in a spiral that forces simultaneous deleveraging across positions—and Mitchell and Pulvino (2012) document the empirical counterpart, showing that arbitrage crashes propagate across strategies as capital withdraws.

Each of the three arbitrage strategies we study has an established literature, but none has been analysed in the way we propose. The inflation-linked bond basis was documented by Fleckenstein et al. (2014) in the context of US Inflation-Linked Treasury Bonds (TIPSs), who found an average mispricing of 54.5 basis points between TIPSs and nominal Treasuries over the 2004–2009 period, with peaks exceeding 200 basis points during the 2008–2009 financial crisis (Figure 1). This basis has since compressed substantially, oscillating in a 0–30 bps band post-2009, reflecting broader institutional participation in the U.S. inflation-linked market—and, as Fleckenstein et al. (2014)

show, the basis narrows as hedge fund capital flows into the trade, providing direct evidence for the slow-moving capital channel. In their Section VII, they further document that changes in the TIPS–Treasury mispricing are correlated with changes in the corporate CDS–Bond basis and the CDX index–component basis, establishing the cross-strategy co-movement that we extend to the European setting in Section IV. In contrast, the analogous basis on Italian BTP Italia bonds persists to date. To the best of our knowledge, the BTP Italia has not been the subject of any prior academic study, and its inflation-linked basis has not previously been analysed as a systematic arbitrage strategy. Its persistence is particularly striking because the BTP Italia eliminates the ballooning risk that characterises TIPSs and other conventional inflation-linked sovereigns,<sup>1</sup> and because its retail-dominated investor base creates a structural segmentation that insulates the basis from the institutional capital flows that closed the U.S. mispricing.

Moving to the second strategy we analyse, the CDS–Bond basis is the difference between the CDS spread and the bond credit spread for the same reference entity, a relationship that should be zero by no arbitrage in a frictionless market. It was first documented by Hull et al. (2004) and Longstaff et al. (2005) in its pre-crisis positive form. Mitchell and Pulvino (2012) show that during 2007–2009, tightening financial constraints drove the basis deeply negative, and Bai and Collin-Dufresne (2019) provide a comprehensive analysis of its cross-sectional determinants and its connection to limits of arbitrage. However, this literature focuses predominantly on U.S. corporate bond markets, and, to our knowledge, very little attention has been paid to the euro-area corporate counterpart, which is the second focus of this study.

This brings us to the third strategy of interest, the CDS index skew, which is defined as the deviation of the traded index spread from its intrinsic, no-arbitrage value implied by the single-name constituents. In a U.S. context, this strategy has been studied by Junge and Trolle (2015), who use it as a market-wide liquidity measure, and by Boyarchenko et al. (2016), who document that

---

<sup>1</sup>Conventional inflation-linked bonds adjust the principal upward with realised inflation; in deflationary scenarios or when inflation expectations spike, the redemption value can grow to many multiples of par (the so-called *ballooning* risk), introducing path-dependent valuation risk. The BTP Italia instead pays inflation revaluation as part of each semiannual coupon and redeems at par regardless of the inflation path, eliminating this source of uncertainty. The persistence of a positive arbitrage in the BTP Italia basis is therefore especially surprising: the instrument has *less* risk than a conventional inflation-linked bond, not more.

post-crisis regulatory capital requirements have made basis trades between the index and single-name CDS “significantly less economical,” leading to persistently wider skews; both papers focus on the U.S. CDX index. To the best of our knowledge, neither the negative basis trade on European investment-grade corporates in the iTraxx Main universe nor the index skew across the four iTraxx sub-indices has been studied as a systematic arbitrage strategy in the academic literature.

Beyond being the first study to analyse these three strategies jointly—and, for the BTP Italia, the first to analyse it at all—our paper extends the Duarte et al. (2007) framework in three respects:

1. Each of our three strategies has a deterministic payoff at maturity: if the position can be held to maturity, the profit is certain regardless of the path of market variables in the interim. The strategies in Duarte et al. (2007), by contrast—swap spreads, yield curve trades, mortgage hedging, volatility selling, capital structure trades—do not share this property: their payoffs depend on models, mean-reversion assumptions, or stochastic processes (prepayment, realised volatility) that leave the outcome uncertain even at maturity. This yields a cleaner test of the alpha-versus-risk question: when the terminal payoff is certain, a positive average return cannot be compensation for payoff risk, and any surviving alpha points to limits to arbitrage rather than to a missing risk premium.
2. We subject the alpha estimates to a progressive battery of factor controls: three classical benchmark specifications from the fixed-income hedge fund literature (Duarte et al., 2007; Fung and Hsieh, 2004; Brooks et al., 2020), a full-sample principal component analysis following Connor and Korajczyk (1986, 1988) on the candidate factor universe, and the Adaptive Elastic Net of Zou and Zhang (2009) with stability selection (Meinshausen and Bühlmann, 2010), which identifies a sparse set of risk drivers for each strategy individually. The alpha that survives this sequence of increasingly demanding tests is not compensation for any of the identified systematic risk exposures. Because these benchmark loadings are estimated in sample, we re-estimate each benchmark out of sample, forming the factor hedge from past-only loadings; the alpha survives, confirming that it is an abnormal return implementable in real time rather than an arte-



fact of in-sample factor combination. Using the conditional framework of Christopherson et al. (1998) we show that it rises systematically during periods of intermediary funding stress.

3. We test the cross-strategy implications of slow-moving capital systematically, drawing on Duffie (2010), Brunnermeier and Pedersen (2009), Mitchell and Pulvino (2012), and Fleckenstein et al. (2014): whereas the existing literature has studied each of these arbitrages in isolation, we examine all three jointly in the European market and ask whether their mispricings co-move, whether the co-movement intensifies in stress, and whether a common factor tied to intermediary capital—rather than to macroeconomic fundamentals—explains it. The European setting provides a discriminating test of the mechanism: all three arbitrages are run by the same type of institutional arbitrageur, but the underlying instruments differ in who holds them. The CDS–Bond and iTraxx legs are held and arbitrated by leveraged institutions; the BTP Italia basis, by contrast, rests on an inflation-linked bond held predominantly by unlevered retail households, who face no margin calls and do not unwind their positions under funding stress. We relate the resulting evidence to the structural limits to arbitrage that keep the mispricings open, including the post-crisis regulatory capital costs documented by Boyarchenko et al. (2016).

We find that all three strategies generate positive and economically significant excess returns net of transaction costs, with annualised Sharpe ratios from 0.71 (BTP Italia) to 1.71 (CDS–Bond Basis), positive skewness, and contained drawdowns. The alpha survives controls based on three widely used benchmark frameworks (estimated both in and out of sample), on rolling principal components, and on the Adaptive Elastic Net for the BTP Italia and CDS–Bond strategies; the iTraxx skew is absorbed by its selected factors unconditionally but reappears as a positive residual during high-stress periods and serves as the leading mispricing in the cross-strategy hierarchy, Granger-causing both the CDS–Bond ( $p = 0.011$ ) and the BTP Italia ( $p = 0.007$ ) mispricings. The residual alpha rises during periods of intermediary funding stress, in a manner consistent with the slow-moving capital mechanism of Duffie (2010). The co-movement of the three mispricings is governed by a common factor that loads on intermediary balance-sheet variables—the He et al. (2017) capital ratio, the Libor–OIS spread, and the Amihud and Mendelson (2015) illiquid-

ity measure—and not on macroeconomic fundamentals (a placebo regression on macro indicators yields uniformly insignificant coefficients). The persistence of the mispricing levels partially mirrors the funding-dependence of each arbitrage trade: the CDS–Bond basis, which requires continuous repo financing of the cash bond, has the longest half-life at approximately three months, roughly one-third longer than the BTP Italia and the iTraxx skew, which mean-revert within two months and are essentially identical to each other. In monthly mispricing changes, the institutional pair (CDS–Bond and iTraxx) co-moves ever more tightly as stress builds, with correlations rising monotonically from  $-0.21$  in low-stress to  $+0.32$  in high-stress regimes. The retail-held BTP Italia basis moves the other way: its correlation with the CDS–Bond basis is significantly negative on the full sample at  $-0.30$ , deepening to  $-0.47$  in high-stress months. This opposing behaviour provides independent evidence of investor-base segmentation: a single intermediary-capital shock pushes the two institutional mispricings wider and the unlevered, retail-held BTP Italia tighter.

The paper is organized as follows. Section I describes the three strategies, their construction, and their performance. Section II evaluates the alpha against three classical benchmark frameworks. Section III extends the analysis to a candidate universe of 85 factors using PCA and the Adaptive Elastic Net. Section IV tests the cross-strategy predictions of the slow-moving capital literature. Section V concludes.

## **I Data, Strategy Construction, and Performance**

This section introduces the three European fixed-income arbitrage strategies that the paper studies: the inflation-linked basis on Italian government bonds (BTP Italia), the negative basis between corporate bonds and their credit default swaps (CDS–Bond Basis), and the CDS index skew—the difference between the traded iTraxx spread and the intrinsic spread implied by its single-name constituents (iTraxx Combined). Section A describes the data and the construction of each trade, with particular attention to the two features of the BTP Italia—the absence of ballooning risk and the retail bondholder base—on which the identification strategy of Section IV rests. Section B aggregates individual trades into investable indices, compares equal-weighted and signal-weighted

schemes, and introduces the regime-dependent transaction costs charged throughout the paper. Section C measures the performance of the resulting indices and subjects it to manipulation-proof and volatility-managed diagnostics.

## A Data and Strategy Construction

Data are sourced from the MOT market operated by Borsa Italiana (BTP Italia and nominal BTP prices), Bloomberg (inflation swap rates and corporate bond spreads), CMA/CME Group (single-name CDS spreads), and J.P. Morgan DataQuery (iTraxx index and constituent data, cross-validated against Barclays Live). The iTraxx skew sample begins in June 2004 with the launch of the first on-the-run series of the iTraxx Europe Main 5-year index, the CDS–Bond sample begins in September 2008, and the BTP Italia sample begins in April 2012, the month after the first issuance. All three samples extend to May 2025. Following Duarte et al. (2007), each individual trade is modelled as a separate fund that opens when the basis crosses an entry threshold and closes when the basis reverts through an exit threshold or the underlying instrument matures. Entry and exit rules are summarised in Table A.I; trade-level statistics are reported in Table A.II. Over the full sample, the strategies generate 260 trades for BTP Italia, 931 for CDS–Bond Basis, and 206 for iTraxx Combined. Figure 2 plots the market-level basis time series for each strategy.

**BTP Italia: The Inflation-Linked Basis.** The BTP Italia is an Italian government bond indexed to domestic inflation, measured by the ISTAT FOI consumer price index (excluding tobacco). It differs in two economically meaningful ways from conventional inflation-linked sovereigns such as the U.S. TIPS studied by Fleckenstein et al. (2014).

The first is the absence of *ballooning risk*. A conventional inflation-linked bond accumulates the inflation adjustment in the principal over the life of the bond, so that the notional outstanding at time  $t$  equals  $100 \times I_t$ , where  $I_t$  is the cumulative inflation factor since issuance. In a credit event, the recovery rate (RR) is applied to the original face value of 100, not to the inflation-adjusted notional: an investor holding a TIPS with  $I_t = 1.25$  recovers  $RR \times 100$  and loses the entire accu-

mulated revaluation of 25 percentage points. The BTP Italia eliminates this exposure. The capital revaluation is paid out semiannually alongside the coupon, so the inflation factor resets to unity at every coupon date and the notional exposed to credit risk remains at par throughout the life of the bond. The two legs of the arbitrage therefore carry identical sovereign credit exposure, which isolates a pure relative-value component in the basis and removes the differential-default contamination that affects the TIPS–Treasury basis in stress periods. Appendix A.1 reports the technical specification of the indexation coefficient and of the deflation floors on coupon and principal.

The second feature is the investor base. The Italian Treasury introduced the BTP Italia in 2012 explicitly as a retail instrument, structuring its primary placement in two sequential phases: an initial phase reserved exclusively for retail investors, which typically absorbs the bulk of each issuance, followed by a shorter institutional phase. Investors who purchase at issuance and hold to maturity receive an additional loyalty bonus (the *premio fedeltà*) that further incentivises buy-and-hold behaviour. The BTP Italia is therefore held predominantly by households with long holding periods and no regulatory balance-sheet constraints. This segmentation matters because, as we document in Section IV, it is what makes the BTP Italia basis behave differently from the two institutional credit strategies during stress episodes: retail holders face no margin calls and do not unwind their positions when intermediary capital withdraws.

We construct the BTP Italia basis following the replication logic of Fleckenstein et al. (2014), adapted to the Italian instrument. The inflation-linked cash flows of the BTP Italia are converted into fixed-rate equivalents using the term structure of zero-coupon inflation swaps, yielding a fixed-rate equivalent yield  $y_t^{\text{IL}}$ . The basis is the difference between this yield and the yield to maturity of the maturity-matched nominal BTP  $y_t^{\text{NOM}}$ :

$$\text{Basis}_t^{\text{BTP}} = y_t^{\text{IL}} - y_t^{\text{NOM}}. \quad (1)$$

A positive basis indicates that the BTP Italia, once converted to fixed, yields more than the nominal bond, and vice versa; the trade is executed accordingly through an asset swap (ASW) on the

BTP Italia against an offsetting position in the maturity-matched nominal BTP. Both the BTP Italia and the nominal BTP are accepted as general collateral in the euro repo market, so the two legs carry identical funding costs, which rules out funding-cost differentials as an explanation for the observed basis.

**CDS–Bond Basis: The Negative Basis Trade.** The CDS–Bond basis for issuer  $i$  at date  $t$  is defined as the difference between the CDS spread and the z-spread of a reference corporate bond:

$$\text{Basis}_{i,t} = \text{CDS spread}_{i,t} - \text{z-spread}_{i,t}, \quad (2)$$

where the z-spread is the constant spread that, when added to the bootstrapped euro zero-coupon swap curve, equates the discounted bond cash flows to the observed market price. When the basis is negative, credit protection via CDS is cheaper than the credit spread embedded in the bond price. The negative basis trade exploits this gap: the arbitrageur purchases the bond (earning the z-spread) and simultaneously buys CDS protection (paying the lower CDS spread), locking in a positive carry that persists until convergence or bond maturity. We restrict the strategy to the negative basis trade; the reverse trade (short bond, sell protection) requires sourcing the corporate bond through a reverse repo, which for corporate names is subject to recall risk—particularly during periods of stress, when the securities lender may recall the borrowed bond and force premature unwinding of the position (Bai and Collin-Dufresne, 2019). In a default event, the bondholder recovers  $\text{RR} \times 100$  on par while the CDS contract pays  $(1 - \text{RR}) \times 100$ , so the combined position returns par regardless of realised recovery—provided the CDS maturity is no shorter than the bond’s, which we ensure by matching hedge tenors to bond residual life (Appendix A.2). The tradable universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Europe Main 5-year index, which ensures liquidity and provides a transparent filter for investment-grade credit quality. When a name exits the on-the-run index at a roll date, no new trades are initiated on that name; existing positions are carried to convergence or maturity. All CDS are centrally cleared through LCH or ICE, eliminating direct counterparty credit risk on the protection leg.

**iTraxx Combined: The CDS Index Skew.** The CDS index skew is the difference between the traded spread of an iTraxx index and its intrinsic spread—the no-arbitrage level implied by the constituents’ CDS curves under standard market conventions:

$$\text{Skew}_t = S_t^{\text{index}} - S_t^{\text{intrinsic}}. \quad (3)$$

A positive skew indicates that the index trades wide relative to the sum of its parts, and vice versa.<sup>2</sup> Three forces sustain a non-zero equilibrium skew. First, the index is far more liquid than its single-name constituents, especially in stress (Junge and Trolle, 2015). Second, the two legs face different sources of demand: institutional investors use the index as the natural macro credit hedge, while dealer banks trade single names for more granular hedging needs, including the hedging of counterparty CVA exposure—forces that can push the two legs in non-synchronised directions and make the skew bidirectional. Third, executing the replicating basket of single-name positions carries operational costs that the index trade does not. Our strategy trades the skew across the four iTraxx sub-indices—Main, Senior Financials, Subordinated Financials, and Crossover—restricting new positions to the on-the-run series at the time of entry. Entry thresholds are calibrated to the historical volatility of each sub-index skew (Table A.I). Unlike the CDS–Bond basis, the iTraxx skew is a derivative-on-derivative trade with no cash leg financed in repo. The two legs are traded under a single netting set—either centrally cleared at the same CCP or bilaterally with the same dealer under an ISDA Master Agreement and CSA—so that long and short positions offset for daily variation margin and balance-sheet purposes.

## B Index Construction and Weighting Schemes

Following Duarte et al. (2007), individual trades are treated as separate fictional hedge funds, each with its own entry date, exit date, and daily return stream. The strategy index on any given day is the weighted average of the returns of the funds that are active on that day. The equal-weighted

---

<sup>2</sup>Each constituent’s survival curve is bootstrapped from its CDS quotes, and the index’s intrinsic spread is the single running spread that reprices the resulting portfolio under the standard quoting convention (flat hazard rate, 40% recovery; see O’Kane, 2008).

(EW) scheme of Duarte et al. (2007) assigns identical weight  $1/K_t$  to every active trade:

$$r_t^{\text{EW}} = \frac{1}{K_t} \sum_{k=1}^{K_t} r_{k,t}. \quad (4)$$

The signal-weighted (SW) scheme, inspired by the curvature signal of Rebonato and Ronzani (2021), weights each trade proportionally to the absolute distance between the basis at entry and its historical equilibrium level,

$$r_t^{\text{SW}} = \sum_{k=1}^{K_t} \frac{|b_{k,0} - \theta_{t_k}|}{\sum_{j=1}^{K_t} |b_{j,0} - \theta_{t_j}|} r_{k,t}, \quad (5)$$

where  $b_{k,0}$  is the basis at the entry date  $t_k$  of trade  $k$ ,  $\theta_{t_k}$  is the expanding-window average of the cross-sectional basis from the start of the sample up to  $t_k$ , and  $r_{k,t}$  is the signed return of trade  $k$ , positive when the position profits, with the direction—long or short—fixed at entry. The absolute value makes the weight reflect the magnitude of the dislocation rather than its sign, which  $r_{k,t}$  already carries. The rationale follows Rebonato and Ronzani (2021): a basis that deviates further from its historical equilibrium represents a richer opportunity and warrants a larger allocation conditional on convergence. Signal-weighted indices dominate equal-weighted indices on both Sharpe and manipulation-proof performance for all three strategies (Appendix A.3, Table A.IV), and we adopt the SW scheme as the baseline throughout the paper; EW results in the appendix confirm that all findings are robust to this choice.

Market conditions are classified into three regimes based on the level of the iTraxx Europe Main 5-year spread: LOW (below 60 bps), MEDIUM (60 to 100 bps), and HIGH (100 bps and above). The same classification recurs in Section II, where we decompose alpha by regime. Transaction costs are applied at the individual trade level as entry and exit fees proportional to DV01, and fee levels are regime-dependent: bid-offer spreads and fees rise monotonically from LOW to HIGH, reflecting the empirical widening of dealer quotes during market dislocation. This tiered structure makes costs conservative precisely in the stress periods when arbitrage returns are largest. The full

fee schedule is reported in Table A.III; from this point forward, all strategy returns are net of these regime-dependent costs.

### C Performance Analysis

Table I reports monthly summary statistics for the three strategy indices. All strategy returns are excess returns by construction: the BTP Italia trade is self-financing (long and short legs are both repoed at the same general collateral rate), the CDS–Bond trade earns the z-spread net of the CDS premium (where the z-spread is itself measured over the swap curve, which proxies for the funding cost), and the iTraxx skew trade requires no capital beyond initial margin. Panel A describes the basis levels: all three series are persistent, with monthly first-order autocorrelations between 0.66 and 0.83, consistent with the slow-moving capital literature (Duffie, 2010; Brunnermeier and Pedersen, 2009; Mitchell et al., 2007) that we examine formally in Section IV. Panel B describes the strategy returns. The CDS–Bond basis produces the highest annualised return, 4.61% ( $t = 5.87$ ), and Sharpe ratio, 1.56, with an annualised volatility of 2.96%. iTraxx Combined returns 1.91% p.a. ( $t = 4.92$ ) with a Sharpe ratio of 1.25 over the longest sample of the three. BTP Italia earns 1.59% p.a. ( $t = 3.52$ ) with a Sharpe ratio of 0.98 over the shortest. All three strategies exhibit positive return skewness—0.81 for BTP Italia, 0.79 for the CDS–Bond basis, and 1.01 for iTraxx Combined—which cuts against the popular characterisation of fixed-income arbitrage as “picking up nickels in front of a steamroller” (Duarte et al., 2007). Maximum drawdowns are moderate and short-lived, as Figure 3 illustrates: the deepest drawdown is 4.4% for the CDS–Bond basis during the 2008 to 2009 credit crisis, 3.0% for BTP Italia, and 2.5% for iTraxx Combined, and all three are recovered within months.

**Manipulation-Proof Performance.** Sharpe ratios are vulnerable to manipulation by strategies whose dynamic trading rules alter the shape of the return distribution, which is precisely the structure that defines the indices we construct. Table II therefore reports the Manipulation-Proof Performance Measure (MPPM) of Goetzmann et al. (2007) for risk-aversion parameters  $\rho \in \{2, 3, 4\}$ :



a single-valued score that is independent of dollar scale, cannot be enhanced by information-free trading or leverage, and is consistent with standard financial market equilibrium—the appropriate benchmark for strategies, like ours, that use dynamic entry and exit rules.

The MPPM is nearly invariant to the risk-aversion level. For BTP Italia, the measure is 1.50, 1.48, and 1.47% per year at  $\rho = 2, 3, 4$ ; for CDS–Bond Basis, 4.43, 4.39, and 4.35; for iTraxx Combined, 1.88, 1.87, and 1.86. Because the concave transformation  $(1 + r_t)^{1-\rho}$  penalises strategies that earn their Sharpe ratio through occasional large losses, this stability—read together with the positive return skewness documented above—shows that the strategies’ risk-adjusted performance does not rely on tail events: an investor with  $\rho = 4$  evaluates these strategies almost identically to an investor with  $\rho = 2$ .

**Volatility-Managed Portfolios.** A second concern the literature has raised about fixed-income arbitrage is that unconditional Sharpe ratios may average over a time-varying risk-return trade-off, with periods of attractive compensation alternating with periods in which risk is poorly priced. Moreira and Muir (2017) propose a simple test. The managed portfolio scales exposure inversely to the previous month’s realised variance,

$$f_{t+1}^\sigma = \frac{c}{\hat{\sigma}_t^2} f_{t+1}, \quad (6)$$

where  $\hat{\sigma}_t^2$  is the realised variance over month  $t$  and the scaling constant  $c$  matches the unconditional standard deviation of the unmanaged portfolio. A significant managed-portfolio alpha relative to buy-and-hold is diagnostic of a time-varying ratio  $\mu_t/\sigma_t^2$ : an investor earns additional risk-adjusted returns by taking less risk precisely when volatility is high, which can only happen if the risk-return trade-off itself varies over time.

Table A.VI reports the results. The CDS–Bond basis produces a positive managed-portfolio alpha of 1.39% per year for the SW index, significant at the 10% level, consistent with a time-varying funding-liquidity mechanism: when volatility is low, intermediary balance sheets are healthy and arbitrage capital is available to exploit the negative basis; when volatility spikes, capital withdraws

and the trade-off deteriorates. For BTP Italia and iTraxx Combined, the managed-portfolio alphas are slightly negative ( $-0.22\%$  and  $-0.16\%$  p.a., both insignificant), so the realised-variance signal identifies no exploitable variation in the risk-return trade-off for these two strategies. The contrast between the one strategy that finances a cash bond and the two that do not is itself informative: Section II shows that, once benchmark factor exposures are controlled for, the funding-liquidity pattern documented here for the CDS–Bond basis generalises into a regime effect common to all three strategies.

## II Benchmark Factor Models

This section tests whether the alpha of the three strategies survives the canonical benchmark spanning regressions of the fixed-income arbitrage literature—Duarte et al. (2007), Fung and Hsieh (2004), and Brooks et al. (2020)—and whether the residual alpha varies with intermediary funding stress, as the slow-moving capital framework of Duffie (2010) predicts. Section A describes the three specifications and the construction of their euro-denominated counterparts. Section B presents the regression results and three complementary evaluations: the alpha by stress regime, the conditional alpha model of Christopherson et al. (1998), and an out-of-sample re-estimation in which the benchmark hedge is formed from past-only loadings.

### A Benchmark Specifications

Three benchmark specifications are widely used in the empirical literature on fixed-income arbitrage and active fixed-income management, and they provide the natural reference points against which our strategies should be evaluated.

Duarte et al. (2007), in the paper that motivates our research design, test whether fixed-income arbitrage excess returns are compensation for exposures to equity, banking-sector, and term-structure

risk:

$$r_{i,t} = \alpha_i + \beta_1 Mkt_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 RS_t + \beta_6 RI_t + \beta_7 RB_t + \beta_8 R2_t + \beta_9 R5_t + \beta_{10} R10_t + \varepsilon_{i,t}, \quad (7)$$

where  $Mkt$ ,  $SMB$ ,  $HML$ , and  $UMD$  are the Fama–French–Carhart equity factors (market, size, value, and momentum),  $RS$  is the S&P bank stock index excess return,  $RI$  and  $RB$  are A/BAA-rated industrial and bank bond index excess returns, and  $R2$ ,  $R5$ ,  $R10$  are the CRSP Fama bond portfolios at two-, five-, and ten-year maturities.

Fung and Hsieh (2004) propose a seven-factor *asset-based style* model that tests whether hedge fund returns are explained by equity, fixed-income, and trend-following exposures:

$$r_{i,t} = \alpha_i + \beta_1 S\&P_t + \beta_2 SC-LC_t + \beta_3 BdOpt_t + \beta_4 FXOpt_t + \beta_5 ComOpt_t + \beta_6 10Y_t + \beta_7 CredSpr_t + \varepsilon_{i,t}, \quad (8)$$

where  $S\&P$  is the S&P 500 excess return,  $SC-LC$  is the small-minus-large-cap total return,  $BdOpt$ ,  $FXOpt$ , and  $ComOpt$  are the three trend-following factors constructed as lookback straddle returns on bond, currency, and commodity futures,  $10Y$  is the excess return on a 10-year government bond total-return index, and  $CredSpr$  is the return spread between BAA-rated corporate bonds and 10-year government bonds.

Brooks et al. (2020) take a different angle, decomposing the returns of active fixed-income managers into eight *traditional risk premia* organised along three channels of risk-taking—duration, credit, and volatility selling:

$$r_{i,t} = \alpha_i + \beta_1 Term_t + \beta_2 GlTerm_t + \beta_3 GlAgg_t + \beta_4 InflLink_t + \beta_5 CorpCred_t + \beta_6 EmDebt_t + \beta_7 EmCurr_t + \beta_8 USTVol_t + \varepsilon_{i,t}, \quad (9)$$

where  $Term$ ,  $GlTerm$ ,  $GlAgg$ , and  $InflLink$  are the total returns on the Treasury, the currency-hedged global Treasury, global aggregate, and global inflation-linked bond indices, each in excess of cash,

*CorpCred* is a high-yield corporate bond index in excess of cash, *EmDebt* is the excess return on a hard-currency emerging-market debt index, *EmCurr* is the return on an equal-weighted basket of emerging-market currencies, and *USTVol* is the return on delta-hedged straddles on 10-year Treasury futures.

Two ingredients of the original specifications are replaced with tradable counterparts. In Fung and Hsieh (2004), the bond and credit factors are yield and spread *changes*; a yield change is not the return on an investable position, so we use the corresponding total-return series (the *10Y* and *CredSpr* definitions above). In Brooks et al. (2020), *CorpCred* blends high yield with leveraged loans; loan indices are quote-based and not investable at the monthly horizon, so we retain the high-yield leg only. With every regressor a realised excess return, the intercept in each specification keeps its interpretation as a Jensen alpha.

Each benchmark is estimated in two parallel panels. The first uses the original U.S.-denominated factors as specified by the respective authors, which is the natural way to replicate each framework as it was first published. The second uses euro equivalents, since all three strategies trade European instruments and may therefore be better explained by European factor exposures. The euro panel is the primary specification and is reported in the main text; the US-factor estimates are reported in Appendix A.5.

For the euro panel, the Fama–French factors—used directly in Duarte et al. (2007) and indirectly in Fung and Hsieh (2004)—are sourced from the Kenneth French data library; because these are reported in USD, we convert them to EUR following Glück et al. (2021). For long factors (market excess return):

$$F_t^{\text{EUR}} = \frac{1 + F_t^{\text{USD}} + r_{f,t}^{\text{USD}}}{1 + r_t^{\text{USD/EUR}}} - 1 - r_{f,t}^{\text{EUR}}, \quad (10)$$

and for long-short factors (SMB, HML, UMD):

$$LS_t^{\text{EUR}} = \frac{LS_t^{\text{USD}}}{1 + r_t^{\text{USD/EUR}}}, \quad (11)$$

where  $F_t^{\text{USD}}$  and  $LS_t^{\text{USD}}$  are the original factors in US dollars,  $r_{f,t}^{\text{USD}}$  and  $r_{f,t}^{\text{EUR}}$  are the US and euro risk-free rates (Fama–French T-bill rate and one-month German bill, respectively), and  $r_t^{\text{USD/EUR}}$  is the period return on the dollar-to-euro exchange rate. All remaining factors enter the euro panel with native European counterparts: European bank stocks, euro-area corporate bond indices, and German government bonds in the Duarte specification; euro-area term, credit, and inflation-linked premia in Brooks et al.; and the 10-year Bund total-return index and the BBB–Bund return spread in Fung and Hsieh. The Fung and Hsieh trend-following lookback straddle factors are currency-independent by construction (D. Hsieh, private communication) and enter both panels identically. The complete factor list is reported in Appendix A.8.1.

All specifications are estimated by ordinary least squares with Newey–West HAC standard errors (Newey and West, 1987), with the truncation lag set by the automatic rule of Newey and West (1994),  $\lfloor 4(T/100)^{2/9} \rfloor$ , which equals four lags for every sample in this paper. Variance inflation factors for each specification are reported in Appendix A.7. Two groups of regressors are highly collinear—the maturity-sorted bond portfolios  $R2$ ,  $R5$ ,  $R10$  in Duarte and the global bond benchmarks in Brooks et al.—which inflates the standard errors of those loadings but leaves inference on the intercept unaffected.

## B Results and Discussion

**Factor Model Regressions.** Tables III, IV, and V report the full euro-panel regression results; the US-factor counterparts are Tables A.VII, A.VIII, and A.IX in Appendix A.5. Across all three frameworks, all three strategies, and both currency denominations, the estimated intercept is positive and significant at the 1% level. The annualised alphas are economically meaningful: +4.0 to +4.6% p.a. for the CDS–Bond basis, +1.6 to +2.2% for iTraxx Combined, and +1.6 to +2.0% for BTP Italia.

The adjusted  $R^2$  is low across all specifications, ranging from zero to 18%, so the standard factor sets leave most of the return variation unexplained—a question we take up in Section III. The significant loadings that do emerge differ across strategies and are economically intuitive. The

CDS–Bond basis loads on credit and rates returns: the A-rated industrial bond return is positive and significant in the Duarte specification, and the 10-year government bond and credit-spread returns are positive and significant in Fung–Hsieh, in both currency denominations—the natural exposures of a long-credit, cash-financed position. BTP Italia and iTraxx Combined load instead on equity factors, with small negative coefficients: market and size for BTP Italia, and size, value, and momentum for iTraxx Combined, in both the Duarte and the Fung–Hsieh specifications. The Brooks et al. specification, which uses broad risk premia rather than individual asset returns, produces fewer significant loadings but delivers comparable alpha estimates, reinforcing that the intercepts are not an artefact of any single factor set.

**Cross-Model Comparison and Regime Analysis.** Table VI consolidates the alpha estimates from the three frameworks. Panel A reports the unconditional alphas side by side: for each strategy, all three frameworks deliver positive and significant alpha, and the estimates are consistent across specifications despite their different factor structures.

The discrete regime decomposition—alpha estimated separately in the LOW, MEDIUM, and HIGH stress regimes on the iTraxx Europe Main 5-year spread, with the same 60 and 100 basis-point cutoffs that govern the transaction-cost tiers of Section B (Appendix A.2.3)—is reported for each framework in Appendix A.6. In every cell of those tables, alpha rises monotonically from the LOW to the HIGH regime, and the HIGH-regime alpha is three to seven times the LOW-regime alpha across the nine strategy–framework combinations. The mechanism is the one Duffie (2010) formalises: when intermediary capital is abundant, arbitrage forces close mispricings quickly and compress the residual premium; when it is scarce—as it is during episodes of widening credit spreads—arbitrageurs retreat or are forced to unwind, mispricings widen, and the premium for those who can still deploy capital rises. This regime dependence motivates the cross-strategy analysis of Section IV, where we test whether the same intermediary balance-sheet shocks drive co-movement of mispricings across the three strategies.

**Conditional Alpha.** Panel B of Table VI replaces the discrete regime split with the conditional performance evaluation framework of Christopherson et al. (1998), in which the alpha varies continuously with a predetermined, publicly observable state variable. We estimate the full conditional model

$$r_{i,t} = \alpha_0 + \alpha_1 z_{t-1} + (\beta + \delta z_{t-1})' \mathbf{X}_t + \varepsilon_t, \quad (12)$$

where  $\mathbf{X}_t$  collects the factor returns of the chosen benchmark and  $z_{t-1}$  is the one-month-lagged level of the iTraxx Europe Main 5-year spread, standardised to mean zero and unit variance over the estimation sample. Lagging the state variable keeps it in the investor's information set, as the conditional framework requires, and conditioning the loadings through  $\delta$  prevents time-varying betas from masquerading as conditional alpha. The intercept  $\alpha_0$  is then the alpha at the average stress level, and  $\alpha_1$  is the additional alpha per one-standard-deviation increase in funding stress. Because fixed-bandwidth HAC  $t$ -tests over-reject in conditional models at these sample sizes (Kiefer and Vogelsang, 2005; Lazarus et al., 2018), significance of  $\alpha_1$  is assessed by a moving-block residual bootstrap with the null imposed ( $B = 9,999$ , block length nine months); HAC inference, reported as a robustness check, gives the same pattern.

The estimates of  $\alpha_1$  in Panel B are positive in all nine strategy–framework cells, and the intercept  $\alpha_0$  remains significant in every case, so a baseline alpha persists at the average stress level. For the two institutionally held strategies,  $\alpha_1$  is significant in every framework: a one-standard-deviation increase in the iTraxx Main spread adds 1.7 to 3.1 percentage points of annual alpha for the CDS–Bond basis and 0.9 to 1.5 for iTraxx Combined. For BTP Italia,  $\alpha_1$  is significant only under the Duarte specification—a weaker stress sensitivity consistent with the retail holder base that Section IV develops. Across the nine cells, the implied alpha at  $z = +1\sigma$  is two to nine times the alpha at  $z = -1\sigma$ : the bulk of the unconditional alpha is earned when funding stress is elevated, the signature of the slow-moving capital mechanism.

**Out-of-Sample Alpha.** A residual concern, central to the recent literature on performance evaluation (Moreira and Muir, 2017; Cederburg et al., 2020), is that the benchmark loadings in Ta-

bles III–V are estimated in sample, so the surviving alpha could reflect an in-sample-optimal combination of the factors rather than an abnormal return an investor could have earned in real time. Panel C of Table VI addresses this directly. For each strategy and each benchmark, we re-estimate the loadings on an expanding past-only window in the recursive out-of-sample design of Welch and Goyal (2008), form the factor hedge from the lagged slopes, and define the out-of-sample alpha as the mean of the realised hedged return  $e_{i,t} = r_{i,t} - \hat{\beta}'_{i,t-1} \mathbf{f}_t$ , tested with the Newey and West (1994) HAC correction. The expanding window is initialised with a 60-month minimum, so that the first fit has at least five observations per parameter under the ten-factor Duarte et al. (2007) specification, and windows with near-singular factor designs are skipped. The out-of-sample alpha remains positive and significant for all three strategies and all three frameworks. The point estimates are close to the in-sample alphas of Panel A, with only modest attenuation, and the ranking across strategies is preserved, with the CDS–Bond basis the largest. The rolling-window counterpart, which additionally relaxes the assumption of constant loadings, is reported in Appendix A.6.4 and delivers the same conclusion.

**Rolling Alpha.** A natural concern with regime-based or conditional analyses is that the estimated relationship between alpha and stress may be driven by a single sub-period—the global financial crisis of 2008 to 2009, the European sovereign debt crisis of 2011 to 2012, or the COVID-19 shock of March 2020—rather than by a stable feature of the data. Figure 4 plots rolling 36-month alpha estimates for each strategy under each of the three benchmark specifications. The rolling alpha is positive across essentially the whole sample for all three strategies, so the estimates are not the product of a single favourable sub-period, and the three benchmark estimates move broadly together within each panel, reinforcing the cross-model robustness of Panel A. Detailed sub-period splits for each framework are reported in Appendix A.6.

Taken together, the five sets of results in this section establish two findings that are central to the rest of the paper. First, the alpha is robust: it survives three widely used benchmark frameworks, in both currency denominations, in every sub-period we examine, and when each benchmark hedge is



re-estimated out of sample. Second, the alpha is not constant: it rises systematically when intermediary capital is scarce, whether stress enters as a discrete regime or as a continuous predetermined state, and the high-stress alpha is a multiple of the calm-market alpha. The first finding says the observed excess returns are not explained by the systematic exposures the prior literature has identified; the second points to an economic mechanism but not yet to the specific risk factors behind the residual variation. Section III takes up that question on a much larger candidate factor universe.

### III Factor Selection and Spanning Regressions

The benchmark factor models of Section II are designed to capture known systematic exposures, but they leave open the question of whether a broader set of risk factors can better explain strategy excess returns and absorb the residual alpha. Starting from the candidate universe of 75 tradable risk factors described in Section A, we proceed in two steps. Section B extracts a small number of latent factors by principal components, following Connor and Korajczyk (1986, 1988), and tests whether these diffuse combinations span strategy excess returns. Section C turns to a supervised alternative—best-subset selection (Bertsimas et al., 2016)—that picks the individual factors most relevant for each strategy from the same candidate set. As in Section II, each layer is subjected to the conditional alpha model (Section D) and to an out-of-sample re-estimation of the hedge, and Section E consolidates the alpha estimates across all layers of the paper.

#### A Candidate Factor Universe

We assemble a candidate set of 75 risk factors from a comprehensive review of the empirical asset pricing, intermediary, and macro-finance literatures. Each factor is selected on the basis of a published theoretical or empirical link to fixed-income arbitrage returns, intermediary balance-sheet constraints, or broad market risk conditions. The factors are organised into six economic categories: credit risk (Panel A), liquidity and funding (Panel B), equity (Panel C), volatility (Panel D), macro-financial (Panel E), and interest rates and term structure (Panel F). The complete list with definitions, data sources, and references is reported in Table A.XV (Appendix A.8.1). This candi-

date universe is the common input to both the PCA of Section B and the best-subset selection of Section C.

Every candidate is tradable: each series is the excess return on an implementable position—an index, a long-short portfolio, a derivative overlay, or a DV01-neutral curve trade. State variables enter through traded counterparts rather than as raw proxies; the intermediary capital factor of He et al. (2017), for example, enters as the excess equity return on the primary-dealer holding companies (correlation 0.92 with the underlying capital ratio), and aggregate liquidity enters through the traded liquidity portfolio of Pástor and Stambaugh (2003). The restriction matters for interpretation: with every regressor a realised excess return, the intercept  $\alpha_i$  in the spanning regressions of this section is a Jensen alpha—the return on an implementable strategy hedged with implementable positions—rather than the residual of a projection on non-traded state variables.

## B Principal Component Analysis

When the number of candidate factors is large relative to the sample size, regressing strategy returns on the full set is unreliable: the strong collinearity of the candidates inflates the variance of the coefficient estimates and overfits the sample. The standard response is to summarise the common variation in a small number of principal components and test whether these diffuse combinations span strategy returns; estimating latent factors from a large panel and using them to measure the abnormal return of a managed position is the performance-measurement framework of Connor and Korajczyk (1986, 1988).

We estimate the components from the full estimation sample rather than from a trailing window. For an in-sample spanning test—as distinct from a return *forecast*—using the entire sample to estimate the loadings is both standard and desirable: it delivers the most precise estimate of the common-factor space, and with the cross-sectional dimension  $N$  and the time dimension  $T$  both large it yields consistent and asymptotically normal second-stage estimates of  $\alpha_i$  and  $\beta_i$  (Giglio and Xiu, 2021). We standardise each factor to zero mean and unit variance over the estimation sample and retain the first  $K = 8$  components, the number selected by the Bai and Ng (2002)  $IC_{p2}$

criterion (the less parsimonious  $IC_{p1}$  indicates thirteen). The eight components capture 64.7% of the variance in the balanced factor panel (Figure 5), and robustness of the alpha to  $K \in \{5, 8, 11, 13\}$  is reported in Appendix A.9.1.

The spanning regression takes the form

$$r_{i,t} = \alpha_i + \beta_i' \mathbf{PC}_t + \varepsilon_{i,t},$$

where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and  $\mathbf{PC}_t$  is the  $K \times 1$  vector of contemporaneous principal component scores. Inference uses Newey–West HAC standard errors (Newey and West, 1987). Because the regressors are estimated principal components, the second-stage standard errors are in principle subject to a generated-regressor adjustment; under  $N, T \rightarrow \infty$  with  $\sqrt{T}/N \rightarrow 0$  the adjustment is asymptotically negligible (Bai and Ng, 2006), and the out-of-sample design below, which re-estimates the components on past-only data, provides the finite-sample check.

Table VII reports the estimates. The intercept remains positive and significant at the 1% level for all three strategies: +1.44% p.a. for BTP Italia, +4.78% for CDS–Bond Basis, and +2.21% for iTraxx Combined. The adjusted  $\bar{R}^2$  is low throughout, reaching at most 13.4%: the diffuse principal components do not span strategy excess returns. Table VIII reports, for each retained component, the candidate factors with the largest absolute loading. The components admit a clear economic reading: PC1 is dominated by credit and equity-market factors, PC2 by global rates and bond returns, and PC3 by swap spreads and dealer CDS.

**Out-of-Sample Alpha.** As in Section II, the full-sample loadings could reflect an in-sample-optimal combination of the factors rather than a return an investor could have earned in real time (Moreira and Muir, 2017; Cederburg et al., 2020). Panel A of Table IX addresses this directly. At each evaluation date we re-estimate the principal components *and* the spanning betas on a past-only window—expanding (initialised at 60 months) as the baseline and a 60-month rolling window as a robustness check—and define the out-of-sample alpha as the mean of the hedged return, tested with the Newey and West (1994) HAC correction. Because the components are themselves re-

estimated within each window, the design is invariant to the sign and rotation indeterminacy of principal components. The out-of-sample alpha remains positive and significant at the 1% level for all three strategies: +1.61% p.a. for BTP Italia, +4.55% for CDS–Bond Basis, and +1.85% for iTraxx Combined under the expanding window, close to the in-sample estimates of Table VII, so the alpha is not an artefact of full-sample factor estimation.

Because PCA is an unsupervised method—it extracts factors that maximise variance in the candidate set, not factors that explain variation in strategy returns—it cannot identify strategy-specific risk drivers. This motivates the supervised selection of the next subsection.

### C Best-Subset Selection

**Preprocessing.** The candidate factors are preprocessed for selection. Six candidates are near-duplicates of another factor in the universe, with absolute correlation above 0.95; one member of each pair is removed, leaving  $p = 69$  candidates in the selection design. Each factor is standardised to unit  $L_2$ -norm, so that which factors enter is invariant to scale; point estimates and inference are subsequently computed on the original units. Even at  $p = 69$ , the space of possible factor subsets— $2^{69} \approx 6 \times 10^{20}$ —rules out exhaustive search and calls for a selection method with an explicit cardinality control.

**Selection.** For each strategy we solve the best-subset problem of Bertsimas et al. (2016),

$$\min_{\alpha, \beta} \frac{1}{T} \sum_{t=1}^T (r_t - \alpha - \beta' \mathbf{f}_t)^2 \quad \text{subject to} \quad \|\beta\|_0 \leq k, \quad (13)$$

computed with the polynomial-time splicing algorithm of Zhu et al. (2020). The  $\ell_0$  constraint has two properties that penalised methods lack and that matter in our design. First, a factor that enters is not shrunk, so among a cluster of correlated candidates the constraint keeps the most informative member and drains its near-duplicates, instead of splitting a common signal across the cluster or shrinking it toward zero as the LASSO does. Second, the cardinality is controlled directly: we fix  $k = 8$  for parsimony and cross-strategy comparability, rather than selecting the count by

an information criterion or cross-validation, whose objective is nearly flat—and whose argmin is therefore ill-determined—on alpha-dominated series with weak factor signal. Table A.XIX (Appendix A.10.1) re-estimates the selection at  $k \in \{5, 8, 10\}$ : the selected sets are broadly nested and the in-sample fit plateaus, so the conclusions do not hinge on the choice. All-subsets model comparison has asset-pricing precedent in Barillas and Shanken (2018).

**Post-Selection Inference.** Given the selected set  $\hat{S}_i$ , we re-estimate the spanning regression by unrestricted OLS on the original (un-standardised) factors,

$$r_{i,t} = \alpha_i + \beta_i' \mathbf{f}_{\hat{S}_i,t} + \varepsilon_{i,t}, \quad (14)$$

with Newey–West HAC standard errors at the Newey and West (1994) automatic lag. Because the set is selected on the same data, we complement the analytic  $t$ -tests with a stationary block bootstrap (Politis and Romano, 1994) of the post-selection regression (9,999 replications, expected block length nine months) and report the percentile 95% confidence interval for alpha; post-double-selection (Belloni et al., 2014), which guards the alpha against omitted-control bias from the selection step, is one of the cross-checks in Appendix A.10.2.

**Results.** Table X reports the post-selection estimates. Alpha is significant at the 1% level for all three strategies: +1.6% p.a. for BTP Italia (bootstrap CI [0.8, 2.2]), +4.8% for CDS–Bond Basis (CI [2.9, 6.6]), and +2.7% for iTraxx Combined (CI [1.3, 3.5]). With eight factors each, the models deliver adjusted  $\bar{R}^2$  of 22%, 21%, and 18%—above any benchmark framework of Section II and the PCA above—yet absorb none of the three alphas.

The selected sets differ across strategies in ways that mirror each trade’s structure. The BTP Italia set is built around the German swap-spread curve (the 2-, 5-, and 10-year DV01-neutral swap-spread trades), traded liquidity, and defensive equity and bond styles; the significant loadings are the fixed-income defensive factor ( $-0.21$ ,  $t = -4.3$ ) and quality ( $+0.04$ ,  $t = 1.9$ ). The CDS–Bond set maps onto the intermediary channel: the strategy loads negatively on the prime-broker CDS fac-

tor ( $-0.22, t = -2.3$ )—it pays off when dealer credit deteriorates—and positively on the default premium ( $+0.11, t = 3.2$ ) and the industrial bond return ( $+0.22, t = 3.2$ ), the natural exposures of a leveraged long-credit position, with smaller loadings on high yield, emerging-market debt, currency trend, and the curvature trade. The iTraxx set is drawn from credit-cycle, volatility, and style factors: value ( $-0.03, t = -2.2$ ), betting-against-beta ( $-0.03, t = -3.4$ ), fixed-income momentum ( $-0.04, t = -2.7$ ), and the one-month variance-swap return ( $-0.99, t = -2.5$ ), with the credit-cycle and implied-volatility candidates entering but insignificant. No factor is shared by all three strategies, and any two sets overlap in at most one factor: the alpha that survives is not compensation for a single common exposure, and its origin is strategy-specific.

**Out-of-Sample Alpha.** Panel B of Table IX applies the recursive design of Section B to the selected models: the best-subset factor set is frozen, only the hedge loadings are re-estimated on the past-only window, and the realised hedged return is tested with the HAC correction—the real-time abnormal return of the selected model, with no per-window re-selection. The out-of-sample alpha is positive and significant at the 1% level for all three strategies under both the expanding and the rolling scheme, and of the same order as the in-sample estimates of Table X.

Two further checks confirm that the selection itself is not the source of the result. Re-selecting with the adaptive elastic net of Zou and Zhang (2009)—a convex, oracle penalised estimator with data-driven cardinality—and with post-double-selection recovers substantially overlapping factor sets and the same alpha verdict (Appendix A.10.2).

## D Conditional Alpha

Table XI decomposes alpha across the three stress regimes on the iTraxx Main 5-year level—LOW, MEDIUM, and HIGH, with the same 60 and 100 basis-point cutoffs used throughout the paper—under both the PCA and the best-subset controls. In every cell, alpha rises monotonically from the LOW to the HIGH regime. The CDS–Bond basis shows the sharpest gradient, and BTP Italia the familiar pattern of an alpha concentrated away from calm markets: under the PCA controls it

risks from +0.4% (insignificant) in LOW to +3.3% in HIGH, and under the best-subset controls from +0.5% to +3.0%. For iTraxx Combined the alpha is significant in every regime and roughly doubles from LOW to HIGH under both specifications.

The Christopherson et al. (1998) conditional model of Section II—the full specification of equation (12), with the lagged standardised stress state conditioning both the alpha and the loadings, and  $\alpha_1$  tested by the moving-block bootstrap—is re-estimated on each factor layer in Appendices A.9.3 and A.10.3. Under the PCA controls,  $\alpha_1$  is significant at the 1% level for all three strategies. Under the best-subset controls, it remains large and significant for the CDS–Bond basis (+2.55% per standard deviation,  $t = 4.27$ ), while for BTP Italia and iTraxx Combined the continuous slope is absorbed once the loadings themselves are allowed to vary with stress—the regime gradient of Table XI survives, but it is carried by the conditional loadings rather than by a conditional intercept. Sub-period stability of the alpha estimates is confirmed in Appendix A.9.2, and rolling 36-month alpha estimates with regime shading are plotted in Appendix A.10.4.

## E Alpha Across Layers

Table XII consolidates the answer to the question this section and the previous one pose. Each cell reports the alpha and adjusted  $\bar{R}^2$  from one layer of controls: the three benchmark frameworks of Section II, the principal components, and the best-subset selection. Moving down the table, the explanatory power rises—from  $\bar{R}^2$  between 0% and 15% under the benchmarks to 18–22% under the best-subset models—while the alpha barely moves: +1.6 to +2.0% p.a. for BTP Italia, +4.0 to +4.8% for the CDS–Bond basis, and +1.8 to +2.7% for iTraxx Combined, significant at the 1% level in every cell. A supervised search over 75 tradable candidates, free to pick whichever eight factors best explain each strategy, absorbs no more of the alpha than the fixed benchmarks do.

Two conclusions follow. First, the excess returns documented in Section C are not compensation for systematic risk exposures, whether those exposures are specified ex ante, extracted as latent components, or selected by the data. Second, the alpha that survives is state-dependent, rising when intermediary funding stress is elevated, under every layer of controls. The natural question

is why such a premium persists—and whether the three mispricings that generate it are connected through the balance sheets of the intermediaries that trade them. Section IV takes up that question.

## IV Cross-Strategy Interdependencies

The preceding sections have established that all three strategies generate alpha that survives both classical benchmark and data-driven factor controls, and that this alpha rises during periods of intermediary funding stress. A natural question follows: are the three mispricings linked to one another, and if so, what economic mechanism ties them together? The slow-moving capital literature introduced above—Duffie (2010), Brunnermeier and Pedersen (2009), Mitchell and Pulvino (2012), Boyarchenko et al. (2016), and Fleckenstein et al. (2014)—yields five testable predictions about how arbitrage mispricings behave when they draw on a common pool of slow-moving intermediary capital: (P1) mispricings serviced by a common pool of intermediary capital should co-move (Duffie, 2010); (P2) the co-movement should intensify when that capital is scarce (Duffie, 2010; Brunnermeier and Pedersen, 2009); (P3) mispricing changes in one strategy should contain information for predicting those of others, with effects propagating asymmetrically along the liquidity gradient (Mitchell and Pulvino, 2012; Fleckenstein et al., 2014); (P4) the persistence of mispricing levels should reflect the funding-dependence of each arbitrage trade, with strategies that require continuous balance-sheet funding displaying the longest half-lives (Boyarchenko et al., 2016); and (P5) a common factor extracted from the co-movement should be explained by intermediary balance-sheet variables rather than by macroeconomic fundamentals (Duffie, 2010; Brunnermeier and Pedersen, 2009).

A feature of our three-strategy setting gives these tests their identifying power. All three arbitrages are run by the same type of leveraged institutional arbitrageur, but the instruments differ in who holds them: the CDS–Bond and iTraxx legs trade among institutions, whereas the BTP Italia inflation-linked bond is held predominantly by retail investors. This heterogeneity in the holder base is the source of identification we exploit throughout—it lets us separate the slow-moving-capital mechanism, under which co-movement depends on sharing a capital pool, from a common



macroeconomic driver, under which all three mispricings would move together regardless of who holds the instruments.

We devote one subsection to each prediction, stating in each case the prediction, the test, and the verdict where the evidence is presented. Section [A](#) first defines the mispricing series and the stress regimes. Sections [B–F](#) then test P1 through P5 in turn. Section [G](#) interprets the evidence as a whole, explains why these arbitrage opportunities persist in post-crisis European credit markets, and collects the verdicts in a summary mapping.

## **A Mispricing Construction and Stress Regimes**

Following Duffie ([2010](#)), we work with mispricing magnitudes  $m_{i,t}$  rather than strategy returns, because it is the level of the mispricing—the distance of the basis from its no-arbitrage value—that measures the degree to which capital is failing to close the gap. For each strategy,  $m_{i,t}$  is defined as the absolute value of the basis, sign-adjusted so that an increase  $\Delta m > 0$  denotes a widening of the mispricing for all three strategies. The BTP Italia mispricing is the cross-sectional mean of the absolute basis across all outstanding issues with at least six months to maturity; the CDS–Bond mispricing is the cross-sectional mean of the absolute value of all negative single-name bases in the iTraxx Main universe on each day; and the iTraxx Combined mispricing is the cross-sectional mean of the absolute skew across the four on-the-run sub-indices. All three series are aggregated to monthly frequency. Market conditions are classified into three regimes—LOW, MEDIUM, and HIGH—based on the iTraxx Europe Main 5-year spread, with thresholds at 60 and 100 basis points. These are the same thresholds that define the stress-dependent transaction-cost tiers applied to every return series in the paper ([Appendix A.2.3](#)) and that demarcate the HIGH stress regime in the benchmark and conditional-alpha analyses of [Sections II and III](#).

## **B P1: Cross-Market Co-Movement**

The first prediction is that mispricings serviced by a common pool of intermediary capital should co-move: when the capital that closes one arbitrage is withdrawn, the same withdrawal leaves the

others unhedged, so their mispricings widen together. We test this with pairwise Pearson correlations of mispricing changes  $\Delta m$ , computed unconditionally and within each stress regime (Table XIV, Panel B).

The prediction holds for the pair whose underlying instruments are all institutionally held and traded—the CDS–Bond basis and the iTraxx skew, hereafter the “institutional pair.” Their mispricing changes correlate at +0.19 unconditionally, significant at the ten percent level, and the correlation rises monotonically across regimes from  $-0.21$  in the LOW regime to  $+0.32$  in the HIGH regime, with the LOW-to-HIGH movement significant at the one percent level under a Fisher  $z$ -test. The Forbes and Rigobon (2002) correction for the heteroskedastic increase in mispricing volatility under stress attenuates the shift but leaves it significant at the ten percent level, so the intensification reflects a genuine increase in the cross-market link rather than the mechanical inflation of correlations that volatility differences alone would produce. When dealer balance sheets are healthy the two markets price almost independently; when balance sheets are depleted, common funding shocks force the simultaneous unwinding of positions in both, and the link turns strongly positive. P1 is confirmed for the institutional pair.

The BTP Italia pairs move in the opposite direction, and it is worth being explicit that this does not contradict P1. The CDS–Bond versus BTP Italia correlation is  $-0.30$  unconditionally, significant at the one percent level, and deepens from  $-0.13$  in LOW to  $-0.47$  in HIGH, the latter significantly different from zero at the five percent level. The BTP Italia versus iTraxx correlation is negative but indistinguishable from zero on the full sample ( $-0.10$ ), and reverses sign across regimes, from  $+0.15$  in LOW to  $-0.14$  in HIGH. The dynamic conditional correlations estimated by a bivariate GARCH–DCC model (Appendix A.11.1, Figure A.2) show that the negative sign of both BTP Italia pairs is a persistent feature of the data—the conditional correlation between CDS–Bond and BTP Italia remains below zero in every month of the sample, with mean  $-0.29$ , and that between BTP Italia and iTraxx is negative throughout, with mean  $-0.10$ —not an artefact of averaging over offsetting positive and negative episodes.

P1 does not predict that all mispricings co-move—only that those drawing on a common capital pool do. The BTP Italia inflation-linked bond is retail-held and does not draw on the institutional pool, so P1 predicts no positive co-movement, and there is none. This is exactly the holder-base heterogeneity that identifies the mechanism: the sign of the co-movement turns on whether the capital pool is shared—positive within the institutional pool, negative against the retail-held basis—rather than being common to all three, as a macroeconomic driver would require. We take up the economics of the negative sign in Section G.

### C P2: Stress Amplification

The second prediction sharpens the first: the co-movement should not merely be present but should intensify as intermediary capital becomes scarce. The regime correlations of the previous subsection already provide one reading—co-movement within the institutional pair rises from  $-0.21$  in LOW to  $+0.32$  in HIGH—but they treat stress as a discrete state and cannot say whether transmission rises continuously with it, nor separate the incremental effect of stress from the average level of transmission. We therefore test the prediction parametrically with an interaction regression:

$$\Delta m_{i,t} = \alpha + \beta_0 \Delta m_{j,t} + \beta_1 (\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t, \quad (15)$$

where  $z_t$  is the standardised iTraxx Main 5-year spread, a proxy for intermediary funding stress. The coefficient  $\beta_1$  captures the incremental transmission during stress:  $\beta_1 > 0$  means that a given shock to  $\Delta m_j$  has a larger effect on  $\Delta m_i$  when balance sheets are strained. The theory is explicit on the mechanism: Duffie (2010) notes that a depleted intermediary is “less able or predisposed to absorb supply and demand shocks,” and Brunnermeier and Pedersen (2009) show that the destabilising effects of funding constraints are “largest when traders are near their constraint.”

Table XVI reports the results. Within the institutional pair, the interaction coefficient on iTraxx mispricing carries the predicted positive sign, and the direct effect of stress on the CDS–Bond basis is significant in both interaction regressions, at the one percent level ( $\gamma = +1.60$ ) when

BTP Italia mispricing is the explanatory variable and at the five percent level ( $\gamma = +0.95$ ) when iTraxx mispricing is. Three pieces of evidence converge to support P2 within the institutional pair. First, the regime correlation rises monotonically across stress states (Section B). Second, the Forbes–Rigobon correction confirms that this rise reflects a genuine increase in the cross-market link rather than a heteroskedasticity artefact. Third, the within-pair interaction term  $\beta_1$  is correctly signed and, in the CDS–Bond  $\rightarrow$  iTraxx direction, significant at the ten percent level ( $\beta_1 = +0.08$ ). All three readings point the same way, and P2 is confirmed for the institutional pair. For the pairs involving BTP Italia the interaction coefficient is significantly negative ( $\beta_1 = -0.64$  for CDS–Bond  $\rightarrow$  BTP Italia, significant at the one percent level): stress widens, rather than amplifies, the link between the institutional pair and the retail-held basis—the mirror image of P2 that the holder-base heterogeneity of Section G explains.

#### D P3: Lead–Lag Transmission

The third prediction concerns timing: if capital moves slowly across markets, a mispricing change in one strategy should help predict future changes in another, and the transmission should run from more liquid to less liquid markets as capital is reallocated across the dealer network. We test predictability with spanning regressions in the design of Fleckenstein et al. (2014),

$$\Delta m_{i,t} = \alpha + \sum_{j \neq i} \sum_{l=0}^L \beta_{j,l} \Delta m_{j,t-l} + \varepsilon_t, \quad (16)$$

with  $L = 2$  lags and Newey–West HAC standard errors. The objects of interest are the lagged coefficients: a significant loading on  $\Delta m_{j,t-1}$  means that last month’s mispricing in strategy  $j$  predicts this month’s in strategy  $i$ , a genuine lead–lag relation rather than contemporaneous co-movement.

Table XV reports the results. The significant coefficients trace a liquidity pecking order: one-month-lagged iTraxx changes predict the widening of the CDS–Bond mispricing with coefficient  $+0.71$  at the ten percent level, while CDS–Bond changes enter the BTP Italia equation contemporaneously with coefficient  $-0.42$  at the five percent level. The two institutional mispricings thus

lead the BTP Italia basis with a delay of zero to one month, the timing dictated by the speed at which dealers can rotate inventory across markets.

The spanning regressions establish predictability but not direction. To fix the direction we embed the three  $\Delta m$  series in a first-order VAR and test for Granger causality. The hierarchy is sharp. The iTraxx index Granger-causes both the CDS–Bond basis ( $p = 0.011$ ) and the BTP Italia basis ( $p = 0.007$ ), and the link survives at two and three lags. Neither direction reverses: BTP Italia does not Granger-cause either institutional mispricing at conventional levels, nor does CDS–Bond Granger-cause iTraxx. The directional link from iTraxx to the CDS–Bond basis strengthens to the one-percent level on the longer bivariate sample available for those two series alone. Shocks thus originate in the most liquid, standardised index market, where dealer inventory adjusts fastest, and reach both the less liquid cash basis and the retail-held inflation-linked basis with a delay. The generalised impulse responses of Pesaran and Shin (1998), invariant to the ordering of variables in the VAR, are consistent with the same picture (Figure 7). A one-standard-deviation shock to the iTraxx index produces a positive response in the CDS–Bond basis that peaks at one month, and a negative response in the BTP Italia basis that peaks at one month; both decay within two to three months, with confidence bands that narrow rapidly toward zero. Shocks in the reverse direction—originating in either of the less liquid bases and propagating back to the iTraxx skew—are indistinguishable from zero across the full horizon, with bootstrap bands that contain zero throughout. The asymmetric pattern matches the forced-deleveraging mechanism of Mitchell and Pulvino (2012), by which intermediaries adjust their most liquid positions first when funding contracts and their less liquid positions only with a delay. P3 is confirmed: predictability, direction, and dynamic response all run from the liquid index to the less liquid bases. Robustness to lag length ( $p = 1, 2, 3$ ) appears in Appendix A.11.2.

## **E P4: Persistence and Funding-Dependence**

The fourth prediction links the persistence of each mispricing to how funding-dependent its trade is: the more a strategy relies on continuously financing a cash position on a dealer balance sheet,

the longer a shock to its mispricing should take to decay, because the capital that closes it arrives only slowly. Duffie (2010) makes this quantitative—the half-life of the reversal that follows a shock is decreasing in the mean rate at which arbitrage capital reaches the trade—so the cross-strategy ordering of half-lives should track the ordering of funding frictions. We estimate first-order autocorrelations of the mispricing levels,

$$m_{i,t} = c_i + \phi_i m_{i,t-1} + u_{i,t}, \quad (17)$$

with Newey–West standard errors (six monthly lags), and read off the implied half-life  $h_i^* = \ln(0.5)/\ln(\phi_i)$ .

Table XVII reports the estimates. The CDS–Bond basis is the most persistent ( $\phi = 0.79$ , half-life  $\approx 3.0$  months), with the BTP Italia and the iTraxx skew essentially identical at  $\phi \approx 0.73$  and half-life  $\approx 2.2$  months, all three coefficients significant at the one percent level. The CDS–Bond basis requires the arbitrageur to hold the cash bond on a repo book and finance it continuously, so a widening persists until fresh balance-sheet capacity arrives—the longest half-life of the three, roughly one-third longer than its counterparts. The two remaining mispricings do not separate: the iTraxx skew is a pure derivative position, but the on-balance-sheet exposure of trading the index against its constituents carries non-trivial repo and counterparty costs that the half-life of the basis itself reflects; the BTP Italia is refreshed by primary auctions reserved to households and stabilised by the loyalty bonus, both of which slow its mean reversion below what a pure-derivative basis would otherwise deliver. P4 is therefore confirmed at the upper end of the funding-intensity spectrum—the cash-bond basis is the most persistent—but not in the finer ordering below it: the differential between the two less funding-intensive strategies is too small to register in monthly persistence. The skew would, on funding grounds alone, be expected to revert fastest, yet its half-life matches the BTP Italia’s.

The impulse responses corroborate this reading from a second angle. The own-responses on the diagonal of Figure 7 decay within the first month: innovations to  $\Delta m$  dissipate quickly precisely

because the level  $m$  mean-reverts over a multi-month horizon, at a speed governed by funding-dependence. This is the “sharp reaction ... and a subsequent and more extended reversal” that Duffie (2010) identifies as the signature of slow-moving capital: the impact is absorbed within a month, while the level reverts over the horizon set by the rate of capital arrival.

## **F P5: The Common Factor**

The fifth prediction concerns the source of the co-movement. If slow-moving intermediary capital is the binding constraint, the common factor behind the three mispricings should load on intermediary balance-sheet variables and not on macroeconomic fundamentals; Duffie (2010) identifies “the level of capital commonly available to bear losses” as the link tying mispricings across markets, and Brunnermeier and Pedersen (2009) tie the commonality of liquidity to the speculators’ shadow cost of capital. We extract the first principal component (PC1) of the three  $\Delta m$  series and regress it on two sets of variables (Table XVIII): intermediary stress proxies, and a placebo set of macroeconomic indicators that should not explain PC1 if the slow-moving capital interpretation is correct.

The contrast is sharp. In Panel A, one proxy per channel of the intermediary literature—the He et al. (2017) capital ratio (−4.29, significant at the five percent level), the Libor–OIS spread (+3.34, significant at the one percent level), the European prime-broker CDS spread (−0.018, significant at the five percent level), and the Amihud and Mendelson (2015) illiquidity measure (+1.73, significant at the one percent level)—all carry the theoretically predicted sign, so that a fall in intermediary capital, a rise in funding costs, and an increase in market illiquidity all predict wider mispricings; the model explains 23.1% of the variation in the common factor, jointly significant at the one percent level, and Panel B confirms the same picture with an extended proxy set ( $\bar{R}^2 = 23.5\%$ , jointly significant at the one percent level). Panel C runs the placebo: regressed on macroeconomic and real-activity uncertainty, long-term inflation expectations, and economic policy uncertainty, the common factor yields four uniformly insignificant coefficients, a model  $\bar{R}^2$  that falls to 11.2%, and a joint  $F$ -statistic that fails to reject at conventional levels. A factor

loading on macroeconomic fundamentals would have reversed this ranking. P5 is confirmed: the force driving the three mispricings to co-move is the state of intermediary balance sheets, not of the macroeconomy.

## **G Discussion: Why These Arbitrages Persist**

**Holder-Base Heterogeneity as the Source of Identification.** The evidence of the preceding subsections is unified by the single structural feature introduced at the outset: although all three strategies are run by the same type of institutional arbitrageur, the instruments differ in who holds them, and it is this heterogeneity that identifies the mechanism. The CDS–Bond basis and the iTraxx skew trade in instruments whose holder base is overwhelmingly institutional—hedge funds, dealer proprietary desks, and leveraged accounts—so that a spike in volatility inflates the Value-at-Risk of the combined position, exhausts the desk’s risk budget, and forces simultaneous liquidation across both books (Brunnermeier and Pedersen, 2009), pushing both bases wider at once. The BTP Italia is different: only its inflation-linked leg is structurally retail-dominated. Introduced in 2012 with a primary auction reserved to households and a loyalty bonus (*premio fedeltà*) for buy-and-hold investors, and serving as the simplest inflation hedge available to an Italian household, the bond is held by preferred-habitat investors in the sense of Vayanos and Vila (2021): they use no leverage, face no margin calls, and do not unwind when dealer capital is scarce. The arbitrageur of the BTP Italia basis is itself institutional and faces the same funding environment as the others—so the strategy is fully part of the experiment, not a bystander to it—but the absence of forced selling on the bondholder side keeps the basis out of the institutional fire sales, so that stress drives it opposite to the institutional pair rather than alongside it. This is what gives the design its identifying power: a common macroeconomic factor would move all three mispricings in the same direction regardless of who holds the instruments, whereas the sign of the co-movement turns entirely on whether the holder base is shared—the pattern documented under P1 and P2.



**Why the Mispricings Persist.** Why have these opportunities not been competed away? The answer lies in structural limits to arbitrage that differ across strategies but share a common root in the scarcity of intermediary capital. For the BTP Italia, the binding constraint is the segmentation itself: institutional arbitrageurs can exploit the basis, but the retail holders who form the natural counterparty neither monitor nor arbitrage it (Vayanos and Vila, 2021). For the CDS–Bond basis, the limits are those of Bai and Collin-Dufresne (2019) and Mitchell and Pulvino (2012): the trade must fund the corporate bond on a dealer balance sheet, the repo market for European investment-grade corporates is thin and subject to recall risk, and the counterparty risk on the CDS leg consumes regulatory capital—so capacity is procyclical and the mispricing countercyclical, widening precisely when the trade is most attractive but least feasible. For the iTraxx skew, the constraint is post-crisis regulation: Boyarchenko et al. (2016) show that the supplementary leverage ratio and other Basel III requirements raised the capital charge on index-versus-constituent trades—as required leverage ratios rose from roughly 2–3% pre-crisis to 5–6% post-crisis, the return on equity fell by a factor of two to three—rendering them “significantly less economical” and producing “potentially ‘new normal’ levels for both the CDS-bond and the CDX-CDS bases.” The persistence of alpha documented in Sections II and III is therefore not a puzzle calling for a risk-based explanation; it is the equilibrium of a market in which the cost of intermediary capital has been permanently raised, and slow-moving capital, in the sense of Duffie (2010), keeps mispricings open longer than frictionless models would predict (Mitchell et al., 2007).

### **Summary: The Five Predictions.**

**P1.** Mispricings drawing on a common capital pool co-move. The institutional pair correlates at +0.19, while the retail-held BTP Italia, outside that pool, correlates significantly negatively with the CDS–Bond basis at −0.30 unconditionally (Table XIV). The sign turns on the holder base, exactly as the common-capital mechanism requires. Confirmed.

- P2.** The co-movement intensifies in stress. Institutional co-movement rises monotonically from  $-0.21$  in LOW to  $+0.32$  in HIGH, a LOW-to-HIGH increase significant at the one percent level under a Fisher  $z$ -test (Tables XIV, XVI). Confirmed.
- P3.** Mispricing changes transmit along the liquidity gradient. The iTraxx index Granger-causes both other mispricings ( $p = 0.007$  and  $0.011$ ), with impulse responses tracing the same liquid-to-illiquid path (Table XV, Figure 7). Confirmed.
- P4.** Persistence reflects funding-dependence. The CDS–Bond basis, the only strategy requiring continuous cash-bond financing on a dealer balance sheet, has a half-life about one-third longer than the iTraxx skew and the BTP Italia (Table XVII); the two less funding-intensive strategies are essentially identical. Partially confirmed.
- P5.** The common factor is intermediary capital, not macro fundamentals. PC1 loads on balance-sheet proxies with the predicted signs ( $\bar{R}^2 = 23.1\%$ , jointly significant at the one percent level) but not on a macroeconomic placebo set ( $\bar{R}^2 = 11.2\%$ , fails to reject at conventional levels) (Table XVIII). Confirmed.

Four of the five predictions find unambiguous support in the data; only P4—the cross-strategy ordering of persistences—is partially supported, holding at the upper end of the funding-intensity spectrum but not within the institutional pair. Taken together, the evidence locates the geography of slow-moving capital not in asset-class labels but in the network of intermediaries that service these markets—and ultimately in who holds the bonds.

Having documented both the existence of significant alpha across three distinct European fixed-income arbitrage strategies and the economic mechanism—slow-moving intermediary capital, amplified by post-crisis regulation and modulated by investor-base segmentation—that sustains these mispricings, we draw our conclusions in Section V.

## V Conclusion

We document positive and economically significant alpha in three European fixed-income arbitrage strategies—the BTP Italia inflation-linked basis, the corporate CDS–bond basis, and the iTraxx index skew—and show that this alpha is not compensation for the identified systematic risk exposures, whether the controls are entered as classical benchmark factors or as data-driven selections from a large candidate universe, and it survives when each benchmark hedge is re-estimated out of sample from past-only loadings. The mispricings are linked by a common factor that loads on intermediary balance-sheet variables (and not on macroeconomic fundamentals), and their persistence at the level of each series reflects the funding-dependence of the arbitrage trade: the CDS–Bond basis, the only strategy requiring continuous repo financing of the cash bond, is the most persistent, while the BTP Italia and the iTraxx skew mean-revert more rapidly and are essentially identical to each other. The three correlations exhibit an asymmetric pattern that mirrors the structural divide between institutional and retail holder bases: positive and rising in stress within the institutional pair (CDS–Bond and iTraxx), and negative across holder bases (BTP Italia vs. either institutional strategy). The slow-moving capital literature (Duffie, 2010; Brunnermeier and Pedersen, 2009; Mitchell and Pulvino, 2012), combined with the post-crisis regulatory environment documented by Boyarchenko et al. (2016), provides an economic explanation for why these mispricings persist.

The findings have implications for two strands of the literature. For the limits-to-arbitrage literature, the European setting provides a natural test of the slow-moving capital mechanism in a market with a structurally heterogeneous holder base, and the investor-base-segmentation pattern in cross-strategy correlations offers independent evidence that the geography of arbitrage capital—rather than asset-class labels—determines cross-market co-movement. For the fixed-income asset pricing literature, supervised selection on a large candidate universe recovers strategy-specific risk drivers that the standard benchmark factor sets miss, yet economically significant alpha persists

for the BTP Italia and CDS–Bond strategies—indicating that the residual return is genuinely unspanned rather than compensation for an overlooked systematic factor.

## References

- Amihud, Yakov, and Haim Mendelson, 2015, The pricing of illiquidity as a characteristic and as risk, *Multinational Finance Journal* 19, 149–168.
- Asness, Clifford S, Andrea Frazzini, and Lasse Heje Pedersen, 2019, Quality minus junk, *Review of Accounting Studies* 24, 34–112.
- Asness, Clifford S, Tobias J Moskowitz, and Lasse Heje Pedersen, 2013, Value and momentum everywhere, *The journal of finance* 68, 929–985.
- Asvanunt, Attakrit, and Scott Richardson, 2017, The credit risk premium, *Journal of Fixed Income* 26, 6–24.
- Bai, Jennie, and Pierre Collin-Dufresne, 2019, The cds-bond basis, *Financial Management* 48, 417–439.
- Bai, Jushan, and Serena Ng, 2002, Determining the number of factors in approximate factor models, *Econometrica* 70, 191–221.
- Bai, Jushan, and Serena Ng, 2006, Confidence intervals for diffusion index forecasts and inference for factor-augmented regressions, *Econometrica* 74, 1133–1150.
- Barillas, Francisco, and Jay Shanken, 2018, Comparing asset pricing models, *The Journal of Finance* 73, 715–754.
- Belloni, Alexandre, Victor Chernozhukov, and Christian Hansen, 2014, Inference on treatment effects after selection among high-dimensional controls, *The Review of Economic Studies* 81, 608–650.
- Bertsimas, Dimitris, Angela King, and Rahul Mazumder, 2016, Best subset selection via a modern optimization lens, *The annals of statistics* 813–852.
- Bongaerts, Dion, Frank De Jong, and Joost Driessen, 2011, Derivative pricing with liquidity risk: Theory and evidence from the credit default swap market, *Journal of Finance* 66, 203–240.

- Boyarchenko, Nina, Pooja Gupta, Nick Steele, and Jacqueline Yen, 2016, Trends in credit market arbitrage, Technical report, Staff Report.
- Brooks, Jordan, Tony Gould, and Scott A Richardson, 2020, Active fixed income illusions, *Journal of Fixed Income* 29.
- Brunnermeier, Markus K, and Lasse Heje Pedersen, 2009, Market liquidity and funding liquidity, *The review of financial studies* 22, 2201–2238.
- Carhart, Mark M, 1997, On persistence in mutual fund performance, *The Journal of finance* 52, 57–82.
- Carr, Peter, and Liuren Wu, 2009, Variance risk premiums, *Review of Financial Studies* 22, 1311–1341.
- Cederburg, Scott, Michael S O’Doherty, Feifei Wang, and Xuemin Sterling Yan, 2020, On the performance of volatility-managed portfolios, *Journal of financial Economics* 138, 95–117.
- Christopherson, Jon A, Wayne E Ferson, and Debra A Glassman, 1998, Conditioning manager alphas on economic information: Another look at the persistence of performance, *The Review of Financial Studies* 11, 111–142.
- Connor, Gregory, and Robert A Korajczyk, 1986, Performance measurement with the arbitrage pricing theory: A new framework for analysis, *Journal of Financial Economics* 15, 373–394.
- Connor, Gregory, and Robert A Korajczyk, 1988, Risk and return in an equilibrium apt: Application of a new test methodology, *Journal of Financial Economics* 21, 255–289.
- Cremers, Martijn, Matthias Fleckenstein, and Priyank Gandhi, 2021, Treasury yield implied volatility and real activity, *Journal of Financial Economics* 140, 412–435.
- Duarte, Jefferson, Francis A Longstaff, and Fan Yu, 2007, Risk and return in fixed-income arbitrage: Nickels in front of a steamroller?, *The Review of Financial Studies* 20, 769–811.
- Duffie, Darrell, 2010, Presidential address: Asset price dynamics with slow-moving capital, *The Journal of finance* 65, 1237–1267.

- Eraker, Bjørn, and Yue Wu, 2017, Explaining the negative returns to volatility claims: An equilibrium approach, *Journal of Financial Economics* 125, 72–98.
- Fama, Eugene F, and Kenneth R French, 1993, Common risk factors in the returns on stocks and bonds, *Journal of financial economics* 33, 3–56.
- Fama, Eugene F, and Kenneth R French, 2017, International tests of a five-factor asset pricing model, *Journal of financial Economics* 123, 441–463.
- Fleckenstein, Matthias, Francis A Longstaff, and Hanno Lustig, 2014, The tips-treasury bond puzzle, *the Journal of Finance* 69, 2151–2197.
- Forbes, Kristin J, and Roberto Rigobon, 2002, No contagion, only interdependence: measuring stock market comovements, *The journal of Finance* 57, 2223–2261.
- Frazzini, Andrea, and Lasse Heje Pedersen, 2014, Betting against beta, *Journal of financial economics* 111, 1–25.
- Fung, William, and David A Hsieh, 2001, The risk in hedge fund strategies: Theory and evidence from trend followers, *The review of financial studies* 14, 313–341.
- Fung, William, and David A Hsieh, 2004, Hedge fund benchmarks: A risk-based approach, *Financial Analysts Journal* 60, 65–80.
- Giglio, Stefano, and Dacheng Xiu, 2021, Asset pricing with omitted factors, *Journal of Political Economy* 129, 1947–1990.
- Glück, Maximilian, Benjamin Hübel, and Hendrik Scholz, 2021, Currency conversion of fama–french factors: How and why, *Journal of portfolio management* 47, 157–175.
- Goetzmann, William, Jonathan Ingersoll, Matthew Spiegel, and Ivo Welch, 2007, Portfolio performance manipulation and manipulation-proof performance measures, *The Review of Financial Studies* 20, 1503–1546.
- Han, Bing, Avanidhar Subrahmanyam, and Yi Zhou, 2017, The term structure of credit spreads, firm fundamentals, and expected stock returns, *Journal of Financial Economics* 124, 147–171.

- Han, Bing, and Yi Zhou, 2015, Understanding the term structure of credit default swap spreads, *Journal of Empirical Finance* 31, 18–35.
- Haubrich, Joseph, George Pennacchi, and Peter Ritchken, 2012, Inflation expectations, real rates, and risk premia: Evidence from inflation swaps, *The Review of Financial Studies* 25, 1588–1629.
- He, Zhiguo, Bryan Kelly, and Asaf Manela, 2017, Intermediary asset pricing: New evidence from many asset classes, *Journal of Financial Economics* 126, 1–35.
- Hull, John, Mirela Predescu, and Alan White, 2004, The relationship between credit default swap spreads, bond yields, and credit rating announcements, *Journal of banking & finance* 28, 2789–2811.
- Ilmanen, Antti, Ronen Israel, Rachel Lee, Tobias J. Moskowitz, and Ashwin Thapar, 2021, How do factor premia vary over time? A century of evidence, *Journal of Investment Management* 19.
- Junge, Benjamin, and Anders B Trolle, 2015, Liquidity risk in credit default swap markets, Research Paper 13-65, Swiss Finance Institute.
- Kiefer, Nicholas M, and Timothy J Vogelsang, 2005, A new asymptotic theory for heteroskedasticity-autocorrelation robust tests, *Econometric Theory* 21, 1130–1164.
- Klingler, Sven, and Suresh Sundaresan, 2019, An explanation of negative swap spreads: Demand for duration from underfunded pension plans, *The Journal of Finance* 74, 675–710.
- Koijen, Ralph S. J., Tobias J. Moskowitz, Lasse Heje Pedersen, and Evert B. Vrugt, 2018, Carry, *Journal of Financial Economics* 127, 197–225.
- Lazarus, Eben, Daniel J Lewis, James H Stock, and Mark W Watson, 2018, HAR inference: Recommendations for practice, *Journal of Business & Economic Statistics* 36, 541–559.
- Litterman, Robert B, and José A Scheinkman, 1991, Common factors affecting bond returns, *Journal of Fixed Income* 1, 54–61.
- Longstaff, Francis A, Sanjay Mithal, and Eric Neis, 2005, Corporate yield spreads: Default risk or liquidity? new evidence from the credit default swap market, *The journal of finance* 60, 2213–2253.



- Martin, Ian, 2017, What is the expected return on the market?, *The Quarterly Journal of Economics* 132, 367–433.
- Meinshausen, Nicolai, and Peter Bühlmann, 2010, Stability selection, *Journal of the Royal Statistical Society Series B: Statistical Methodology* 72, 417–473.
- Mitchell, Mark, Lasse Heje Pedersen, and Todd Pulvino, 2007, Slow moving capital, *American Economic Review* 97, 215–220.
- Mitchell, Mark, and Todd Pulvino, 2012, Arbitrage crashes and the speed of capital, *Journal of Financial Economics* 104, 469–490.
- Moreira, Alan, and Tyler Muir, 2017, Volatility-managed portfolios, *The Journal of Finance* 72, 1611–1644.
- Newey, Whitney K, and Kenneth D West, 1987, A simple positive semi-definite heteroscedasticity and autocorrelation consistent matrix estimation, *Econometrica* 55, 703–708.
- Newey, Whitney K, and Kenneth D West, 1994, Automatic lag selection in covariance matrix estimation, *The review of economic studies* 61, 631–653.
- O’Kane, Dominic, 2008, *Modelling single-name and multi-name credit derivatives* (John Wiley & Sons).
- Pástor, L’uboš, and Robert F Stambaugh, 2003, Liquidity risk and expected stock returns, *Journal of Political economy* 111, 642–685.
- Pelizzon, Lorian, Marti G. Subrahmanyam, Davide Tomio, and Jun Uno, 2016, Sovereign credit risk, liquidity, and European Central Bank intervention: Deus ex machina?, *Journal of Financial Economics* 122, 86–115.
- Pesaran, H Hashem, and Yongcheol Shin, 1998, Generalized impulse response analysis in linear multivariate models, *Economics letters* 58, 17–29.
- Politis, Dimitris N, and Joseph P Romano, 1994, The stationary bootstrap, *Journal of the American Statistical association* 89, 1303–1313.

- Rebonato, Riccardo, and Riccardo Ronzani, 2021, Is convexity efficiently priced? evidence from international swap markets, *Journal of Empirical Finance* 63, 392–413.
- Siriwardane, Emil N, 2019, Limited investment capital and credit spreads, *Journal of Finance* 74, 2303–2347.
- Vayanos, Dimitri, and Jean-Luc Vila, 2021, A preferred-habitat model of the term structure of interest rates, *Econometrica* 89, 77–112.
- Welch, Ivo, and Amit Goyal, 2008, A comprehensive look at the empirical performance of equity premium prediction, *The Review of Financial Studies* 21, 1455–1508.
- Zhu, Junxian, Canhong Wen, Jin Zhu, Heping Zhang, and Xueqin Wang, 2020, A polynomial algorithm for best-subset selection problem, *Proceedings of the National Academy of Sciences* 117, 33117–33123.
- Zou, Hui, and Hao Helen Zhang, 2009, On the adaptive elastic-net with a diverging number of parameters, *Annals of statistics* 37, 1733.

## A Appendix

### A.1 BTP Italia: Indexation Coefficient and Deflation Floors

The BTP Italia indexes both coupons and principal to the Italian consumer price index for blue- and white-collar worker households, excluding tobacco (FOI ex tabacchi) published by Istat. At each coupon date  $t$ , the indexation coefficient  $CI_t$  is the ratio of the relevant FOI reference index to its value at the previous coupon date  $t - 6m$ , so that  $CI_t$  measures the cumulative inflation over the six-month coupon period. The semiannual coupon and the capital revaluation payment are computed as

$$C_t = \frac{1}{2} c N \tilde{CI}_t, \quad R_t = N (\tilde{CI}_t - 1), \quad (18)$$

where  $c$  is the fixed real annual coupon rate set at issuance,  $N$  is the subscribed nominal capital, and  $\tilde{CI}_t$  is the *modified* indexation coefficient that incorporates a semiannual deflation floor:

$$\tilde{CI}_t = \max(CI_t, 1). \quad (19)$$

If the FOI ex tabacchi falls between consecutive coupon dates, the indexation coefficient is reset to one for that semester, the coupon reverts to the fixed real rate  $c/2 \cdot N$ , and no capital revaluation is paid; in the following semester, the base index is reset to the higher of the previous semester's level and the highest reading observed in earlier semesters. The principal is redeemed at par at maturity ( $N$ ), regardless of the cumulative inflation path. Two features of this design are relevant for the arbitrage. First, the semiannual reset of  $\tilde{CI}_t$  to unity at every coupon date means that the notional exposed to default risk is always the original face value  $N$ , eliminating the path-dependent contamination that affects conventional inflation-linked bonds such as the U.S. TIPS or the euro-area BTP€i. Second, the deflation floor on the coupon guarantees a strictly positive nominal payment in every period, which is absent from instruments with floor only at maturity.

## A.2 Trade Construction Details

### A.2.1 Entry/Exit Rules and Thresholds

Table A.I summarises entry and exit thresholds for each strategy. All strategies require a minimum of 6 months to maturity at entry and maintain at least one open trade after the first signal. The CDS–Bond universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Main series; names that exit the index are no longer traded. All CDS are centrally cleared. The iTraxx skew strategy trades only on-the-run series across Main, SnrFin, SubFin, and Xover.

**Table A.I**  
**Entry and Exit Thresholds by Strategy**

All thresholds in bps. CDS–Bond Basis is a one-directional negative basis trade (long bond, buy CDS protection). Minimum 6 months to maturity at entry for all strategies. Trades shorter than 3 trading days excluded ex-post.

| Strategy       | Entry Long | Entry Short | Exit Long | Exit Short |
|----------------|------------|-------------|-----------|------------|
| BTP Italia     | > 40       | < –50       | < 10      | > –10      |
| CDS–Bond Basis | < –40      | –           | > 0       | –          |
| iTraxx Main    | > 5        | < –10       | < –5      | > 10       |
| iTraxx SnrFin  | > 8        | < –15       | < –5      | > 5        |
| iTraxx SubFin  | > 10       | < –20       | < –10     | > 10       |
| iTraxx Xover   | > 15       | < –25       | < –15     | > 10       |

### A.2.2 Trade-Level Statistics

Table A.II reports trade-level statistics for each strategy, including the number of trades, average duration, average profit per trade, and the distribution of exit reasons.

**Table A.II**  
**Trade-Level Statistics by Strategy**

This table reports trade-level statistics for each arbitrage strategy. P&L is the cumulative return per trade in percentage points. Duration is in calendar days. Exit reasons may not sum to 100% due to rounding.

|                          | BTP-Italia | iTraxx <i>combined</i> | CDS-Bond Basis |
|--------------------------|------------|------------------------|----------------|
| Total trades             | 260        | 206                    | 931            |
| Long                     | 224        | 146                    | 931            |
| Short                    | 36         | 60                     | 0              |
| Avg duration (days)      | 386        | 225                    | 240            |
| Median duration (days)   | 212        | 97                     | 119            |
| Avg P&L per trade (%)    | 2.079      | 1.491                  | 2.463          |
| Median P&L per trade (%) | 1.960      | 1.395                  | 1.927          |
| <i>Exit reasons (%)</i>  |            |                        |                |
| Target hit               | 92.3       | 97.1                   | 90.7           |
| Maturity                 | 6.9        | 0.5                    | 1.0            |
| End of sample            | 0.8        | 2.4                    | 8.4            |

### A.2.3 Transaction Costs

Transaction costs are computed at the individual trade level and applied as entry and exit fees proportional to DV01 (bonds) or duration (CDS). Fee levels are regime-dependent: at each trade date, the prevailing level of the iTraxx Main 5Y index determines which of three tiers applies (Table A.III). Thresholds are set at 60 and 100 bps, reflecting the empirical widening of bid-ask spreads in stressed markets. For trades held to maturity, the exit fee is waived. All return series in subsequent sections are net of these costs.

**Table A.III**  
**Transaction Fee Schedule (bps per unit DV01 or duration)**

The fee is applied as entry  $DV01 \times \text{fee (bps)}$  at entry plus exit  $DV01 \times \text{fee (bps)}$  at exit; the exit fee is waived for trades held to maturity. The regime is determined by the iTraxx Main 5Y level at the trade date: LOW ( $<60$  bps), MEDIUM (60–100 bps), HIGH ( $\geq 100$  bps).

|                | LOW<br>(Main < 60) | MEDIUM<br>(60–100) | HIGH<br>(Main > 100) |
|----------------|--------------------|--------------------|----------------------|
| BTP Italia     | 3.0                | 5.0                | 8.0                  |
| CDS–Bond Basis | 4.0                | 7.0                | 10.0                 |
| iTraxx Main    | 0.5                | 0.75               | 1.0                  |
| iTraxx SnrFin  | 0.5                | 0.75               | 1.0                  |
| iTraxx SubFin  | 1.0                | 2.0                | 3.0                  |
| iTraxx Xover   | 3.0                | 5.0                | 7.0                  |

### A.3 Signal-Weighted vs Equal-Weighted Comparison

Table A.IV compares the signal-weighted (SW, Rebonato and Ronzani, 2021) and equal-weighted (EW, Duarte et al., 2007) indices. SW is adopted as the baseline in the main text; EW results confirm robustness to the weighting scheme. The equal-weighted counterpart of the manipulation-proof performance measure (Table II) is reported in Table A.V; the volatility-managed results for both weighting schemes are in Table A.VI.

**Table A.IV**  
**Equal-Weighted Strategy Returns**

This table reports monthly percentage excess returns, net of transaction costs, for the equal-weighted indices, in which each active trade receives weight  $1/K_i$ ; it is the equal-weighted counterpart of Panel B of Table I, whose signal-weighted indices are the baseline of the paper. The mean and standard deviation are annualized and the Sharpe ratio is their ratio; kurtosis is excess kurtosis; the  $t$ -statistic for the mean is corrected for serial correlation (Newey–West); and %Neg is the proportion of negative monthly excess returns.

| Strategy        | $n$ | Mean<br>(% p.a.) | $t$ -Stat | Std Dev<br>(% p.a.) | Min<br>(%) | Max<br>(%) | Skew | Kurt | %Neg | AC(1) | Sharpe |
|-----------------|-----|------------------|-----------|---------------------|------------|------------|------|------|------|-------|--------|
| BTP Italia      | 163 | 1.34             | 3.01      | 1.69                | -1.25      | 1.92       | 0.82 | 1.88 | 39.9 | 0.06  | 0.79   |
| CDS–Bond Basis  | 207 | 3.84             | 5.71      | 2.62                | -3.37      | 3.54       | 0.49 | 5.28 | 28.0 | -0.08 | 1.46   |
| iTraxx Combined | 243 | 1.37             | 5.30      | 1.05                | -1.44      | 1.11       | 0.26 | 3.68 | 39.9 | -0.04 | 1.30   |

**Table A.V**  
**Manipulation-Proof Performance Measure by Risk Aversion (Equal-Weighted)**

MPPM is the manipulation-proof performance measure of Goetzmann et al. (2007), computed from monthly returns;  $\rho$  is the coefficient of relative risk aversion and the risk-free rate is 1-month Euribor. Returns are net of transaction costs.

| Strategy        | $\rho = 2$<br>(% p.a.) | $\rho = 3$<br>(% p.a.) | $\rho = 4$<br>(% p.a.) |
|-----------------|------------------------|------------------------|------------------------|
| BTP Italia      | 1.24                   | 1.22                   | 1.21                   |
| CDS–Bond Basis  | 3.66                   | 3.63                   | 3.60                   |
| iTraxx Combined | 1.35                   | 1.34                   | 1.34                   |

#### A.4 Volatility-Managed Portfolios

Table A.VI reports the volatility-managed portfolio results of Moreira and Muir (2017) for the three strategies, for both signal-weighted (Panel A) and equal-weighted (Panel B) indices.

**Table A.VI**  
**Moreira–Muir Volatility-Managed Portfolio Performance**

Following Moreira and Muir (2017). The volatility-managed strategy scales the position inversely to the previous month's realized variance,  $f_{t+1}^\sigma = (c/\sigma_t^2) f_{t+1}$ , with  $c$  chosen to match buy-and-hold volatility. Alpha is the intercept from regressing volatility-managed on buy-and-hold monthly returns, with Newey–West standard errors (4 lags). Panel A uses signal-weighted indices; Panel B, equal-weighted. Returns are net of transaction costs. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Strategy                        | Buy-and-Hold |          |        | Vol-Managed |          |        | Alpha<br>(% p.a.) | $t(\alpha)$ |
|---------------------------------|--------------|----------|--------|-------------|----------|--------|-------------------|-------------|
|                                 | Ret          | Vol      | Sharpe | Ret         | Vol      | Sharpe |                   |             |
|                                 | (% p.a.)     | (% p.a.) |        | (% p.a.)    | (% p.a.) |        |                   |             |
| <i>Panel A: Signal-Weighted</i> |              |          |        |             |          |        |                   |             |
| BTP Italia                      | 1.52         | 1.60     | 0.95   | 0.37        | 1.60     | 0.23   | -0.22             | -0.50       |
| CDS–Bond Basis                  | 4.52         | 2.88     | 1.57   | 3.08        | 2.88     | 1.07   | 1.39*             | 1.90        |
| iTraxx Combined                 | 1.91         | 1.53     | 1.25   | 0.84        | 1.53     | 0.55   | -0.16             | -0.58       |
| <i>Panel B: Equal-Weighted</i>  |              |          |        |             |          |        |                   |             |
| BTP Italia                      | 1.27         | 1.67     | 0.76   | 0.21        | 1.67     | 0.13   | -0.33             | -0.73       |
| CDS–Bond Basis                  | 3.73         | 2.53     | 1.48   | 2.45        | 2.53     | 0.97   | 0.93              | 1.38        |
| iTraxx Combined                 | 1.36         | 1.05     | 1.29   | 0.60        | 1.05     | 0.57   | -0.01             | -0.09       |

## A.5 Benchmark Regressions with US Factors

For completeness, the three benchmark regressions are re-estimated using the original US factor proxies; the euro-area results in the main text (Tables III, IV, and V) remain the primary specification.

**Table A.VII**  
**Duarte Factor Model Regressions (US Factors)**

This table reports estimates of the regression  $r_{i,t} = \alpha_i + \beta_1 Mkt_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 R_{S,t} + \beta_6 R_{I,t} + \beta_7 R_{B,t} + \beta_8 R_{2,t} + \beta_9 R_{5,t} + \beta_{10} R_{10,t} + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the ten Duarte et al. (2007) US equity and bond risk factors.  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC).  $Mkt$ ,  $SMB$ ,  $HML$ ,  $UMD$ : the Fama–French market, size, value, and momentum factors.  $R_S$ : U.S. bank stock excess return.  $R_I$ ,  $R_B$ : A/BAA-rated U.S. industrial and bank bond excess returns.  $R_2$ ,  $R_5$ ,  $R_{10}$ : maturity-sorted U.S. Treasury portfolio excess returns.  $\bar{R}^2$  is the adjusted  $R^2$ . Sample: BTP Italia 163 months, CDS–Bond Basis 207 months, iTraxx Combined 243 months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                        | $\alpha$ (% p.a.) | $\beta_{Mkt}$       | $\beta_{SMB}$     | $\beta_{HML}$    | $\beta_{UMD}$     | $\beta_{R_S}$     | $\beta_{R_I}$     | $\beta_{R_B}$    | $\beta_{R_2}$    | $\beta_{R_5}$     | $\beta_{R_{10}}$ | $\bar{R}^2$ |
|------------------------|-------------------|---------------------|-------------------|------------------|-------------------|-------------------|-------------------|------------------|------------------|-------------------|------------------|-------------|
| <i>BTP Italia</i>      | 2.02***<br>(4.17) | −0.05***<br>(−3.49) | −0.02*<br>(−1.72) | −0.02<br>(−1.60) | 0.01<br>(0.44)    | 0.03***<br>(3.19) | 0.03<br>(0.37)    | −0.09<br>(−1.19) | −0.02<br>(−0.07) | −0.23<br>(−0.62)  | 0.21<br>(0.97)   | 0.11        |
| <i>CDS–Bond Basis</i>  | 4.58***<br>(5.87) | −0.03<br>(−0.90)    | 0.01<br>(0.45)    | −0.00<br>(−0.11) | −0.01<br>(−1.01)  | 0.00<br>(0.16)    | 0.25***<br>(3.35) | −0.06<br>(−1.63) | 0.19<br>(0.40)   | −0.70*<br>(−1.89) | 0.26<br>(1.34)   | 0.18        |
| <i>iTraxx Combined</i> | 1.86***<br>(4.71) | 0.00<br>(0.31)      | −0.02*<br>(−1.68) | −0.02<br>(−1.38) | −0.01*<br>(−1.92) | −0.00<br>(−0.09)  | 0.02<br>(0.50)    | −0.02<br>(−1.04) | 0.22<br>(0.74)   | −0.07<br>(−0.31)  | 0.01<br>(0.10)   | 0.02        |



**Table A.VIII**  
**Fung & Hsieh Factor Model Regressions (US Factors)**

This table reports estimates of the regression  $r_{i,t} = \alpha_i + \beta_1 S\&P_t + \beta_2 SC-LC_t + \beta_3 BdOpt_t + \beta_4 FXOpt_t + \beta_5 ComOpt_t + \beta_6 10Y_t + \beta_7 CredSpr_t + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the seven Fung & Hsieh (2004) US hedge fund risk factors.  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC). S&P: S&P 500 excess return. SC–LC: Russell 2000 minus S&P 500 (small-minus-large) equity spread. BdOpt, FXOpt, ComOpt: trend-following returns from lookback straddles on government-bond, currency, and commodity futures. 10Y: excess return on a 10-year U.S. Treasury total-return index. CredSpr: return spread between BAA U.S. corporate bonds and 10-year U.S. Treasuries.  $\bar{R}^2$  is the adjusted  $R^2$ . Sample: BTP Italia 163 months, CDS–Bond 207 months, iTraxx 243 months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                   | $\alpha$ (% p.a.) | $\beta_{S\&P}$    | $\beta_{SC-LC}$    | $\beta_{BdOpt}$ | $\beta_{FXOpt}$    | $\beta_{ComOpt}$ | $\beta_{10Y}$     | $\beta_{CredSpr}$ | $\bar{R}^2$ |
|-------------------|-------------------|-------------------|--------------------|-----------------|--------------------|------------------|-------------------|-------------------|-------------|
| <i>BTP Italia</i> | 1.84***<br>(3.82) | -0.02*<br>(-1.94) | -0.02*<br>(-1.89)  | 0.00<br>(0.90)  | -0.00<br>(-0.00)   | 0.00<br>(0.86)   | -0.02<br>(-0.50)  | -0.04<br>(-1.05)  | 0.10        |
| <i>CDS–Bond</i>   | 4.14***<br>(5.05) | -0.01<br>(-0.64)  | 0.01<br>(0.46)     | 0.00<br>(1.17)  | -0.01**<br>(-1.97) | 0.00<br>(0.08)   | 0.13***<br>(3.33) | 0.14***<br>(3.40) | 0.11        |
| <i>iTraxx</i>     | 1.75***<br>(4.44) | 0.01<br>(0.65)    | -0.02**<br>(-2.02) | 0.00<br>(0.39)  | 0.00<br>(1.03)     | 0.00<br>(0.52)   | 0.03<br>(1.11)    | 0.01<br>(0.51)    | 0.02        |

**Table A.IX**  
**Active Fixed Income Factor Model Regressions (US Factors)**

This table reports estimates of the regression  $r_{i,t} = \alpha_i + \beta_1 Term_t + \beta_2 Global\ Term_t + \beta_3 Global\ Agg_t + \beta_4 Infl.\ Linkers_t + \beta_5 Corp\ Credit_t + \beta_6 Emerg\ Debt_t + \beta_7 Emerg\ Curr_t + \beta_8 UST\ Vol_t + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the eight Active Fixed Income risk premia of Brooks et al. (2020).  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC). Term: U.S. Treasury excess return over cash. Global Term: Bloomberg Global Treasury Index (EUR-hedged), in excess of cash. Global Agg: excess return on the Bloomberg Global Aggregate Bond Index. Infl. Linkers: global inflation-linked bonds in excess of cash. Corp Credit: U.S. high-yield corporate bond index, in excess of cash. Emerg Debt: excess return on hard-currency emerging-market debt. Emerg Curr: emerging-market currency basket versus the U.S. dollar, in excess of USD cash. UST Vol: return on a one-month variance swap (delta-hedged ATM straddle) on the 10-year U.S. Treasury future.  $\bar{R}^2$  is the adjusted  $R^2$ . Sample: BTP Italia 163 months, CDS–Bond Basis 207 months, iTraxx Combined 243 months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                        | $\alpha$ (% p.a.) | $\beta_{Term}$   | $\beta_{Global\ Term}$ | $\beta_{Global\ Agg}$ | $\beta_{Infl.\ Linkers}$ | $\beta_{Corp\ Credit}$ | $\beta_{Emerg\ Debt}$ | $\beta_{Emerg\ Curr}$ | $\beta_{UST\ Vol}$ | $\bar{R}^2$ |
|------------------------|-------------------|------------------|------------------------|-----------------------|--------------------------|------------------------|-----------------------|-----------------------|--------------------|-------------|
| <i>BTP Italia</i>      | 1.99***<br>(4.56) | 0.25**<br>(2.50) | 0.09<br>(0.28)         | -0.35<br>(-0.90)      | -0.06*<br>(-1.85)        | -0.04<br>(-0.79)       | 0.06<br>(1.25)        | -0.01<br>(-0.35)      | -11.05<br>(-0.57)  | 0.11        |
| <i>CDS–Bond Basis</i>  | 3.98***<br>(5.49) | -0.20<br>(-1.22) | -0.15<br>(-0.25)       | 0.47<br>(0.66)        | -0.05<br>(-0.60)         | -0.04<br>(-0.80)       | 0.13***<br>(3.32)     | -0.01<br>(-0.40)      | 27.60<br>(0.97)    | 0.11        |
| <i>iTraxx Combined</i> | 1.86***<br>(4.26) | 0.07<br>(0.87)   | -0.22<br>(-0.90)       | 0.26<br>(1.01)        | -0.02<br>(-0.48)         | 0.06**<br>(2.04)       | -0.08**<br>(-2.17)    | 0.01<br>(0.52)        | 34.05*<br>(1.90)   | 0.06        |

## A.6 Subperiod and Rolling Window Analysis

This appendix reports sub-period splits, stress-regime decompositions, and out-of-sample robustness for the benchmark factor models of Section II. Each table contains Panel A (full sample, first half, second half) and Panel B (alpha by LOW/MEDIUM/HIGH stress regime on the iTraxx Main 5Y level, thresholds at 60 and 100 bps).

### A.6.1 Duarte et al. (2007)

Table A.X reports sub-period splits and stress regime alpha for the Duarte et al. (2007) specification.

**Table A.X**  
**Sub-Period and Regime Analysis: Duarte et al. (2007)**

Annualized alpha (% p.a.) of the three strategies under the Duarte et al. (2007) specification, EUR factors, monthly frequency. Panel A: OLS with Newey–West HAC standard errors (4 lags) on time sub-samples. Panel B: regime alphas from common-beta dummy intercepts  $r_t = \sum_g \alpha_g D_{g,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$ , with  $g \in \{\text{LOW, MEDIUM, HIGH}\}$  defined on the iTraxx Main 5Y level (LOW < 60 bps, MEDIUM [60, 100), HIGH  $\geq 100$ ).  $N$  = months per sub-sample or regime.  
\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| <i>Panel A: Sub-period alpha (% p.a.)</i> |            |      |     |                 |      |     |                |      |     |
|-------------------------------------------|------------|------|-----|-----------------|------|-----|----------------|------|-----|
| Period                                    | BTP Italia |      |     | iTraxx Combined |      |     | CDS–Bond Basis |      |     |
|                                           | $\alpha$   | $t$  | $N$ | $\alpha$        | $t$  | $N$ | $\alpha$       | $t$  | $N$ |
| Full Sample                               | +1.62***   | 3.29 | 163 | +2.19***        | 4.90 | 243 | +4.24***       | 5.27 | 207 |
| First Half                                | +1.11*     | 1.96 | 81  | +2.10***        | 4.56 | 121 | +5.76***       | 4.39 | 103 |
| Second Half                               | +1.79**    | 2.33 | 82  | +2.60***        | 3.41 | 122 | +2.50***       | 2.84 | 104 |

| <i>Panel B: Alpha by stress regime (% p.a.)</i> |            |      |     |                 |      |     |                |      |     |
|-------------------------------------------------|------------|------|-----|-----------------|------|-----|----------------|------|-----|
| Regime                                          | BTP Italia |      |     | iTraxx Combined |      |     | CDS–Bond Basis |      |     |
|                                                 | $\alpha$   | $t$  | $N$ | $\alpha$        | $t$  | $N$ | $\alpha$       | $t$  | $N$ |
| LOW                                             | +0.78      | 1.24 | 62  | +0.97**         | 2.22 | 91  | +2.15**        | 2.23 | 63  |
| MEDIUM                                          | +1.97***   | 3.15 | 80  | +2.69***        | 3.65 | 99  | +4.01***       | 3.53 | 94  |
| HIGH                                            | +3.05*     | 1.70 | 21  | +3.27***        | 4.89 | 53  | +7.11***       | 4.50 | 50  |

### A.6.2 Brooks et al. (2020) — Active Fixed Income

Table A.XI reports the same analysis under the Brooks et al. (2020) specification.

**Table A.XI**  
**Sub-Period and Regime Analysis: Brooks et al. (2020)**

Annualized alpha (% p.a.) of the three strategies under the Active FI (Brooks et al. (2020)) specification, EUR factors, monthly frequency. Panel A: OLS with Newey–West HAC standard errors (4 lags) on time sub-samples. Panel B: regime alphas from common-beta dummy intercepts  $r_t = \sum_g \alpha_g D_{g,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$ , with  $g \in \{\text{LOW, MEDIUM, HIGH}\}$  defined on the iTraxx Main 5Y level (LOW < 60 bps, MEDIUM [60, 100), HIGH  $\geq$  100).  $N$  = months per sub-sample or regime. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| <i>Panel A: Sub-period alpha (% p.a.)</i> |            |      |     |                 |      |     |                |      |     |
|-------------------------------------------|------------|------|-----|-----------------|------|-----|----------------|------|-----|
| Period                                    | BTP Italia |      |     | iTraxx Combined |      |     | CDS–Bond Basis |      |     |
|                                           | $\alpha$   | $t$  | $N$ | $\alpha$        | $t$  | $N$ | $\alpha$       | $t$  | $N$ |
| Full Sample                               | +1.98***   | 3.83 | 163 | +1.84***        | 4.40 | 243 | +4.37***       | 5.49 | 207 |
| First Half                                | +1.26      | 1.63 | 81  | +1.69***        | 3.88 | 121 | +6.38***       | 4.71 | 103 |
| Second Half                               | +2.15***   | 3.11 | 82  | +1.84***        | 2.92 | 122 | +1.90***       | 3.24 | 104 |

| <i>Panel B: Alpha by stress regime (% p.a.)</i> |            |      |     |                 |      |     |                |      |     |
|-------------------------------------------------|------------|------|-----|-----------------|------|-----|----------------|------|-----|
| Regime                                          | BTP Italia |      |     | iTraxx Combined |      |     | CDS–Bond Basis |      |     |
|                                                 | $\alpha$   | $t$  | $N$ | $\alpha$        | $t$  | $N$ | $\alpha$       | $t$  | $N$ |
| LOW                                             | +1.03**    | 2.02 | 62  | +0.79**         | 2.05 | 91  | +2.54***       | 2.85 | 63  |
| MEDIUM                                          | +2.31***   | 3.74 | 80  | +2.15***        | 3.11 | 99  | +4.09***       | 3.54 | 94  |
| HIGH                                            | +3.95*     | 1.85 | 21  | +3.21***        | 3.99 | 53  | +7.61***       | 4.76 | 50  |

### A.6.3 Fung & Hsieh (2004)

Table A.XII reports results for the Fung and Hsieh (2004) specification.

**Table A.XII**  
**Sub-Period and Regime Analysis: Fung & Hsieh (2004)**

Annualized alpha (% p.a.) of the three strategies under the Fung & Hsieh (2004) specification, EUR factors, monthly frequency. Panel A: OLS with Newey–West HAC standard errors (4 lags) on time sub-samples. Panel B: regime alphas from common-beta dummy intercepts  $r_t = \sum_g \alpha_g D_{g,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$ , with  $g \in \{\text{LOW}, \text{MEDIUM}, \text{HIGH}\}$  defined on the iTraxx Main 5Y level (LOW < 60 bps, MEDIUM [60, 100), HIGH  $\geq 100$ ).  $N$  = months per sub-sample or regime. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| <i>Panel A: Sub-period alpha (% p.a.)</i> |            |      |     |                 |      |     |                |      |     |
|-------------------------------------------|------------|------|-----|-----------------|------|-----|----------------|------|-----|
| Period                                    | BTP Italia |      |     | iTraxx Combined |      |     | CDS–Bond Basis |      |     |
|                                           | $\alpha$   | $t$  | $N$ | $\alpha$        | $t$  | $N$ | $\alpha$       | $t$  | $N$ |
| Full Sample                               | +1.81***   | 4.08 | 163 | +1.84***        | 4.49 | 243 | +3.98***       | 5.05 | 207 |
| First Half                                | +1.56**    | 2.04 | 81  | +1.96***        | 4.40 | 121 | +5.62***       | 4.34 | 103 |
| Second Half                               | +1.66***   | 3.00 | 82  | +1.73***        | 3.12 | 122 | +2.08***       | 2.68 | 104 |

| <i>Panel B: Alpha by stress regime (% p.a.)</i> |            |      |     |                 |      |     |                |      |     |
|-------------------------------------------------|------------|------|-----|-----------------|------|-----|----------------|------|-----|
| Regime                                          | BTP Italia |      |     | iTraxx Combined |      |     | CDS–Bond Basis |      |     |
|                                                 | $\alpha$   | $t$  | $N$ | $\alpha$        | $t$  | $N$ | $\alpha$       | $t$  | $N$ |
| LOW                                             | +0.95      | 1.63 | 62  | +0.50           | 1.44 | 91  | +2.27**        | 2.39 | 63  |
| MEDIUM                                          | +2.05***   | 3.81 | 80  | +2.23***        | 3.16 | 99  | +3.65***       | 3.19 | 94  |
| HIGH                                            | +3.33*     | 1.74 | 21  | +3.34***        | 4.21 | 53  | +6.70***       | 4.12 | 50  |

#### A.6.4 Out-of-Sample Alpha: Rolling-Window Robustness

Table A.XIII re-estimates the out-of-sample alpha of Panel C of Table VI with the loadings fitted on a rolling 60-month window, which relaxes the constant-loadings assumption of the expanding scheme. The layout mirrors Panel C, so the two schemes can be compared cell by cell; the expanding estimates are not reproduced here.

**Table A.XIII**  
**Out-of-Sample Alpha: Rolling-Window Robustness**

Out-of-sample alpha of the three benchmark hedges (EUR factors) with the loadings re-estimated on a rolling 60-month past-only window, which relaxes the constant-loadings assumption of the expanding scheme of Panel C of Table VI; the layout mirrors Panel C. At each month  $t$ ,  $\hat{\beta}_{i,t-1}$  is estimated by OLS of the strategy excess return  $r_{i,t}$  on the benchmark factors  $\mathbf{f}_t$  over  $[t-60, t)$ , and the realized abnormal return is  $e_{i,t} = r_{i,t} - \hat{\beta}_{i,t-1}' \mathbf{f}_t$ ; the in-window intercept (the alpha under measurement) is excluded from the hedge, and rank-deficient or ill-conditioned windows (condition index  $> 10^3$ ) are skipped.  $\alpha$  is the annualized mean of  $e_{i,t}$  (% p.a.), tested with Newey–West HAC standard errors; IR is the annualized information ratio  $\bar{e}_i / \sigma(e_i) \times \sqrt{12}$ ;  $N$  is the number of out-of-sample months, common to the three benchmarks within each strategy. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                        | $N$ | Duarte et al. (2007) |      |       | Brooks et al. (2020) |      |       | Fung & Hsieh (2004) |      |       |
|------------------------|-----|----------------------|------|-------|----------------------|------|-------|---------------------|------|-------|
|                        |     | $\alpha$             | $t$  | IR    | $\alpha$             | $t$  | IR    | $\alpha$            | $t$  | IR    |
| <i>BTP Italia</i>      | 103 | +1.24**              | 2.21 | +0.75 | +1.94***             | 3.15 | +1.06 | +1.56***            | 2.61 | +0.94 |
| <i>iTraxx Combined</i> | 183 | +2.43***             | 3.97 | +1.18 | +1.66***             | 2.61 | +0.89 | +2.01***            | 3.51 | +1.06 |
| <i>CDS–Bond Basis</i>  | 147 | +3.35***             | 3.54 | +1.15 | +3.54***             | 3.92 | +1.20 | +3.11***            | 3.42 | +1.05 |

## A.7 VIF Diagnostics

Table A.XIV reports Variance Inflation Factors for the EUR factor specifications used in the benchmark regressions of Section II. VIF is computed on the factor matrix for each strategy’s sample period; because the factors are identical across strategies, differences in VIF reflect only the different estimation windows.

All equity and bond-market factors exhibit VIF below 10, confirming acceptable multicollinearity levels. The Duarte specification shows elevated VIF for the term-structure factors R2, R5, and R10 (VIF  $\approx 15$ –55), which is expected given the high correlation structure of yield-curve returns and does not affect inference on alpha. The Fung & Hsieh specification shows VIF below 5 for all factors. The Active FI specification shows VIF below 5 for most factors, with elevated values for Global\_Term and Global\_Aggregate (VIF  $\approx 40$ –50) reflecting the strong correlation between global bond benchmarks.

**Table A.XIV**  
**Variance Inflation Factors — Benchmark Factor Models, EUR Factors**

Variance inflation factors of the EUR factors of the three benchmark specifications, computed on each factor matrix over each strategy's estimation sample; conventional caution thresholds are 5 and 10. Elevated VIF for  $R_2$ ,  $R_5$ ,  $R_{10}$  (Panel A) reflects the inherent correlation of maturity-sorted government bond returns; elevated VIF for Global\_Term and Global\_Aggregate (Panel C) reflects the strong correlation between global bond benchmarks. Neither affects inference on alpha.

| Factor                                  | BTP Italia | CDS–Bond Basis | iTraxx Combined |
|-----------------------------------------|------------|----------------|-----------------|
| <b>Panel A: Duarte et al. (2007)</b>    |            |                |                 |
| Mkt-RF                                  | 4.86       | 5.27           | 4.97            |
| SMB                                     | 1.11       | 1.18           | 1.13            |
| HML                                     | 3.15       | 3.14           | 3.01            |
| UMD                                     | 1.72       | 1.85           | 1.81            |
| RS                                      | 7.29       | 8.25           | 8.10            |
| RI                                      | 16.07      | 5.23           | 5.08            |
| RB                                      | 11.83      | 2.82           | 2.77            |
| R2                                      | 16.02      | 14.42          | 15.93           |
| R5                                      | 55.30      | 51.71          | 55.21           |
| R10                                     | 26.56      | 24.19          | 24.01           |
| <b>Panel B: Fung &amp; Hsieh (2004)</b> |            |                |                 |
| SNPMRF                                  | 2.22       | 2.49           | 2.22            |
| SCMLC                                   | 1.07       | 1.15           | 1.10            |
| PTFSBD                                  | 1.45       | 1.40           | 1.37            |
| PTFSFX                                  | 1.46       | 1.50           | 1.46            |
| PTFSCOM                                 | 1.19       | 1.27           | 1.24            |
| R10                                     | 3.28       | 2.60           | 2.45            |
| BAAMTSY                                 | 4.10       | 4.36           | 4.04            |
| <b>Panel C: Brooks et al. (2020)</b>    |            |                |                 |
| Term                                    | 6.61       | 6.03           | 5.53            |
| Global_Term                             | 60.48      | 51.95          | 50.57           |
| Global_Aggregate                        | 66.17      | 55.82          | 53.25           |
| Inflation_Linkers                       | 5.04       | 4.10           | 3.84            |
| Corporate_Credit                        | 4.01       | 3.75           | 3.64            |
| Emerging_Debt                           | 8.22       | 6.04           | 5.60            |
| Emerging_Currency                       | 2.12       | 2.29           | 2.12            |
| UST_Volatility                          | 1.53       | 1.47           | 1.42            |

## A.8 Candidate Factor Universe

### A.8.1 Complete Factor List with Sources

Table A.XV reports the full set of candidate risk factors used in the penalised regressions of Section III, together with their data sources, transformations, and economic category.

**Table A.XV**  
**Candidate Risk Factors**

| Factor ID                           | Description                                                                                                                                                                              | Reference                                               |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Panel A: Credit Risk Factors</b> |                                                                                                                                                                                          |                                                         |
| BTP_BUND                            | BTP-Bund spread factor (10-year). Duration-matched excess return of Italian over German government bonds, long the Italian and short the German 7-10Y total-return index.                | Pelizzon et al. (2016)                                  |
| BTP_BUND_2Y                         | BTP-Bund spread factor (2-year). Duration-matched excess return of Italian over German government bonds at the short end, long the Italian and short the German 1-3Y total-return index. | Pelizzon et al. (2016)                                  |
| CDX_IG                              | Credit risk factor (U.S. investment grade). Excess return on a fully-collateralised sell-protection position in the 5-year CDX North American Investment Grade index.                    | Asvanunt and Richardson (2017); Bongaerts et al. (2011) |
| CREDIT_EU                           | Bond credit risk factor. Return spread between euro-area BBB and AAA corporate bond total-return indices.                                                                                | Fama and French (1993)                                  |
| CREDIT_US                           | Bond credit risk factor. Return spread between BAA and AAA U.S. corporate bond total-return indices.                                                                                     | Fama and French (1993)                                  |
| CRED_SPR_EU                         | Bond credit spread factor. Return spread between euro-area BBB corporate bonds and 10-year German Bunds.                                                                                 | Fung and Hsieh (2001)                                   |
| CRED_SPR_US                         | Bond credit spread factor. Return spread between BAA U.S. corporate bonds and 10-year U.S. Treasuries.                                                                                   | Fung and Hsieh (2001)                                   |
| DEF_EU                              | Bond default risk factor. Total return spread between long-term European corporate bonds and long-term German Bunds.                                                                     | Fama and French (1993)                                  |
| DEF_US                              | Bond default risk factor. Total return spread between long-term U.S. corporate bonds and long-term U.S. government bonds.                                                                | Fama and French (1993)                                  |
| EMERG_DEBT                          | Excess return on hard-currency emerging market debt.                                                                                                                                     | Brooks et al. (2020)                                    |
| EMERG_FX                            | Return on a basket of emerging market currencies versus the USD (MSCI Emerging Markets Currency Index; currency weights from the MSCI EM Index), in excess of USD cash.                  | Brooks et al. (2020)                                    |
| GLOBAL_AGG                          | Excess return on the Bloomberg Global Aggregate Bond Index.                                                                                                                              | Brooks et al. (2020)                                    |
| HY_CORP                             | Sub-investment-grade corporate credit factor: excess return of a EUR-hedged Pan-European high-yield corporate bond index over cash.                                                      | Asvanunt and Richardson (2017)                          |

*Note:* Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 75 factors.

**Table A.XV (continued)**

| <b>Factor ID</b>                  | <b>Description</b>                                                                                                                                                                                                                           | <b>Reference</b>                                        |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| ITRX_MAIN                         | Credit risk factor (European investment grade). Excess return on a fully-collateralised sell-protection position in the 5-year iTraxx Europe Main index.                                                                                     | Asvanunt and Richardson (2017); Bongaerts et al. (2011) |
| PB_CDS_1Y_EU                      | Intermediary funding factor (Europe, short tenor). Excess return on an equally weighted basket of 1-year sell-protection positions in the senior CDS of five European prime brokers.                                                         | Siriwardane (2019); He et al. (2017)                    |
| PB_CDS_1Y_US                      | Intermediary funding factor (United States, short tenor). Excess return on an equally weighted basket of 1-year sell-protection positions in the senior CDS of four U.S. prime brokers.                                                      | Siriwardane (2019); He et al. (2017)                    |
| PB_CDS_5Y_EU                      | Intermediary funding factor (Europe). Excess return on an equally weighted basket of 5-year sell-protection positions in the senior CDS of five European prime brokers.                                                                      | Siriwardane (2019); He et al. (2017)                    |
| PB_CDS_5Y_US                      | Intermediary funding factor (United States). Excess return on an equally weighted basket of 5-year sell-protection positions in the senior CDS of four U.S. prime brokers.                                                                   | Siriwardane (2019); He et al. (2017)                    |
| RI_EU                             | Excess return on a European A-rated industrial corporate bond index.                                                                                                                                                                         | Duarte et al. (2007)                                    |
| SLOPE_3S5S_MAIN                   | Credit-curve slope factor. Excess return on a DV01-matched position long protection at the 5-year and short protection at the 3-year point of the iTraxx Europe Main curve.                                                                  | Han et al. (2017); Han and Zhou (2015)                  |
| SNRFIN_MAIN                       | Financial-sector stress factor. Excess return on a DV01-matched position long credit via the iTraxx Senior Financials index and short the iTraxx Europe Main index.                                                                          | He et al. (2017); Siriwardane (2019)                    |
| XOVER_MAIN                        | Credit-cycle factor. Excess return on a beta-neutral position long credit via the 5-year iTraxx Crossover index and short the 5-year iTraxx Europe Main index, the short leg scaled by a rolling 36-month spread-change beta.                | Asvanunt and Richardson (2017)                          |
| <b>Panel B: Liquidity Factors</b> |                                                                                                                                                                                                                                              |                                                         |
| HKM_IC                            | Intermediary capital factor (tradable). Value-weighted equity return on the primary-dealer holding companies, in excess of cash and converted to euro; a tradable proxy (0.92 correlation) for the He-Kelly-Manela capital-ratio innovation. | He et al. (2017)                                        |
| LIQ_V                             | Value-weighted return on a 10–1 portfolio formed by sorting stocks on historical liquidity betas.                                                                                                                                            | Pástor and Staambaugh (2003)                            |
| <b>Panel C: Equity Factors</b>    |                                                                                                                                                                                                                                              |                                                         |
| BAB_EU                            | Betting Against Beta (Europe). Long low-beta and short high-beta European equities, each leg rescaled to unit beta; USD factor converted to euro.                                                                                            | Frazzini and Pedersen (2014)                            |
| CMA_EU                            | Investment factor (Conservative Minus Aggressive). Return spread between European conservative and aggressive investment portfolios.                                                                                                         | Fama and French (2017)                                  |
| COM_CARRY                         | Commodity carry factor. Self-financing long-short return on commodity futures sorted on the slope of the futures curve (carry); converted to euro.                                                                                           | Ilmanen et al. (2021)                                   |

*Note:* Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 75 factors.



**Table A.XV (continued)**

| Factor ID                          | Description                                                                                                                                                                                                                                                                                            | Reference              |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| COM_MOM                            | Commodity momentum factor. Self-financing long-short return on commodity futures sorted on their past 12-month return; converted to euro.                                                                                                                                                              | Ilmanen et al. (2021)  |
| FX_CARRY                           | Currency carry factor. Self-financing long-short return, long high-interest-rate and short low-interest-rate developed-market (G10) currencies; converted to euro.                                                                                                                                     | Ilmanen et al. (2021)  |
| FX_MOM                             | Currency momentum factor. Self-financing long-short return on developed-market (G10) currencies sorted on their past 12-month return; converted to euro.                                                                                                                                               | Ilmanen et al. (2021)  |
| GMOM                               | Global all-asset momentum factor. Long-short momentum return averaged across the eight Value-and-Momentum-Everywhere strategies (stock selection in the U.S., U.K., Europe and Japan, plus equity-index, government-bond, currency and commodity allocations), converted to euro.                      | Asness et al. (2013)   |
| GVAL                               | Global all-asset value factor. Long-short value return averaged across the eight Value-and-Momentum-Everywhere strategies (stock selection in the U.S., U.K., Europe and Japan, plus equity-index, government-bond, currency and commodity allocations), converted to euro.                            | Asness et al. (2013)   |
| HML_EU                             | Value factor (High Minus Low). Return spread between European high and low book-to-market portfolios.                                                                                                                                                                                                  | Fama and French (2017) |
| MKT_EU                             | Market excess return on the European value-weighted equity market portfolio over the 1-month German government bill.                                                                                                                                                                                   | Fama and French (2017) |
| QMJ_EU                             | Quality Minus Junk (Europe). Long high-quality and short low-quality (“junk”) European stocks, where quality aggregates profitability, growth, safety, and payout; USD factor converted to euro.                                                                                                       | Asness et al. (2019)   |
| RB_EU                              | Excess return on a European bank bond index.                                                                                                                                                                                                                                                           | Duarte et al. (2007)   |
| RMW_EU                             | Profitability factor (Robust Minus Weak). Return spread between European robust and weak operating profitability portfolios.                                                                                                                                                                           | Fama and French (2017) |
| RS_EU                              | Excess return on the EuroStoxx Banks index.                                                                                                                                                                                                                                                            | Duarte et al. (2007)   |
| SMB_EU                             | Size factor (Small Minus Big). Return spread between diversified European small- and large-cap portfolios.                                                                                                                                                                                             | Fama and French (2017) |
| UMD_EU                             | Momentum factor (Up Minus Down). Monthly return on a European zero-investment winners-minus-losers momentum portfolio.                                                                                                                                                                                 | Carhart (1997)         |
| <b>Panel D: Volatility Factors</b> |                                                                                                                                                                                                                                                                                                        |                        |
| ATM_IV_CDX                         | Excess return on a three-month variance swap on the CDX Investment Grade index spread: realized variance of the on-the-run index spread minus its at-the-money (100% moneyness) implied variance fixed at the start of the window (equivalently, the delta-hedged ATM payer/receiver straddle return). | Carr and Wu (2009)     |
| ATM_IV_ITRX                        | Excess return on a three-month variance swap on the iTraxx Europe Main index spread: realized variance of the on-the-run index spread minus its at-the-money (100% moneyness) implied variance fixed at the start of the window (equivalently, the delta-hedged ATM payer/receiver straddle return).   | Carr and Wu (2009)     |

*Note:* Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 75 factors.

Table A.XV (continued)

| Factor ID  | Description                                                                                                                                                                                                                                   | Reference             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| IV_BUND    | Excess return on a one-month variance swap on the 10-year German Bund future: realized variance of the future minus its at-the-money implied variance fixed at the start of the month (equivalently, the delta-hedged ATM straddle return).   | Cremers et al. (2021) |
| IV_TSY     | Excess return on a one-month variance swap on the 10-year U.S. Treasury future: realized variance of the future minus its at-the-money implied variance fixed at the start of the month (equivalently, the delta-hedged ATM straddle return). | Cremers et al. (2021) |
| BdOpt      | Return on the bond trend-following factor, constructed from lookback straddles on government bond futures.                                                                                                                                    | Fung and Hsieh (2004) |
| ComOpt     | Return on the commodity trend-following factor, constructed from lookback straddles on commodity futures.                                                                                                                                     | Fung and Hsieh (2004) |
| FXOpt      | Return on the currency trend-following factor, constructed from lookback straddles on major currency futures.                                                                                                                                 | Fung and Hsieh (2004) |
| PTFSIR     | Return on the short-term interest rate trend-following factor, constructed from lookback straddles on 3-month interest rate futures.                                                                                                          | Fung and Hsieh (2001) |
| PTFSSTK    | Return on the equity index trend-following factor, constructed from lookback straddles on stock index futures.                                                                                                                                | Fung and Hsieh (2004) |
| SVS_RET_1M | Excess return on a one-month simple variance swap on the S&P 500: realized simple variance minus the option-implied SVIX <sup>2</sup> strike fixed at the start of the month.                                                                 | Martin (2017)         |
| SVS_RET_3M | Excess return on a simple variance swap on the S&P 500 over a trailing three-month window (realized simple variance minus the SVIX <sup>2</sup> strike).                                                                                      | Martin (2017)         |
| V2X_FUT    | Volatility factor (euro area). Excess return on a constant-maturity (one-month) rolling long position in VSTOXX futures (VSTOXX Short-Term Futures index), in excess of the euro risk-free.                                                   | Eraker and Wu (2017)  |
| VIX_FUT    | Volatility factor (U.S.). Excess return on a constant-maturity (one-month) rolling long position in VIX futures (S&P 500 VIX Short-Term Futures Excess-Return index).                                                                         | Eraker and Wu (2017)  |

## Panel F: Interest Rate Factors

|                 |                                                                                                                                                                                                   |                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 5Y5Y_INFL       | Forward inflation factor (euro area). Mark-to-market excess return on a long-inflation position in a 5y–5y forward zero-coupon inflation swap (annuity-weighted change in the forward breakeven). | Haubrich et al. (2012)          |
| 5Y5Y_INFL_US    | Forward inflation factor (United States), expressed in euro. Mark-to-market excess return on a long-inflation position in a 5y–5y forward zero-coupon inflation swap.                             | Haubrich et al. (2012)          |
| CURV_2S5S10S_EU | Yield-curve curvature factor. Excess return on a DV01-neutral butterfly in German government bond futures, long the 5-year belly (Bobl) and short the 2-year (Schatz) and 10-year (Bund) wings.   | Litterman and Scheinkman (1991) |
| CURV_2S5S10S_US | Yield-curve curvature factor. Excess return on a DV01-neutral butterfly in U.S. Treasury futures, long the 5-year belly and short the 2- and 10-year wings.                                       | Litterman and Scheinkman (1991) |

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 75 factors.

**Table A.XV (continued)**

| <b>Factor ID</b> | <b>Description</b>                                                                                                                                                         | <b>Reference</b>                                     |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| FI_CARRY         | Fixed-income carry factor. Self-financing long-short return on developed-market government bonds sorted on term-structure carry (yield plus roll-down); converted to euro. | Ilmanen et al. (2021)                                |
| FI_DEF           | Fixed-income defensive factor. Self-financing long-short return on developed-market government bonds tilted toward low-risk exposures; converted to euro.                  | Ilmanen et al. (2021)                                |
| FI_MOM           | Fixed-income momentum factor. Self-financing long-short return on developed-market government bonds sorted on their past 12-month return; converted to euro.               | Ilmanen et al. (2021)                                |
| FI_VAL           | Fixed-income value factor. Self-financing long-short return on a cross-section of developed-market government bonds sorted on a value signal; converted to euro.           | Ilmanen et al. (2021)                                |
| GLOBAL_TERM      | Excess return over cash on the Bloomberg Global Treasury Index (EUR-hedged).                                                                                               | Brooks et al. (2020)                                 |
| GOVT_EU          | Euro-area term-premium factor. Excess return over cash on the broad German government bond aggregate (all maturities), the euro-area analogue of the US Term factor.       | Brooks et al. (2020)                                 |
| INFL_LINK        | Excess return on global inflation-linked bonds over cash or nominal Treasuries.                                                                                            | Brooks et al. (2020)                                 |
| R10_EU           | Excess return on the 10-year German Bund.                                                                                                                                  | Duarte et al. (2007)                                 |
| R2_EU            | Excess return on the 2-year German government bond (Schatz).                                                                                                               | Duarte et al. (2007)                                 |
| R5_EU            | Excess return on the 5-year German government bond (Bobl).                                                                                                                 | Duarte et al. (2007)                                 |
| SLOPE_10S30S_EU  | Yield-curve slope factor. Excess return on a DV01-matched steepener in German government bond futures, long the 10-year (Bund) and short the 30-year (Buxl).               | Litterman and Scheinkman (1991)                      |
| SLOPE_2S10S_EU   | Yield-curve slope factor. Excess return on a DV01-matched steepener in German government bond futures, long the 2-year (Schatz) and short the 10-year (Bund).              | Litterman and Scheinkman (1991)                      |
| SLOPE_2S10S_US   | Yield-curve slope factor. Excess return on a DV01-matched steepener in U.S. Treasury futures, long the 2-year and short the 10-year.                                       | Litterman and Scheinkman (1991)                      |
| SS10Y            | Swap-spread factor at the 10-year point. Excess return on a DV01-matched position long the German Bund future and pay-fixed on a 10-year constant-maturity EUR swap.       | Koijen et al. (2018); Klingler and Sundaresan (2019) |
| SS2Y             | Swap-spread factor at the 2-year point. Excess return on a DV01-matched position long the German Schatz future and pay-fixed on a 2-year constant-maturity EUR swap.       | Koijen et al. (2018); Klingler and Sundaresan (2019) |
| SS5Y             | Swap-spread factor at the 5-year point. Excess return on a DV01-matched position long the German Bobl future and pay-fixed on a 5-year constant-maturity EUR swap.         | Koijen et al. (2018); Klingler and Sundaresan (2019) |
| TERM_EU          | Bond term structure factor. Excess total return on long-term German Bunds over the 1-month German government bill return.                                                  | Fama and French (1993)                               |
| TERM_US          | Bond term structure factor. Excess total return on long-term U.S. government bonds over the 1-month U.S. Treasury bill return.                                             | Fama and French (1993)                               |

## A.9 PCA Robustness

### A.9.1 Robustness across the Number of Components

Table A.XVI reports alpha and  $\bar{R}^2$  across the number of retained components  $K$ . Results are robust to the choice of  $K$

**Table A.XVI**  
**PCA Spanning Robustness across the Number of Components**

Full-sample contemporaneous spanning  $r_{i,t} = \alpha_i + \beta_i' \mathbf{PC}_t + \varepsilon_{i,t}$  for  $K$  principal components (baseline  $K = 8$ ). Newey–West HAC  $t$  in parentheses. Expl. Var. is the cumulative variance captured by the  $K$  components. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| $K$ | btp italia        |             | cds bond basis    |             | itraxx combined   |             | Expl. Var. |
|-----|-------------------|-------------|-------------------|-------------|-------------------|-------------|------------|
|     | $\alpha$ (% p.a.) | $\bar{R}^2$ | $\alpha$ (% p.a.) | $\bar{R}^2$ | $\alpha$ (% p.a.) | $\bar{R}^2$ | (%)        |
| 5   | 1.50***<br>(3.21) | 0.089       | 4.79***<br>(5.98) | 0.114       | 2.21***<br>(5.20) | 0.050       | 55.0       |
| 8   | 1.44***<br>(3.24) | 0.088       | 4.78***<br>(5.99) | 0.109       | 2.21***<br>(5.68) | 0.134       | 64.7       |
| 11  | 1.47***<br>(3.50) | 0.086       | 4.77***<br>(5.84) | 0.111       | 2.21***<br>(5.70) | 0.136       | 71.8       |
| 13  | 1.75***<br>(3.97) | 0.144       | 4.76***<br>(5.95) | 0.105       | 2.21***<br>(5.82) | 0.146       | 75.6       |

### A.9.2 Subperiod and Regime Analysis

Half-sample splits and regime-dependent alpha confirm that PCA-based alpha is stable across subperiods and increases during stress.

**Table A.XVII**  
**PCA Subperiod Analysis (Contemporaneous)**

|                   | Full Sample       | First Half       | Second Half      |
|-------------------|-------------------|------------------|------------------|
| <b>BTP Italia</b> |                   |                  |                  |
| $\alpha$ (% p.a.) | 1.44***<br>(3.24) | 1.27**<br>(1.97) | 1.38**<br>(2.37) |
| $\beta_{PC1}$     | -0.031**          | -0.009           | -0.017           |

*continued on next page*

(continued)

|               | Full Sample | First Half | Second Half |
|---------------|-------------|------------|-------------|
|               | (-2.13)     | (-0.46)    | (-0.75)     |
| $\beta_{PC2}$ | -0.025**    | -0.003     | -0.040**    |
|               | (-2.04)     | (-0.18)    | (-2.54)     |
| $\beta_{PC3}$ | 0.030       | -0.003     | 0.099*      |
|               | (1.06)      | (-0.08)    | (1.79)      |
| $\beta_{PC4}$ | 0.049       | 0.114      | 0.039       |
|               | (1.42)      | (1.64)     | (1.09)      |
| $\beta_{PC5}$ | 0.002       | 0.045      | -0.019      |
|               | (0.03)      | (0.62)     | (-0.32)     |
| $\beta_{PC6}$ | -0.025      | -0.033     | 0.009       |
|               | (-0.61)     | (-0.77)    | (0.18)      |
| $\beta_{PC7}$ | -0.038      | -0.051     | 0.021       |
|               | (-1.45)     | (-1.12)    | (0.75)      |
| $\beta_{PC8}$ | -0.022      | -0.023     | -0.031      |
|               | (-1.23)     | (-0.42)    | (-1.21)     |
| $\bar{R}^2$   | 0.088       | 0.009      | 0.147       |
| $N$           | 157         | 78         | 79          |

**CDS–Bond**

|                   |           |          |          |
|-------------------|-----------|----------|----------|
| $\alpha$ (% p.a.) | 4.78***   | 6.06***  | 3.22***  |
|                   | (5.99)    | (3.92)   | (4.36)   |
| $\beta_{PC1}$     | 0.036*    | 0.005    | 0.121*** |
|                   | (1.90)    | (0.31)   | (4.82)   |
| $\beta_{PC2}$     | 0.060***  | 0.075    | 0.035*   |
|                   | (2.87)    | (1.64)   | (1.88)   |
| $\beta_{PC3}$     | -0.086*** | -0.057** | -0.107*  |
|                   | (-2.76)   | (-1.97)  | (-1.73)  |
| $\beta_{PC4}$     | -0.014    | 0.014    | -0.003   |
|                   | (-0.61)   | (0.26)   | (-0.09)  |
| $\beta_{PC5}$     | -0.017    | -0.011   | -0.048   |
|                   | (-0.55)   | (-0.28)  | (-0.86)  |
| $\beta_{PC6}$     | -0.007    | -0.056   | -0.044   |
|                   | (-0.24)   | (-0.84)  | (-1.61)  |
| $\beta_{PC7}$     | 0.042     | 0.062    | 0.127**  |
|                   | (1.31)    | (1.47)   | (2.52)   |
| $\beta_{PC8}$     | 0.029     | 0.049    | 0.086    |
|                   | (0.74)    | (1.14)   | (1.48)   |
| $\bar{R}^2$       | 0.109     | 0.041    | 0.267    |
| $N$               | 197       | 98       | 99       |

**Index Skew**

|                   |          |          |         |
|-------------------|----------|----------|---------|
| $\alpha$ (% p.a.) | 2.21***  | 2.47***  | 1.75*** |
|                   | (5.68)   | (4.61)   | (3.38)  |
| $\beta_{PC1}$     | -0.006   | 0.001    | 0.013   |
|                   | (-0.86)  | (0.17)   | (0.79)  |
| $\beta_{PC2}$     | 0.012    | 0.001    | 0.000   |
|                   | (0.93)   | (0.05)   | (0.01)  |
| $\beta_{PC3}$     | 0.037*** | 0.025*** | 0.120** |
|                   | (3.01)   | (3.04)   | (2.47)  |

continued on next page

(continued)

|               | Full Sample          | First Half         | Second Half        |
|---------------|----------------------|--------------------|--------------------|
| $\beta_{PC4}$ | 0.018<br>(0.94)      | -0.011<br>(-0.62)  | 0.074***<br>(2.73) |
| $\beta_{PC5}$ | 0.042***<br>(2.65)   | 0.021<br>(1.57)    | 0.145***<br>(2.98) |
| $\beta_{PC6}$ | -0.075***<br>(-3.02) | -0.038*<br>(-1.67) | -0.043*<br>(-1.66) |
| $\beta_{PC7}$ | 0.020<br>(1.11)      | 0.002<br>(0.06)    | 0.019<br>(0.79)    |
| $\beta_{PC8}$ | 0.042*<br>(1.68)     | -0.007<br>(-0.27)  | 0.097***<br>(2.79) |
| $\bar{R}^2$   | 0.134                | 0.005              | 0.253              |
| $N$           | 209                  | 104                | 105                |

Note: First/Second Half: temporal split at sample midpoint; stress regimes (LOW/MEDIUM/HIGH) are reported in Table XI.  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC, 4 lags). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

### A.9.3 Conditional Alpha: Christopherson–Ferson–Glassman (PCA)

Table A.XVIII reports the Christopherson–Ferson–Glassman conditional alpha using PCA factors: because the model uses the full continuous variation in the stress proxy rather than a discrete split, it establishes that the regime effect of Table XI does not hinge on the choice of cutoffs; the cutoffs themselves are the 60 and 100 basis-point thresholds that govern the transaction-cost tiers (Appendix A.2.3).

**Table A.XVIII**  
**Conditional Alpha (PCA, Contemporaneous): Christopherson, Ferson, and Glassman (1998) Model**

Christopherson, Ferson, and Glassman (1998) full conditional model,  $r_t = \alpha_0 + \alpha_1 z_{t-1} + (\beta + \delta z_{t-1})' \mathbf{PC}_t + \varepsilon_t$ , with the  $K = 8$  full-sample PCA factors: both the alpha and the factor loadings are conditioned on  $z_{t-1}$ , the one-month-lagged standardized iTraxx Main 5Y spread (a predetermined conditioning variable). Timing: Contemporaneous. Significance of  $\alpha_1$  from a moving-block bootstrap ( $B = 9,999$ ,  $H_0: \alpha_1 = 0$  imposed, block length 9); Newey–West HAC  $t$ -statistics in parentheses; the HAC  $p$ -value is a robustness check. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                    | BTP Italia | CDS–Bond Basis | iTraxx Combined |
|------------------------------------|------------|----------------|-----------------|
| $\alpha_0$ (ann. %)                | +1.32***   | +4.80***       | +2.18***        |
| $\alpha_1$ (% p.a. per $1\sigma$ ) | +1.33***   | +2.24***       | +1.39***        |
| $t$ -stat                          | (3.17)     | (3.50)         | (3.97)          |
| $p$ (bootstrap)                    | 0.0021     | 0.0089         | 0.0015          |
| $p$ (HAC, robustness)              | 0.0015     | 0.0005         | 0.0001          |
| $\alpha(z = +1\sigma)$ ann.        | +2.73      | +6.73          | +3.67           |
| $\alpha(z = -1\sigma)$ ann.        | +0.06      | +2.25          | +0.88           |
| $T$                                | 156        | 196            | 208             |

## A.10 Factor-Selection Robustness

### A.10.1 Sensitivity to the Subset Cardinality

Table A.XIX re-estimates the best-subset selection as the cardinality  $k$  varies around the baseline. The selected sets are broadly nested and the in-sample fit plateaus, so the conclusions do not hinge on the choice of  $k$ .

**Table A.XIX**  
**Best-Subset Support as the Cardinality  $k$  Varies**

Each row reports the best-subset solution of exactly  $k$  factors (Bertsimas, King and Mazumder, 2016) on the full factor panel.  $\bar{R}^2$  is the in-sample adjusted  $R^2$  on the standardized design. The primary specification fixes  $k = 8$ ; the selected set is broadly nested and the fit plateaus across  $k$ , so the conclusions do not hinge on the cardinality.

| $k$                    | #  | $\bar{R}^2$ | Selected factors                                                                                          |
|------------------------|----|-------------|-----------------------------------------------------------------------------------------------------------|
| <i>BTP Italia</i>      |    |             |                                                                                                           |
| 5                      | 5  | 0.25        | LIQ_V, MKT_EU, SMB_EU, FI_DEF, SS5Y                                                                       |
| 8                      | 8  | 0.22        | DEF_US, LIQ_V, SMB_EU, QMJ_EU, FI_DEF, SS2Y, SS5Y, SS10Y                                                  |
| 10                     | 10 | 0.25        | CDX_IG, XOVER_MAIN, DEF_US, LIQ_V, SMB_EU, FI_DEF, 5Y5Y_INFL_US, SS2Y, SS5Y, SS10Y                        |
| <i>CDS–Bond Basis</i>  |    |             |                                                                                                           |
| 5                      | 5  | 0.18        | PB_CDS_1Y_EU, DEF_US, EMERG_DEBT, RI_EU, PTFSFX                                                           |
| 8                      | 8  | 0.21        | PB_CDS_5Y_EU, DEF_US, HY_CORP, EMERG_DEBT, RI_EU, FX_MOM, PTFSFX, CURV_2S5S10S_EU                         |
| 10                     | 10 | 0.22        | PB_CDS_5Y_EU, DEF_US, HY_CORP, EMERG_DEBT, RI_EU, SMB_EU, FX_MOM, PTFSFX, IV_BUND, CURV_2S5S10S_EU        |
| <i>iTraxx Combined</i> |    |             |                                                                                                           |
| 5                      | 5  | 0.16        | BAB_EU, SVS_RET_1M, ATM_IV_ITRX, IV_TSY, SLOPE_10S30S_EU                                                  |
| 8                      | 8  | 0.18        | XOVER_MAIN, HML_EU, BAB_EU, FI_MOM, FI_DEF, SVS_RET_1M, ATM_IV_ITRX, ATM_IV_CDX                           |
| 10                     | 10 | 0.23        | HKM_IC, BAB_EU, GMOM, FI_MOM, SVS_RET_1M, ATM_IV_ITRX, ATM_IV_CDX, SLOPE_10S30S_EU, SLOPE_2S10S_US, SS10Y |

### A.10.2 Factor Selection Across Methods

Table A.XX reports the factor set selected for each strategy by the best-subset primary selector and by two independent cross-checks: the adaptive elastic net and post-double-selection. The selections overlap almost entirely, indicating that the identified exposures are not an artifact of the cardinality constraint or of any single penalization scheme.

**Table A.XX**  
**Selected Factors Across Selection Methods**

The selected factor set for each strategy under best-subset and the adaptive elastic net. Best-subset (Bertsimas, King and Mazumder, 2016) is the primary selector. The adaptive elastic net (Zou and Zhang, 2009) is a convex, oracle penalized estimator with data-driven cardinality; post-double-selection (Belloni, Chernozhukov and Hansen, 2014) guards the alpha against omitted-control bias. Agreement across these methods indicates the selection is not an artifact of the best-subset cardinality constraint.

| Method                 | Selected factors                                                                                                                                                                                                      |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>BTP Italia</i>      |                                                                                                                                                                                                                       |
| Best-subset            | DEF_US, LIQ_V, SMB_EU, QMJ_EU, FI_DEF, SS2Y, SS5Y, SS10Y                                                                                                                                                              |
| Adaptive elastic net   | HY_CORP, UMD_EU, FI_DEF, R2_EU                                                                                                                                                                                        |
| <i>CDS–Bond Basis</i>  |                                                                                                                                                                                                                       |
| Best-subset            | PB_CDS_5Y_EU, DEF_US, HY_CORP, EMERG_DEBT, RI_EU, FX_MOM, PTFSFX, CURV_2S5S10S_EU                                                                                                                                     |
| Adaptive elastic net   | CRED_SPR_EU, EMERG_DEBT, RI_EU, SS5Y                                                                                                                                                                                  |
| <i>iTraxx Combined</i> |                                                                                                                                                                                                                       |
| Best-subset            | XOVER_MAIN, HML_EU, BAB_EU, FI_MOM, FI_DEF, SVS_RET_1M, ATM_IV_ITRX, ATM_IV_CDX                                                                                                                                       |
| Adaptive elastic net   | PB_CDS_5Y_US, SLOPE_3S5S_MAIN, CRED_SPR_EU, DEF_EU, EMERG_DEBT, HKM_IC, HML_EU, BAB_EU, GMOM, FI_MOM, FI_DEF, FX_CARRY, PTFSFX, SVS_RET_1M, ATM_IV_ITRX, IV_TSY, TERM_US, SLOPE_2S10S_EU, SLOPE_2S10S_US, SS5Y, SS10Y |

#### A.10.3 Conditional Alpha: Detailed Results (Best-Subset)

Table A.XXI reports the full Christopherson–Ferson–Glassman conditional alpha model with best-subset-selected factors; as in the PCA case, the continuous conditioning variable makes the regime effect independent of the choice of discrete cutoffs.



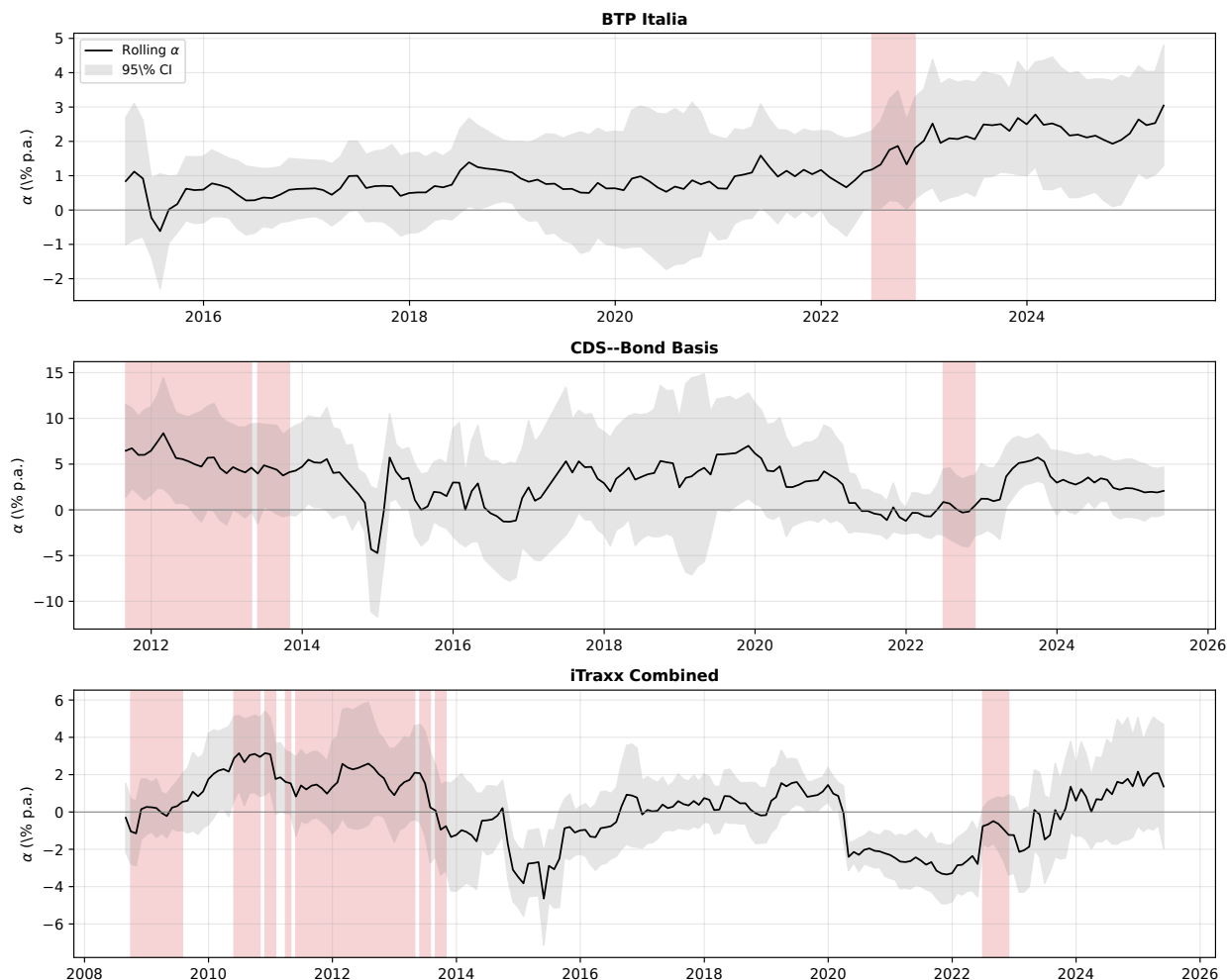
**Table A.XXI**  
**Conditional Alpha: Christopherson, Ferson, and Glassman (1998) Model**

Christopherson, Ferson, and Glassman (1998) full conditional performance model,  $r_t = \alpha_0 + \alpha_1 z_{t-1} + (\beta + \delta z_{t-1})' \mathbf{f}_{\hat{S},t} + \varepsilon_t$ , with the best-subset-selected factors  $\hat{S}$ : both the alpha and the factor loadings are conditioned on  $z_{t-1}$ , the one-month-lagged standardized iTraxx Main 5Y spread (mean zero, unit variance, a predetermined conditioning variable);  $\alpha_1 > 0$  indicates alpha rises with credit stress. Significance of  $\alpha_1$  is from a moving-block bootstrap ( $B = 9,999$ ,  $H_0: \alpha_1 = 0$  imposed, block length 9);  $t$ -statistics in parentheses are Newey–West HAC (lag 4). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                                           | BTP Italia | CDS–Bond Basis | iTraxx Combined |
|-----------------------------------------------------------|------------|----------------|-----------------|
| <i>Panel A: Conditional alpha</i>                         |            |                |                 |
| $\alpha_0$ (ann. %)                                       | +1.56***   | +4.83***       | +2.69***        |
| $\alpha_1$ (% p.a. per $1\sigma$ )                        | +0.41      | +2.55***       | +0.57           |
| $t$ -stat                                                 | (1.20)     | (4.31)         | (1.43)          |
| $p$ (bootstrap)                                           | 0.2891     | 0.0063         | 0.2379          |
| $p$ (HAC, robustness)                                     | 0.2313     | 0.0000         | 0.1522          |
| $\bar{R}^2$ (unconditional)                               | 0.2132     | 0.2116         | 0.1759          |
| $\bar{R}^2$ (conditional)                                 | 0.2997     | 0.2570         | 0.2010          |
| $\alpha(z=+1\sigma)$ (% p.a.)                             | +1.96      | +7.46          | +3.29           |
| $\alpha(z=-1\sigma)$ (% p.a.)                             | +1.13      | +2.37          | +2.14           |
| $T$                                                       | 156        | 196            | 210             |
| <i>Panel B: Conditional betas (<math>\delta_j</math>)</i> |            |                |                 |
| $\text{ATM}_{\text{IV\_CDX}} \times z_t$                  | –          | –              | –0.001          |
| $\text{ATM}_{\text{IV\_ITRX}} \times z_t$                 | –          | –              | –0.266          |
| $\text{BAB}_{\text{EU}} \times z_t$                       | –          | –              | –0.006          |
| $\text{CURV}_{2\text{S5S10S\_EU}} \times z_t$             | –          | +0.016***      | –               |
| $\text{DEF}_{\text{US}} \times z_t$                       | –0.003     | +0.024         | –               |
| $\text{EMERG}_{\text{DEBT}} \times z_t$                   | –          | +0.041*        | –               |
| $\text{FI}_{\text{DEF}} \times z_t$                       | –0.069     | –              | +0.030          |
| $\text{FI}_{\text{MOM}} \times z_t$                       | –          | –              | +0.014          |
| $\text{FX}_{\text{MOM}} \times z_t$                       | –          | –0.001         | –               |
| $\text{HML}_{\text{EU}} \times z_t$                       | –          | –              | –0.008          |
| $\text{HY}_{\text{CORP}} \times z_t$                      | –          | –0.063*        | –               |
| $\text{LIQ}_{\text{V}} \times z_t$                        | –0.014     | –              | –               |
| $\text{PB}_{\text{CDS\_5Y\_EU}} \times z_t$               | –          | +0.094         | –               |
| $\text{PTFSFX} \times z_t$                                | –          | +0.003         | –               |
| $\text{QMJ}_{\text{EU}} \times z_t$                       | –0.015     | –              | –               |
| $\text{RI}_{\text{EU}} \times z_t$                        | –          | –0.044         | –               |
| $\text{SMB}_{\text{EU}} \times z_t$                       | –0.043**   | –              | –               |
| $\text{SS10Y} \times z_t$                                 | +0.016*    | –              | –               |
| $\text{SS2Y} \times z_t$                                  | +0.016     | –              | –               |
| $\text{SS5Y} \times z_t$                                  | –0.022*    | –              | –               |
| $\text{SVS}_{\text{RET\_1M}} \times z_t$                  | –          | –              | –0.094          |
| $\text{XOVER}_{\text{MAIN}} \times z_t$                   | –          | –              | +0.020          |

#### A.10.4 Rolling Alpha

Figure A.1 plots rolling 36-month alpha with regime shading for the best-subset-selected factors. Alpha spikes during stress episodes.



**Figure A.1**

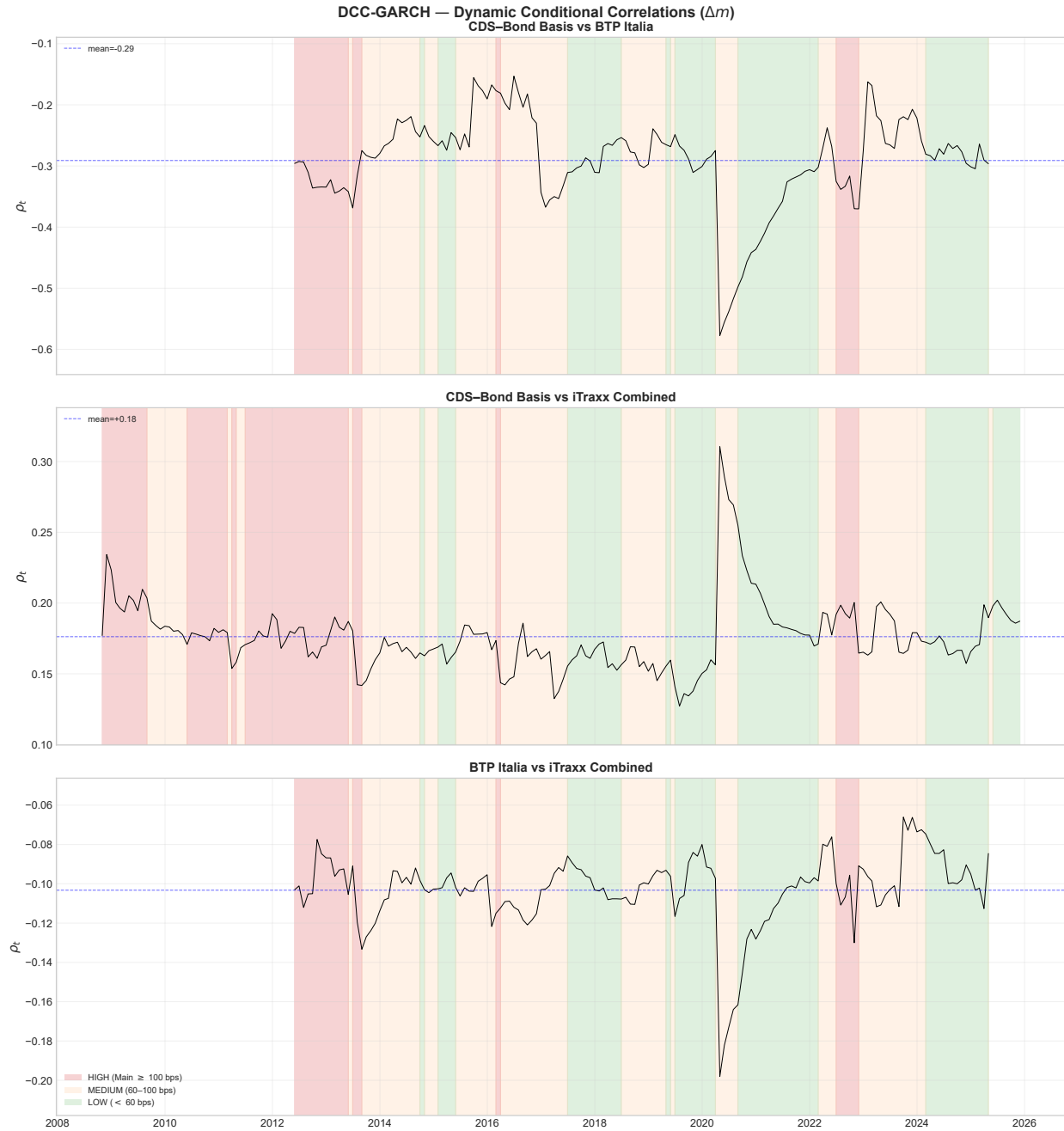
Rolling Alpha with Regime Shading (Best-Subset Factors). Grey bands: 95% confidence intervals. Red shading: iTraxx Main 5Y > 100 bps. Rolling window: 36 months.

## A.11 Cross-Strategy Interdependencies: Additional Tests

This appendix collects robustness checks and supplementary analyses for Section IV. All tests use the same mispricing magnitudes  $m_{i,t}$  and regime definitions as the main text.

### A.11.1 DCC-GARCH

We estimate bivariate DCC-GARCH(1,1) models on each pair of  $\Delta m$  series. Figure A.2 plots the time-varying conditional correlations. Spikes in conditional correlation coincide with stress episodes (COVID-19, ECB hiking cycle), corroborating the regime-dependent results in Section IV.



**Figure A.2**

Dynamic Conditional Correlations (DCC-GARCH). Bivariate DCC(1,1) estimates for each pair of  $\Delta m$ . Shaded areas: HIGH stress regime (iTraxx Main  $> 100$  bps).

### A.11.2 Granger Causality: Lag Robustness

Table A.XXII reports Granger causality tests at lag lengths  $p = 1, 2, 3$ . The main result — iTraxx Combined Granger-causes both the CDS-Bond basis and BTP Italia — is robust across all specifications.

**Table A.XXII**  
**Granger Causality: Robustness to Lag Length**

Each entry is the  $F$ -statistic for the null that the row variable does not Granger-cause the column variable, from a  $\text{VAR}(p)$  on the  $\Delta m$  series estimated separately at  $p = 1, 2, 3$  lags. Lag selection: HQC selects  $p = 1$ ; AIC/FPE select  $p = 2$ ; BIC selects  $p = 0$  (forced to 1). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Cause           | Effect          | $p = 1$ |          | $p = 2$ |          | $p = 3$ |          |
|-----------------|-----------------|---------|----------|---------|----------|---------|----------|
|                 |                 | $F$     | $p$ -val | $F$     | $p$ -val | $F$     | $p$ -val |
| BTP Italia      | CDS–Bond Basis  | 0.05    | 0.8176   | 0.55    | 0.5790   | 1.61    | 0.1858   |
| iTraxx Combined | CDS–Bond Basis  | 6.52**  | 0.0110   | 3.97**  | 0.0195   | 2.14*   | 0.0946   |
| CDS–Bond Basis  | BTP Italia      | 3.08*   | 0.0802   | 1.40    | 0.2465   | 1.53    | 0.2071   |
| iTraxx Combined | BTP Italia      | 7.39*** | 0.0068   | 4.16**  | 0.0162   | 3.04**  | 0.0289   |
| CDS–Bond Basis  | iTraxx Combined | 0.01    | 0.9043   | 0.16    | 0.8537   | 0.13    | 0.9445   |
| BTP Italia      | iTraxx Combined | 0.48    | 0.4897   | 1.22    | 0.2970   | 0.87    | 0.4589   |

## Tables

**Table I**  
**Summary Statistics for Fixed Income Arbitrage Strategies**

This table reports summary statistics for the signal-weighted arbitrage strategies, net of transaction costs. Panel A reports monthly basis levels (in basis points), with  $P_{10}$  and  $P_{90}$  the 10th and 90th percentiles; Panel B reports monthly percentage excess returns. The basis for each strategy is constructed as described in Section A. In Panel B, the mean and standard deviation are annualized and the Sharpe ratio is their ratio; kurtosis is excess kurtosis; the  $t$ -statistic for the mean is corrected for serial correlation (Newey–West, 4 lags); and %Neg is the proportion of negative monthly excess returns. All series end in May 2025; the samples begin in April 2012 (BTP Italia), September 2005 (iTraxx Combined), and September 2008 (CDS–Bond Basis).

| <b>Panel A: Basis Levels (bps)</b> |      |        |         |       |       |       |      |       |          |          |
|------------------------------------|------|--------|---------|-------|-------|-------|------|-------|----------|----------|
| Strategy                           | Mean | Median | Std Dev | Min   | Max   | Skew  | Kurt | AC(1) | $P_{10}$ | $P_{90}$ |
| <i>BTP Italia</i>                  | 44.7 | 46.3   | 28.6    | -66.5 | 101.0 | -1.43 | 3.29 | 0.81  | 14.4     | 73.8     |
| <i>CDS-Bond Basis</i>              | 6.9  | 9.8    | 18.5    | -73.9 | 39.5  | -0.97 | 1.69 | 0.83  | -18.7    | 30.5     |
| <i>iTraxx Combined</i>             | 0.2  | 1.1    | 6.7     | -37.5 | 21.9  | -1.92 | 7.76 | 0.66  | -6.1     | 6.6      |

| <b>Panel B: Strategy Returns (%)</b> |     |                  |           |                     |            |            |      |      |      |       |        |
|--------------------------------------|-----|------------------|-----------|---------------------|------------|------------|------|------|------|-------|--------|
| Strategy                             | $n$ | Mean<br>(% p.a.) | $t$ -Stat | Std Dev<br>(% p.a.) | Min<br>(%) | Max<br>(%) | Skew | Kurt | %Neg | AC(1) | Sharpe |
| <i>BTP Italia</i>                    | 163 | 1.59             | 3.52      | 1.62                | -1.25      | 1.81       | 0.81 | 1.77 | 36.8 | 0.08  | 0.98   |
| <i>CDS Bond Basis</i>                | 207 | 4.61             | 5.87      | 2.96                | -3.33      | 3.66       | 0.79 | 4.26 | 26.1 | -0.05 | 1.56   |
| <i>iTraxx Combined</i>               | 243 | 1.91             | 4.92      | 1.53                | -2.06      | 2.50       | 1.01 | 7.37 | 38.3 | 0.01  | 1.25   |

**Table II**  
**Manipulation-Proof Performance Measure by Risk Aversion**

MPPM is the manipulation-proof performance measure of Goetzmann et al. (2007), computed from monthly returns;  $\rho$  is the coefficient of relative risk aversion and the risk-free rate is 1-month Euribor. Returns are net of transaction costs.

| Strategy        | $\rho = 2$<br>(% p.a.) | $\rho = 3$<br>(% p.a.) | $\rho = 4$<br>(% p.a.) |
|-----------------|------------------------|------------------------|------------------------|
| BTP Italia      | 1.50                   | 1.48                   | 1.47                   |
| CDS–Bond Basis  | 4.43                   | 4.39                   | 4.35                   |
| iTraxx Combined | 1.88                   | 1.87                   | 1.86                   |

**Table III**  
**Duarte Factor Model Regressions**

This table reports estimates of the regression  $r_{i,t} = \alpha_i + \beta_1 Mkt_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 R_{S,t} + \beta_6 R_{I,t} + \beta_7 R_{B,t} + \beta_8 R_{2,t} + \beta_9 R_{5,t} + \beta_{10} R_{10,t} + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the ten Duarte et al. (2007) euro-area equity and bond risk factors.  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC).  $Mkt$ : European value-weighted market excess return over the 1-month German bill;  $SMB$ ,  $HML$ ,  $UMD$ : the Fama–French European size, value, and momentum factors.  $R_S$ : excess return on the EuroStoxx Banks index.  $R_I$ ,  $R_B$ : excess returns on European A-rated industrial and European bank bond indices.  $R_2$ ,  $R_5$ ,  $R_{10}$ : excess returns on European 2-, 5-, and 10-year government bond portfolios.  $\bar{R}^2$  is the adjusted  $R^2$ . Sample: BTP Italia 163 months, CDS–Bond Basis 207 months, iTraxx Combined 243 months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                        | $\alpha$ (% p.a.) | $\beta_{Mkt}$     | $\beta_{SMB}$       | $\beta_{HML}$      | $\beta_{UMD}$     | $\beta_{R_S}$     | $\beta_{R_I}$     | $\beta_{R_B}$    | $\beta_{R_2}$    | $\beta_{R_5}$    | $\beta_{R_{10}}$ | $\bar{R}^2$ |
|------------------------|-------------------|-------------------|---------------------|--------------------|-------------------|-------------------|-------------------|------------------|------------------|------------------|------------------|-------------|
| <i>BTP Italia</i>      | 1.62***<br>(3.29) | -0.04*<br>(-1.79) | -0.06***<br>(-2.86) | -0.03<br>(-1.42)   | 0.02<br>(1.13)    | 0.01<br>(1.01)    | -0.03<br>(-0.27)  | 0.04<br>(0.49)   | 0.03<br>(0.06)   | -0.04<br>(-0.14) | -0.02<br>(-0.16) | 0.11        |
| <i>CDS–Bond Basis</i>  | 4.24***<br>(5.27) | 0.00<br>(0.20)    | 0.04<br>(0.90)      | 0.03<br>(1.11)     | -0.02<br>(-0.88)  | -0.02*<br>(-1.74) | 0.45***<br>(3.94) | -0.06<br>(-1.57) | -0.27<br>(-0.45) | -0.04<br>(-0.10) | -0.10<br>(-0.71) | 0.15        |
| <i>iTraxx Combined</i> | 2.19***<br>(4.90) | -0.02<br>(-1.25)  | -0.03**<br>(-2.04)  | -0.07**<br>(-2.26) | -0.02*<br>(-1.95) | 0.01<br>(1.53)    | 0.05<br>(0.75)    | -0.02<br>(-0.78) | 0.06<br>(0.16)   | 0.05<br>(0.21)   | -0.06<br>(-0.78) | 0.06        |



**Table IV**  
**Fung & Hsieh Factor Model Regressions**

This table reports estimates of the regression  $r_{i,t} = \alpha_i + \beta_1 S\&P_t + \beta_2 SC-LC_t + \beta_3 BdOpt_t + \beta_4 FXOpt_t + \beta_5 ComOpt_t + \beta_6 10Y_t + \beta_7 CredSpr_t + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the seven Fung & Hsieh (2004) euro-area hedge fund risk factors.  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC). S&P: European value-weighted market excess return. SC–LC: the Fama–French European size (small-minus-big) factor. BdOpt, FXOpt, ComOpt: trend-following returns from lookback straddles on government-bond, currency, and commodity futures. 10Y: excess return on a 10-year German government bond total-return index. CredSpr: return spread between euro-area BBB corporate bonds and 10-year German Bunds.  $\bar{R}^2$  is the adjusted  $R^2$ . Sample: BTP Italia 163 months, CDS–Bond 207 months, iTraxx 243 months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                            | $\alpha$ (% p.a.) | $\beta_{S\&P}$     | $\beta_{SC-LC}$     | $\beta_{BdOpt}$ | $\beta_{FXOpt}$   | $\beta_{ComOpt}$ | $\beta_{10Y}$     | $\beta_{CredSpr}$ | $\bar{R}^2$ |
|----------------------------|-------------------|--------------------|---------------------|-----------------|-------------------|------------------|-------------------|-------------------|-------------|
| <i>BTP Italia</i>          | 1.81***<br>(4.08) | −0.03**<br>(−2.53) | −0.06***<br>(−2.70) | 0.00<br>(1.25)  | 0.00<br>(0.03)    | 0.00<br>(1.09)   | −0.01<br>(−0.27)  | 0.02<br>(0.30)    | 0.10        |
| <i>CDS–Bond</i>            | 3.98***<br>(5.05) | −0.02<br>(−1.18)   | 0.05<br>(1.24)      | 0.00<br>(1.01)  | −0.01*<br>(−1.95) | −0.00<br>(−0.42) | 0.17***<br>(3.43) | 0.19***<br>(2.87) | 0.11        |
| <i>iTraxx<br/>Combined</i> | 1.84***<br>(4.49) | 0.00<br>(0.41)     | −0.02<br>(−1.50)    | 0.00<br>(0.60)  | 0.00<br>(1.01)    | 0.00<br>(0.57)   | 0.03<br>(0.83)    | 0.02<br>(0.62)    | 0.00        |

**Table V**  
**Active Fixed Income Factor Model Regressions**

This table reports estimates of the regression  $r_{i,t} = \alpha_i + \beta_1 \text{Term}_t + \beta_2 \text{Global Term}_t + \beta_3 \text{Global Agg}_t + \beta_4 \text{Infl. Linkers}_t + \beta_5 \text{Corp Credit}_t + \beta_6 \text{Emerg Debt}_t + \beta_7 \text{Emerg Curr}_t + \beta_8 \text{UST Vol}_t + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the eight Active Fixed Income risk premia of Brooks et al. (2020).  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC). Term: euro-area government bond excess return over cash. Global Term: Bloomberg Global Treasury Index (EUR-hedged), in excess of cash. Global Agg: excess return on the Bloomberg Global Aggregate Bond Index. Infl. Linkers: global inflation-linked bonds in excess of cash. Corp Credit: EUR-hedged pan-European high-yield corporate bond index, in excess of cash. Emerg Debt: excess return on hard-currency emerging-market debt. Emerg Curr: emerging-market currency basket versus the U.S. dollar, in excess of USD cash. UST Vol: return on a one-month variance swap (delta-hedged ATM straddle) on the 10-year U.S. Treasury future.  $\bar{R}^2$  is the adjusted  $R^2$ . Sample: BTP Italia 163 months, CDS–Bond Basis 207 months, iTraxx Combined 243 months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                        | $\alpha$ (% p.a.) | $\beta_{\text{Term}}$ | $\beta_{\text{Global Term}}$ | $\beta_{\text{Global Agg}}$ | $\beta_{\text{Infl. Linkers}}$ | $\beta_{\text{Corp Credit}}$ | $\beta_{\text{Emerg Debt}}$ | $\beta_{\text{Emerg Curr}}$ | $\beta_{\text{UST Vol}}$ | $\bar{R}^2$ |
|------------------------|-------------------|-----------------------|------------------------------|-----------------------------|--------------------------------|------------------------------|-----------------------------|-----------------------------|--------------------------|-------------|
| <i>BTP Italia</i>      | 1.98***<br>(3.83) | -0.01<br>(-0.28)      | -0.05<br>(-0.15)             | 0.12<br>(0.45)              | -0.06*<br>(-1.84)              | -0.11**<br>(-2.07)           | 0.06<br>(1.20)              | -0.02<br>(-0.67)            | -0.63<br>(-0.03)         | 0.07        |
| <i>CDS–Bond Basis</i>  | 4.37***<br>(5.49) | 0.07<br>(1.63)        | -0.46<br>(-0.77)             | 0.32<br>(0.59)              | -0.05<br>(-0.83)               | -0.03<br>(-0.70)             | 0.14***<br>(2.98)           | 0.01<br>(0.40)              | 15.85<br>(0.58)          | 0.11        |
| <i>iTraxx Combined</i> | 1.84***<br>(4.40) | -0.04*<br>(-1.91)     | -0.23<br>(-0.99)             | 0.45*<br>(1.85)             | 0.01<br>(0.22)                 | 0.02<br>(1.50)               | -0.08**<br>(-2.25)          | 0.01<br>(0.27)              | 38.28**<br>(1.98)        | 0.06        |

**Table VI**  
**Alpha Estimates Across Benchmark Factor Models**

*Note:* Panel A reports annualized alpha (% p.a.) from monthly OLS regressions with Newey–West HAC standard errors (4 lags) using EUR factors. Panel B reports the intercept  $\alpha_0$  and the stress-loading coefficient  $\alpha_1$  from the Christopherson et al. (1998) full conditional model  $r_t = \alpha_0 + \alpha_1 z_{t-1} + (\beta + \delta z_{t-1})' X_t + \varepsilon_t$ , in which both the alpha and the factor loadings are conditioned on  $z_{t-1}$ , the one-month-lagged standardized iTraxx Main 5Y spread (a pre-determined conditioning variable);  $\bar{R}^2$  is the adjusted  $R^2$ . Significance of  $\alpha_1$  is assessed by a moving-block bootstrap ( $B = 9,999$ ,  $H_0: \alpha_1 = 0$  imposed, block length 9). Panel C reports out-of-sample alpha in the recursive design of Welch and Goyal (2008): loadings estimated by OLS on the expanding past-only window  $[1, t]$  (60-month burn-in), and the realized abnormal return  $e_{i,t} = r_{i,t} - \hat{\beta}'_{i,t-1} \mathbf{f}_t$  tested with Newey–West HAC; IR is the annualized information ratio of the hedged return, and  $N$  the number of out-of-sample months, common to the three benchmarks within each strategy (rolling-window counterpart: Appendix A.6.4).  $t$ -statistics in parentheses are Newey–West HAC throughout. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

*Panel A: Full-sample alpha (% p.a.)*

| Strategy               | Duarte et al. (2007) |      |     | Brooks et al. (2020) |      |     | Fung & Hsieh (2004) |      |     |
|------------------------|----------------------|------|-----|----------------------|------|-----|---------------------|------|-----|
|                        | $\alpha$             | $t$  | $N$ | $\alpha$             | $t$  | $N$ | $\alpha$            | $t$  | $N$ |
| <i>BTP Italia</i>      | +1.62***             | 3.29 | 163 | +1.98***             | 3.83 | 163 | +1.81***            | 4.08 | 163 |
| <i>iTraxx Combined</i> | +2.19***             | 4.90 | 243 | +1.84***             | 4.40 | 243 | +1.84***            | 4.49 | 243 |
| <i>CDS–Bond Basis</i>  | +4.24***             | 5.27 | 207 | +4.37***             | 5.49 | 207 | +3.98***            | 5.05 | 207 |

*Panel B: Christopherson, Ferson, and Glassman (1998) conditional alpha (% p.a.)*

|                        | Duarte et al. (2007) |                   |             | Brooks et al. (2020) |                    |             | Fung & Hsieh (2004) |                    |             |
|------------------------|----------------------|-------------------|-------------|----------------------|--------------------|-------------|---------------------|--------------------|-------------|
|                        | $\alpha_0$           | $\alpha_1$        | $\bar{R}^2$ | $\alpha_0$           | $\alpha_1$         | $\bar{R}^2$ | $\alpha_0$          | $\alpha_1$         | $\bar{R}^2$ |
| <i>BTP Italia</i>      | +1.58***<br>(3.78)   | +1.20**<br>(2.25) | 0.23        | +1.74***<br>(3.13)   | +0.85<br>(1.02)    | 0.13        | +1.70***<br>(5.24)  | +0.50<br>(1.22)    | 0.19        |
| <i>iTraxx Combined</i> | +1.94***<br>(4.75)   | +0.93**<br>(2.55) | 0.11        | +1.93***<br>(4.43)   | +1.47***<br>(3.41) | 0.15        | +1.89***<br>(4.85)  | +1.51***<br>(4.66) | 0.05        |
| <i>CDS–Bond Basis</i>  | +4.67***<br>(5.99)   | +2.10**<br>(2.74) | 0.25        | +4.73***<br>(6.50)   | +3.07***<br>(3.80) | 0.15        | +3.96***<br>(5.45)  | +1.73**<br>(2.47)  | 0.17        |

*Panel C: Out-of-sample alpha (% p.a.)*

|                        | $N$ | Duarte et al. (2007) |      |       | Brooks et al. (2020) |      |       | Fung & Hsieh (2004) |      |       |
|------------------------|-----|----------------------|------|-------|----------------------|------|-------|---------------------|------|-------|
|                        |     | $\alpha$             | $t$  | IR    | $\alpha$             | $t$  | IR    | $\alpha$            | $t$  | IR    |
| <i>BTP Italia</i>      | 103 | +1.29**              | 2.24 | +0.78 | +2.01***             | 3.55 | +1.18 | +1.69***            | 2.86 | +1.05 |
| <i>iTraxx Combined</i> | 183 | +2.27***             | 4.71 | +1.37 | +1.78***             | 3.59 | +1.09 | +2.02***            | 4.05 | +1.20 |
| <i>CDS–Bond Basis</i>  | 147 | +3.48***             | 4.29 | +1.25 | +3.75***             | 4.21 | +1.29 | +3.45***            | 3.87 | +1.19 |

**Table VII**  
**PCA Spanning Regressions**

This table reports estimates of the spanning regression  $r_{i,t} = \alpha_i + \beta_1 PC_{1,t} + \beta_2 PC_{2,t} + \dots + \beta_8 PC_{8,t} + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of strategy  $i$  and the regressors are the first 8 principal components of the standardized candidate factor panel, extracted over the full estimation sample (the number of components is selected by the Bai–Ng  $IC_{p2}$  criterion).  $\alpha$  is annualized (% p.a.).  $t$ -statistics in parentheses (Newey–West HAC,  $\lfloor 4(T/100)^{2/9} \rfloor = 4$  lags).  $\bar{R}^2$  is the adjusted  $R^2$ ;  $N$  is the number of monthly observations. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

|                                                            | $\alpha$ (% p.a.) | $\beta_{PC1}$      | $\beta_{PC2}$      | $\beta_{PC3}$       | $\beta_{PC4}$    | $\beta_{PC5}$     | $\beta_{PC6}$       | $\beta_{PC7}$    | $\beta_{PC8}$    | $\bar{R}^2$ | $N$ |
|------------------------------------------------------------|-------------------|--------------------|--------------------|---------------------|------------------|-------------------|---------------------|------------------|------------------|-------------|-----|
| <b>Contemporaneous (<math>PC_t \rightarrow R_t</math>)</b> |                   |                    |                    |                     |                  |                   |                     |                  |                  |             |     |
| <i>BTP Italia</i>                                          | 1.44***<br>(3.24) | -0.03**<br>(-2.13) | -0.02**<br>(-2.04) | 0.03<br>(1.06)      | 0.05<br>(1.42)   | 0.00<br>(0.03)    | -0.02<br>(-0.61)    | -0.04<br>(-1.45) | -0.02<br>(-1.23) | 0.088       | 157 |
| <i>CDS–Bond</i>                                            | 4.78***<br>(5.99) | 0.04*<br>(1.90)    | 0.06***<br>(2.87)  | -0.09***<br>(-2.76) | -0.01<br>(-0.61) | -0.02<br>(-0.55)  | -0.01<br>(-0.24)    | 0.04<br>(1.31)   | 0.03<br>(0.74)   | 0.109       | 197 |
| <i>Index Skew</i>                                          | 2.21***<br>(5.68) | -0.01<br>(-0.86)   | 0.01<br>(0.93)     | 0.04***<br>(3.01)   | 0.02<br>(0.94)   | 0.04***<br>(2.65) | -0.07***<br>(-3.02) | 0.02<br>(1.11)   | 0.04*<br>(1.68)  | 0.134       | 209 |

**Table VIII**  
**Top Factor Loadings by Principal Component**

For each of the first 3 principal components, the 6 candidate factors with the largest absolute loading, with the corresponding eigenvector entry. Loadings are the full-sample eigenvectors of the standardized candidate factor panel; the sign indicates the direction of the relationship.

| PC1         |         | PC2         |         | PC3          |         |
|-------------|---------|-------------|---------|--------------|---------|
| Factor      | Loading | Factor      | Loading | Factor       | Loading |
| CRED_SPR_EU | +0.20   | GLOBAL_AGG  | +0.30   | SNRFIN_MAIN  | +0.30   |
| DEF_US      | +0.20   | GLOBAL_TERM | +0.28   | BTP_BUND_2Y  | +0.21   |
| DEF_EU      | +0.19   | R10_EU      | +0.28   | PB_CDS_5Y_EU | +0.20   |
| CRED_SPR_US | +0.19   | GOVT_EU     | +0.28   | CREDIT_US    | -0.20   |
| ITRX_MAIN   | +0.19   | TERM_EU     | +0.28   | BTP_BUND     | +0.19   |
| HY_CORP     | +0.19   | RI_EU       | +0.27   | FI_VAL       | +0.19   |

**Table IX**  
**Out-of-Sample Alpha: PCA and Best-Subset**

Out-of-sample alpha of the PCA (Panel A) and best-subset (Panel B) hedges, in the recursive design of Welch and Goyal (2008). At each month  $t$  the hedge is estimated on a past-only window—expanding with a 60-month burn-in (headline) or rolling 60-month, which relaxes the constant-loadings assumption—and the realized abnormal return is  $e_{i,t} = r_{i,t} - \hat{\beta}'_{i,t-1} \mathbf{f}_t$ ; the in-window intercept (the alpha under measurement) is excluded from the hedge. In Panel A the principal components are re-estimated within each window, so the design is invariant to the sign and rotation indeterminacy of PCA; in Panel B the best-subset factor set is frozen and only the loadings are re-estimated.  $\alpha$  is the annualized mean of  $e_{i,t}$  (% p.a.), tested with Newey–West HAC standard errors; IR is the annualized information ratio  $\bar{e}_i / \sigma(e_i) \times \sqrt{12}$ ;  $N$  is the number of out-of-sample months. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Strategy               | $N$ | Expanding (60-month burn-in) |      |       | Rolling 60-month |      |       |
|------------------------|-----|------------------------------|------|-------|------------------|------|-------|
|                        |     | $\alpha$                     | $t$  | IR    | $\alpha$         | $t$  | IR    |
| Panel A: PCA           |     |                              |      |       |                  |      |       |
| <i>btp italia</i>      | 97  | +1.61***                     | 2.86 | +0.97 | +1.65***         | 3.00 | +1.02 |
| <i>cds bond basis</i>  | 137 | +4.55***                     | 4.98 | +1.55 | +3.88***         | 3.75 | +1.25 |
| <i>itraxx combined</i> | 149 | +1.85***                     | 3.75 | +1.11 | +1.96***         | 3.97 | +1.13 |
| Panel B: Best-Subset   |     |                              |      |       |                  |      |       |
| <i>btp italia</i>      | 97  | +1.42***                     | 2.90 | +0.89 | +1.91***         | 3.34 | +1.10 |
| <i>cds bond basis</i>  | 137 | +3.90***                     | 4.43 | +1.42 | +3.74***         | 3.90 | +1.32 |
| <i>itraxx combined</i> | 149 | +2.16***                     | 4.24 | +1.32 | +2.02***         | 4.06 | +1.09 |

**Table X**  
**Post-Selection OLS on the Best-Subset Factors**

The table reports OLS estimates of  $r_{i,t} = \alpha_i + \sum_j \beta_{i,j} F_{j,t} + \varepsilon_{i,t}$ , where  $r_{i,t}$  is the monthly excess return of arbitrage strategy  $i$  and  $F_{j,t}$  are the factors selected for that strategy by best-subset selection (Bertsimas, King and Mazumder, 2016). Factors enter in original (un-standardized) units;  $\alpha$  is annualized (% p.a.). The 95% confidence interval for  $\alpha$  is the percentile interval from a stationary bootstrap (9999 replications).  $t$ -statistics use Newey–West HAC standard errors (4 lags). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ . Factor definitions are in Table A.XV.

|                                    | Coefficient | $t$ -stat |
|------------------------------------|-------------|-----------|
| <i>Panel A: BTP Italia</i>         |             |           |
| $\alpha$ (% p.a.)                  | 1.6***      | 4.05      |
| 95% CI (bootstrap)                 | [0.8, 2.2]  |           |
| DEF_US                             | −0.003      | −0.13     |
| LIQ_V                              | −0.012      | −1.18     |
| SMB_EU                             | −0.028      | −1.33     |
| QMJ_EU                             | 0.041*      | 1.91      |
| FI_DEF                             | −0.206***   | −4.03     |
| SS2Y                               | −0.001      | −0.17     |
| SS5Y                               | −0.011      | −0.63     |
| SS10Y                              | 0.000       | 0.00      |
| $T = 157, k = 8, \bar{R}^2 = 0.22$ |             |           |
| <i>Panel B: CDS–Bond Basis</i>     |             |           |
| $\alpha$ (% p.a.)                  | 4.8***      | 5.87      |
| 95% CI (bootstrap)                 | [2.9, 6.6]  |           |
| PB_CDS_5Y_EU                       | −0.223**    | −2.28     |
| DEF_US                             | 0.112***    | 3.21      |
| HY_CORP                            | −0.109**    | −2.20     |
| EMERG_DEBT                         | 0.074**     | 2.17      |
| RI_EU                              | 0.217***    | 3.06      |
| FX_MOM                             | −0.041      | −1.33     |
| PTFSFX                             | −0.006**    | −2.26     |
| CURV_2S5S10S_EU                    | −0.015*     | −1.82     |
| $T = 197, k = 8, \bar{R}^2 = 0.21$ |             |           |
| <i>Panel C: iTraxx Combined</i>    |             |           |
| $\alpha$ (% p.a.)                  | 2.7***      | 5.00      |
| 95% CI (bootstrap)                 | [1.3, 3.5]  |           |
| XOVER_MAIN                         | −0.033      | −1.17     |
| HML_EU                             | −0.028**    | −2.21     |
| BAB_EU                             | −0.034***   | −3.57     |
| FI_MOM                             | −0.044***   | −2.64     |
| FI_DEF                             | −0.016      | −0.50     |
| SVS_RET_1M                         | −0.989**    | −2.48     |
| ATM_IV_ITRX                        | 0.527       | 1.45      |
| ATM_IV_CDX                         | −0.304      | −0.82     |
| $T = 209, k = 8, \bar{R}^2 = 0.18$ |             |           |

**Table XI**  
**Alpha by Stress Regime: PCA and Best-Subset**

Annualized alpha (% p.a.) per stress regime, from a single regression with three intercept dummies and betas common across regimes (no constant),  $r_t = \sum_g \alpha_g D_{g,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$ , Newey–West HAC standard errors (4 lags). PCA: spanning regression on the  $K = 8$  full-sample principal components. Best-Subset: post-selection OLS on the selected factors (Table X). Regimes on the 5-year iTraxx Europe Main spread: LOW < 60, MEDIUM  $\in [60, 100)$ , HIGH  $\geq 100$  bps.  $t$ -statistics in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Strategy               | PCA                |                    |                    | Best-Subset        |                    |                    |
|------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
|                        | LOW                | MEDIUM             | HIGH               | LOW                | MEDIUM             | HIGH               |
| <i>BTP Italia</i>      | +0.37<br>(0.65)    | +1.68***<br>(3.17) | +3.27*<br>(1.82)   | +0.47<br>(1.04)    | +2.06***<br>(3.36) | +3.04**<br>(2.16)  |
| <i>CDS–Bond Basis</i>  | +2.87***<br>(3.12) | +4.44***<br>(3.87) | +7.55***<br>(4.84) | +2.69***<br>(2.99) | +4.69***<br>(3.95) | +7.59***<br>(4.93) |
| <i>iTraxx Combined</i> | +1.03*<br>(1.95)   | +2.23***<br>(4.62) | +3.45***<br>(4.32) | +1.85***<br>(3.36) | +2.87***<br>(4.15) | +3.54***<br>(4.20) |

**Table XII**  
**Alpha Across Empirical Layers: Factor Benchmarks, PCA, and Best-Subset**

Each cell is the annualized alpha (% p.a.) and adjusted  $R^2$  from a spanning regression of the strategy excess return on the factor set of the corresponding layer: the three established hedge-fund factor benchmarks of Section A, the principal-component factors, and the best-subset selection (Bertsimas, King and Mazumder, 2016). A residual alpha that survives all three layers is evidence that the premium is not compensation for the identified systematic exposures. Alphas are annualized; stars denote significance from each layer's own inference. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Method                  | <i>BTP Italia</i> |             | <i>CDS-Bond</i> |             | <i>iTraxx</i> |             |
|-------------------------|-------------------|-------------|-----------------|-------------|---------------|-------------|
|                         | $\alpha$          | $\bar{R}^2$ | $\alpha$        | $\bar{R}^2$ | $\alpha$      | $\bar{R}^2$ |
| Duarte ( $k = 10$ )     | +1.6***           | 0.11        | +4.2***         | 0.15        | +2.2***       | 0.06        |
| Brooks ( $k = 8$ )      | +2.0***           | 0.07        | +4.4***         | 0.11        | +1.8***       | 0.06        |
| Fung-Hsieh ( $k = 7$ )  | +1.8***           | 0.10        | +4.0***         | 0.11        | +1.8***       | 0.00        |
| PCA ( $k = 8$ )         | +1.4***           | 0.09        | +4.8***         | 0.11        | +2.2***       | 0.13        |
| Best-subset ( $k = 8$ ) | +1.6***           | 0.22        | +4.8***         | 0.21        | +2.7***       | 0.18        |



**Table XIII**  
**Alpha Across Factor-Selection Methods**

For each selection method,  $n$  is the number of selected factors and two post-selection alphas of the monthly net-of-cost strategy excess return are reported.  $\alpha^{\text{split}}$ : honest split-sample alpha — factors are selected on the first half of the sample only, and the alpha is the intercept of an OLS on the second half (valid post-selection inference by sample splitting); not available for the AEN rows, whose set is estimated on the full sample.  $\alpha^{\text{OOS}}$ : frozen-set out-of-sample alpha — with the selected set held fixed, loadings are re-estimated each month on an expanding window (60-month burn-in) and the alpha is the mean abnormal return  $y_t - \mathbf{x}'_t \hat{\beta}_{t-1}$ , with the estimated intercept excluded from the fitted value. Alphas are annualized (% p.a.);  $t$ -statistics in parentheses use Newey–West HAC standard errors (4 lags). The  $\ell_0$  problem is solved by the splicing algorithm (abess) polished against greedy forward search; risk categories are the panels of Table A.XV. Selection with zero factors reports the raw strategy alpha. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Method                           | BTP Italia |                         |                       | CDS–Bond Basis |                         |                       | iTraxx Combined |                         |                       |
|----------------------------------|------------|-------------------------|-----------------------|----------------|-------------------------|-----------------------|-----------------|-------------------------|-----------------------|
|                                  | $n$        | $\alpha^{\text{split}}$ | $\alpha^{\text{OOS}}$ | $n$            | $\alpha^{\text{split}}$ | $\alpha^{\text{OOS}}$ | $n$             | $\alpha^{\text{split}}$ | $\alpha^{\text{OOS}}$ |
| Best subset ( $\ell_0$ , $k=8$ ) | 8          | +1.69***<br>(2.67)      | +1.50***<br>(2.91)    | 8              | +2.92***<br>(4.31)      | +3.89***<br>(4.44)    | 8               | +2.49***<br>(3.59)      | +2.01***<br>(4.46)    |
| One per risk category            | 5          | +1.91***<br>(2.83)      | +1.60***<br>(3.41)    | 5              | +2.52***<br>(3.47)      | +3.93***<br>(4.40)    | 5               | +2.18***<br>(3.19)      | +1.91***<br>(3.93)    |
| Adaptive elastic net             | 4          | –                       | +1.48***<br>(3.07)    | 4              | –                       | +4.17***<br>(4.68)    | 21              | –                       | +1.83***<br>(4.49)    |
| AEN, one per category            | 3          | –                       | +1.57***<br>(3.18)    | 2              | –                       | +4.13***<br>(4.68)    | 5               | –                       | +1.91***<br>(3.94)    |
| Relaxed lasso                    | 12         | +1.96***<br>(2.94)      | +1.48***<br>(2.79)    | 2              | +3.11***<br>(3.76)      | +4.16***<br>(4.53)    | 0               | +2.20***<br>(3.00)      | +1.79***<br>(3.36)    |
| Post-lasso (plug-in penalty)     | 1          | +1.96***<br>(2.94)      | +1.57***<br>(3.51)    | 2              | +3.11***<br>(3.76)      | +4.16***<br>(4.53)    | 0               | +2.20***<br>(3.00)      | +1.79***<br>(3.36)    |
| Stability selection (CPSS)       | 1          | +1.96***<br>(2.94)      | +1.57***<br>(3.51)    | 0              | +3.11***<br>(3.76)      | +3.77***<br>(4.04)    | 1               | +2.06***<br>(2.81)      | +1.78***<br>(3.48)    |

**Table XIV**  
**Unconditional and Regime-Dependent Correlations**

$m_{i,t}$  is the market-level mispricing magnitude of strategy  $i$  and  $\Delta m_{i,t}$  its monthly change. Panel A reports correlations of monthly strategy returns; Panel B, of the  $\Delta m$  series. Unconditional: Pearson correlations with Newey–West HAC  $t$ -statistics (4 lags). Regime columns: within-regime Pearson correlations, with regimes on the iTraxx Main 5Y level (LOW < 60, MEDIUM [60, 100), HIGH  $\geq$  100 bps); stars from a two-sided  $t$ -test with  $n - 2$  degrees of freedom. HIGH<sup>FR</sup> applies the Forbes and Rigobon (2002) adjustment for the higher return variance in the HIGH regime, so that the rise in correlation is not a heteroskedasticity artifact. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| <i>Panel A: Strategy Returns</i>  |               |           |           |          |        |                    |
|-----------------------------------|---------------|-----------|-----------|----------|--------|--------------------|
| Pair                              | Unconditional |           | By Regime |          |        |                    |
|                                   | Pearson       | $t$ -stat | LOW       | MED      | HIGH   | HIGH <sup>FR</sup> |
| CDS–Bond Basis vs BTP Italia      | −0.189        | −1.56     | −0.123    | −0.285** | −0.134 | −0.066             |
| CDS–Bond Basis vs iTraxx Combined | −0.020        | −0.17     | −0.079    | −0.019   | −0.038 | −0.017             |
| BTP Italia vs iTraxx Combined     | 0.217***      | 2.59      | 0.151     | 0.251**  | 0.154  | 0.082              |

*Panel B:*

| <i>Mispricing Changes (<math>\Delta m</math>)</i> |               |           |           |           |          |                    |
|---------------------------------------------------|---------------|-----------|-----------|-----------|----------|--------------------|
| Pair                                              | Unconditional |           | By Regime |           |          |                    |
|                                                   | Pearson       | $t$ -stat | LOW       | MED       | HIGH     | HIGH <sup>FR</sup> |
| CDS–Bond Basis vs BTP Italia                      | −0.298***     | −3.01     | −0.134    | −0.324*** | −0.465** | −0.326             |
| CDS–Bond Basis vs iTraxx Combined                 | 0.191*        | 1.70      | −0.208*   | 0.201*    | 0.316**  | 0.135*             |
| BTP Italia vs iTraxx Combined                     | −0.100        | −1.07     | 0.150     | −0.147    | −0.144   | −0.069             |

**Table XV**  
**Cross-Strategy Spanning Regressions ( $\Delta m$ )**

Pairwise regressions  $\Delta m_{i,t} = \alpha + \beta \Delta m_{j,t} + \varepsilon_t$ , where  $\Delta m_{i,t}$  is the monthly change in the mispricing magnitude of strategy  $i$ . Newey–West HAC standard errors (4 lags). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Dependent       | Regressor          | $\beta$  | $t$ -stat | $p$    | $\bar{R}^2$ |
|-----------------|--------------------|----------|-----------|--------|-------------|
| CDS–Bond Basis  | BTP Italia_L0      | −0.148** | -2.20     | 0.0280 | 0.0913      |
|                 | BTP Italia_L1      | −0.010   | -0.21     | 0.8314 |             |
|                 | BTP Italia_L2      | −0.039   | -1.18     | 0.2379 |             |
|                 | iTraxx Combined_L0 | +0.370   | 0.91      | 0.3625 |             |
|                 | iTraxx Combined_L1 | +0.710*  | 1.94      | 0.0528 |             |
|                 | iTraxx Combined_L2 | +0.030   | 0.08      | 0.9366 |             |
| BTP Italia      | iTraxx Combined_L0 | −0.687   | -1.40     | 0.1622 | 0.1000      |
|                 | iTraxx Combined_L1 | −0.950   | -1.38     | 0.1664 |             |
|                 | iTraxx Combined_L2 | +0.391   | 0.47      | 0.6377 |             |
|                 | CDS–Bond Basis_L0  | −0.424** | -2.44     | 0.0147 |             |
|                 | CDS–Bond Basis_L1  | +0.215   | 1.59      | 0.1124 |             |
|                 | CDS–Bond Basis_L2  | +0.072   | 0.63      | 0.5295 |             |
| iTraxx Combined | BTP Italia_L0      | −0.010   | -0.76     | 0.4472 | 0.0103      |
|                 | BTP Italia_L1      | +0.016   | 0.80      | 0.4235 |             |
|                 | BTP Italia_L2      | +0.028*  | 1.90      | 0.0576 |             |
|                 | CDS–Bond Basis_L0  | +0.017   | 0.54      | 0.5921 |             |
|                 | CDS–Bond Basis_L1  | +0.019   | 0.76      | 0.4478 |             |
|                 | CDS–Bond Basis_L2  | +0.008   | 0.40      | 0.6924 |             |

**Table XVI**  
**Stress Amplification: Interaction Regressions**

$\Delta m_{i,t} = \alpha + \beta_0 \Delta m_{j,t} + \beta_1 (\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t$ , where  $\Delta m_{i,t}$  is the monthly change in the mispricing magnitude of strategy  $i$  and  $z_t$  is the contemporaneous standardized iTraxx Main 5Y level. Newey–West HAC standard errors (4 lags). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Dependent       | Variable                       | Coefficient | $t$ -stat | $p$   | $\bar{R}^2$ |
|-----------------|--------------------------------|-------------|-----------|-------|-------------|
| CDS–Bond Basis  | $\Delta m_j$ (BTP Italia)      | −0.144**    | −2.06     | 0.039 | 0.1221      |
|                 | $\Delta m_j \times z_t$        | −0.086      | −1.50     | 0.133 |             |
|                 | $z_t$ (stress)                 | +1.591***   | 2.61      | 0.009 |             |
| BTP Italia      | $\Delta m_j$ (CDS–Bond Basis)  | −0.464***   | −3.20     | 0.001 | 0.1286      |
|                 | $\Delta m_j \times z_t$        | −0.637***   | −2.68     | 0.007 |             |
|                 | $z_t$ (stress)                 | +1.502      | 0.97      | 0.332 |             |
| CDS–Bond Basis  | $\Delta m_j$ (iTraxx Combined) | +0.482      | 1.07      | 0.284 | 0.0531      |
|                 | $\Delta m_j \times z_t$        | +0.016      | 0.08      | 0.939 |             |
|                 | $z_t$ (stress)                 | +0.934**    | 2.11      | 0.035 |             |
| iTraxx Combined | $\Delta m_j$ (CDS–Bond Basis)  | +0.030      | 1.14      | 0.254 | 0.0731      |
|                 | $\Delta m_j \times z_t$        | +0.079*     | 1.72      | 0.086 |             |
|                 | $z_t$ (stress)                 | −0.076      | −0.27     | 0.784 |             |
| BTP Italia      | $\Delta m_j$ (iTraxx Combined) | −0.389      | −0.66     | 0.509 | 0.0154      |
|                 | $\Delta m_j \times z_t$        | −0.618      | −0.92     | 0.360 |             |
|                 | $z_t$ (stress)                 | −0.128      | −0.08     | 0.939 |             |
| iTraxx Combined | $\Delta m_j$ (BTP Italia)      | −0.010      | −0.83     | 0.408 | 0.0201      |
|                 | $\Delta m_j \times z_t$        | −0.020      | −1.00     | 0.316 |             |
|                 | $z_t$ (stress)                 | −0.045      | −0.19     | 0.850 |             |

**Table XVII**  
**AR(1) Persistence of Mispricing Levels**

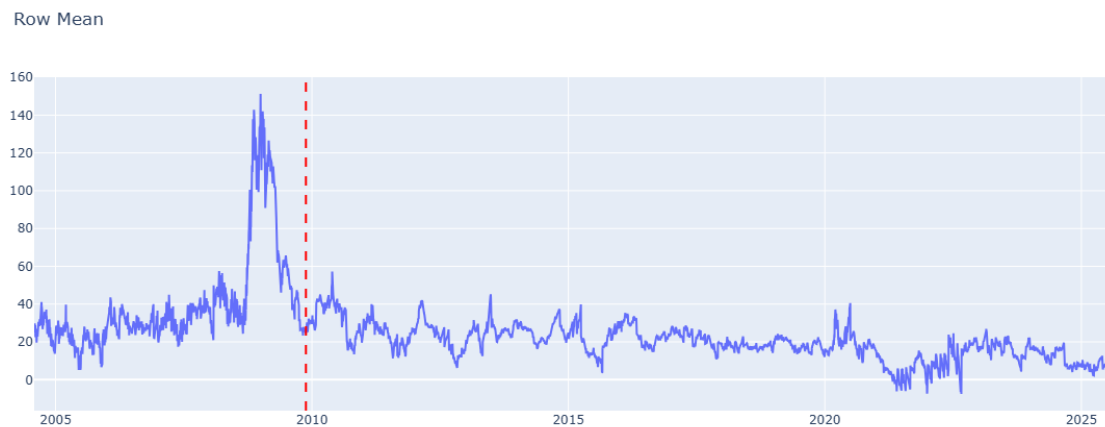
AR(1) regression of the mispricing level  $m_{i,t} = c_i + \phi_i m_{i,t-1} + u_{i,t}$ , where  $m_{i,t}$  is the market-level mispricing magnitude of strategy  $i$ . The half-life is  $h_i^* = \ln(0.5)/\ln(\phi_i)$  months. Newey–West HAC standard errors with six lags. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Strategy        | $\phi$    | HAC $t$ -stat | Half-life (months) | $N$ |
|-----------------|-----------|---------------|--------------------|-----|
| CDS–Bond Basis  | 0.7913*** | 14.52         | 3.0                | 148 |
| BTP Italia      | 0.7272*** | 13.92         | 2.2                | 148 |
| iTraxx Combined | 0.7258*** | 9.96          | 2.2                | 148 |

**Table XVIII**  
**Common Factor (PC1) and Funding Stress**

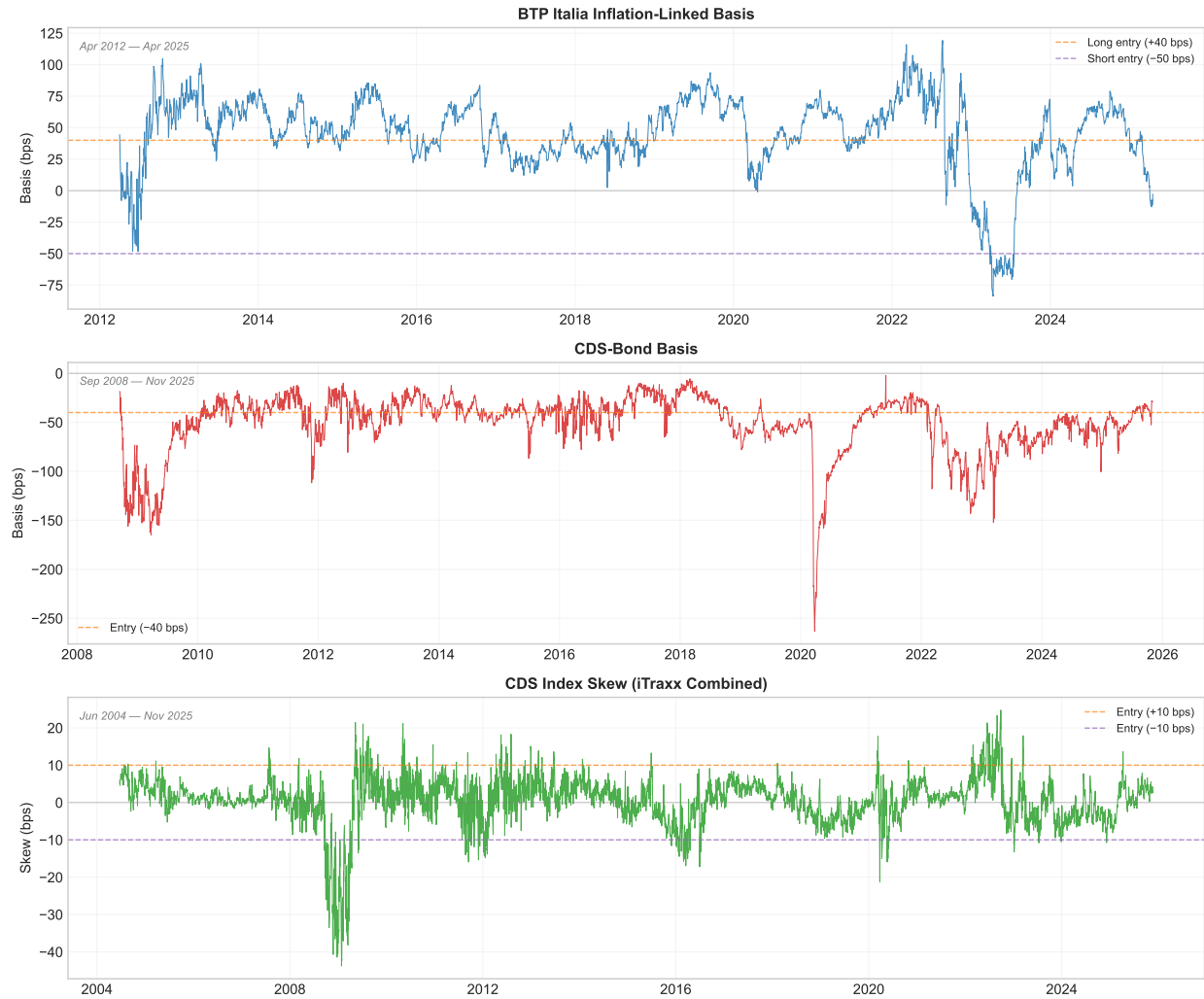
OLS regression of the first principal component of the three  $\Delta m$  series on intermediary stress proxies and on a macro placebo set. Panel A: one proxy per Duffie channel (intermediary capital, funding cost, dealer health, market illiquidity). Panel B: all available funding and liquidity proxies. Panel C: macroeconomic placebo set (macro and real-activity uncertainty, long-term inflation expectations, economic policy uncertainty). Significant coefficients in Panels A and B indicate that the common factor driving correlated mispricings is linked to intermediary balance-sheet conditions (Duffie, 2010; Brunnermeier and Pedersen, 2009; He et al., 2017). The non-significant coefficients in Panel C are consistent with the common factor not being driven by macroeconomic fundamentals. Newey–West HAC standard errors (4 lags). \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

| Factor                                    | Coefficient | $t$ -stat | $p$ -value |
|-------------------------------------------|-------------|-----------|------------|
| <b>Panel A: Core Intermediary Proxies</b> |             |           |            |
| $T = 149, \bar{R}^2 = 0.233$              |             |           |            |
| HKM                                       | −4.278**    | −2.23     | 0.026      |
| LIBOR_OIS                                 | +3.349***   | 3.09      | 0.002      |
| PB_CDS_5Y_EU                              | +0.349*     | 1.92      | 0.054      |
| ILLIQ                                     | +1.766***   | 2.80      | 0.005      |
| <b>Panel C: Macroeconomic Placebo</b>     |             |           |            |
| $T = 151, \bar{R}^2 = 0.106$              |             |           |            |
| $\Delta U M$                              | +0.363      | 0.03      | 0.977      |
| $\Delta U R$                              | +10.313     | 1.22      | 0.223      |
| 5Y5Y_INFL                                 | −0.627      | −0.65     | 0.516      |
| EPU_EU                                    | +0.003      | 1.32      | 0.187      |



**Figure 1**

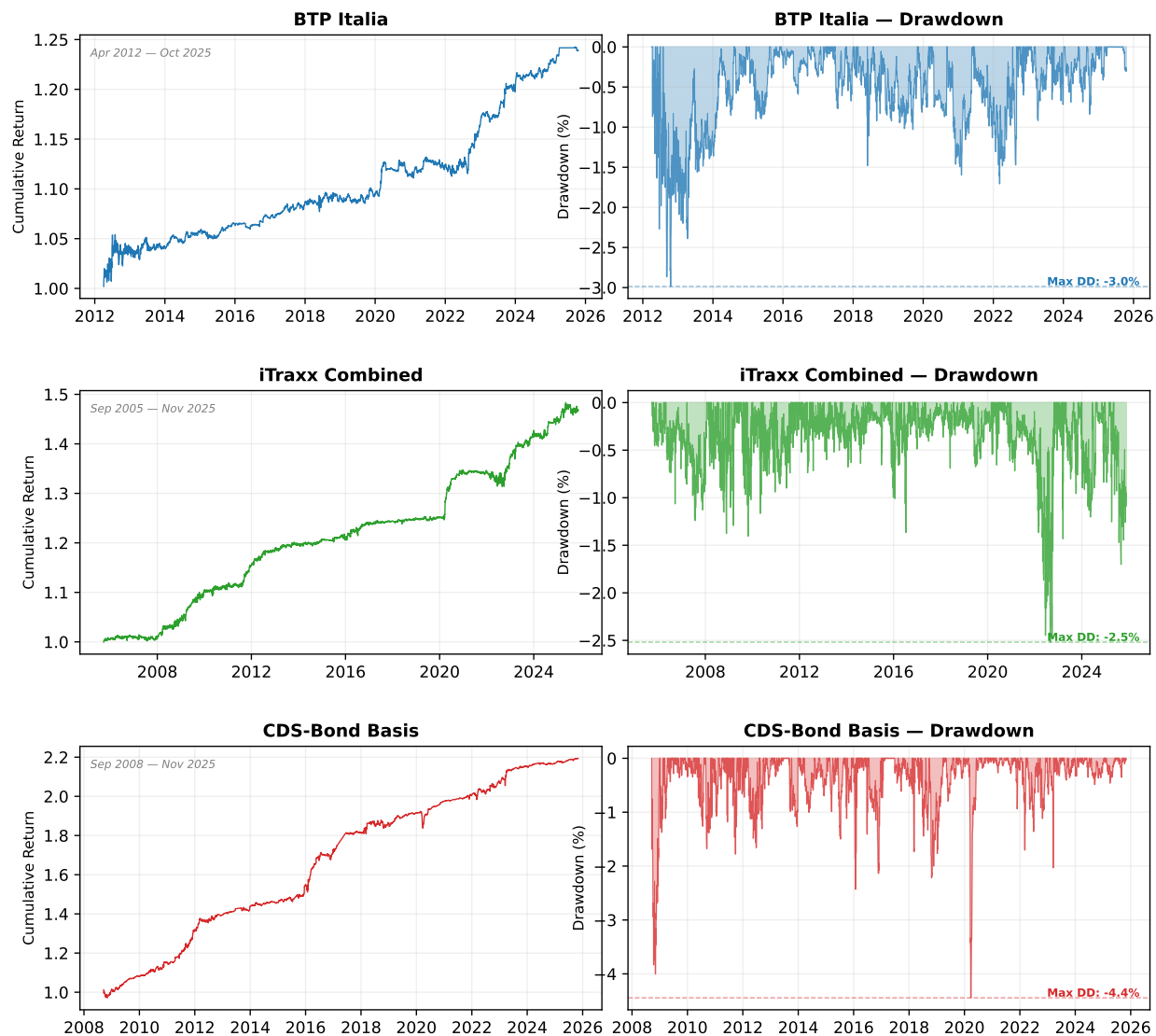
The TIPS–Treasury Basis, 2005–2025. The series is the average TIPS–Treasury mispricing across matched bond pairs, in basis points, following Fleckenstein et al. (2014). The vertical dashed line marks the end of their sample (19 November 2009). The basis exceeded 140 bps during the 2008–09 crisis; that extreme dislocation has not recurred, and the basis has since compressed, turning slightly negative in 2021–22.



**Figure 2**

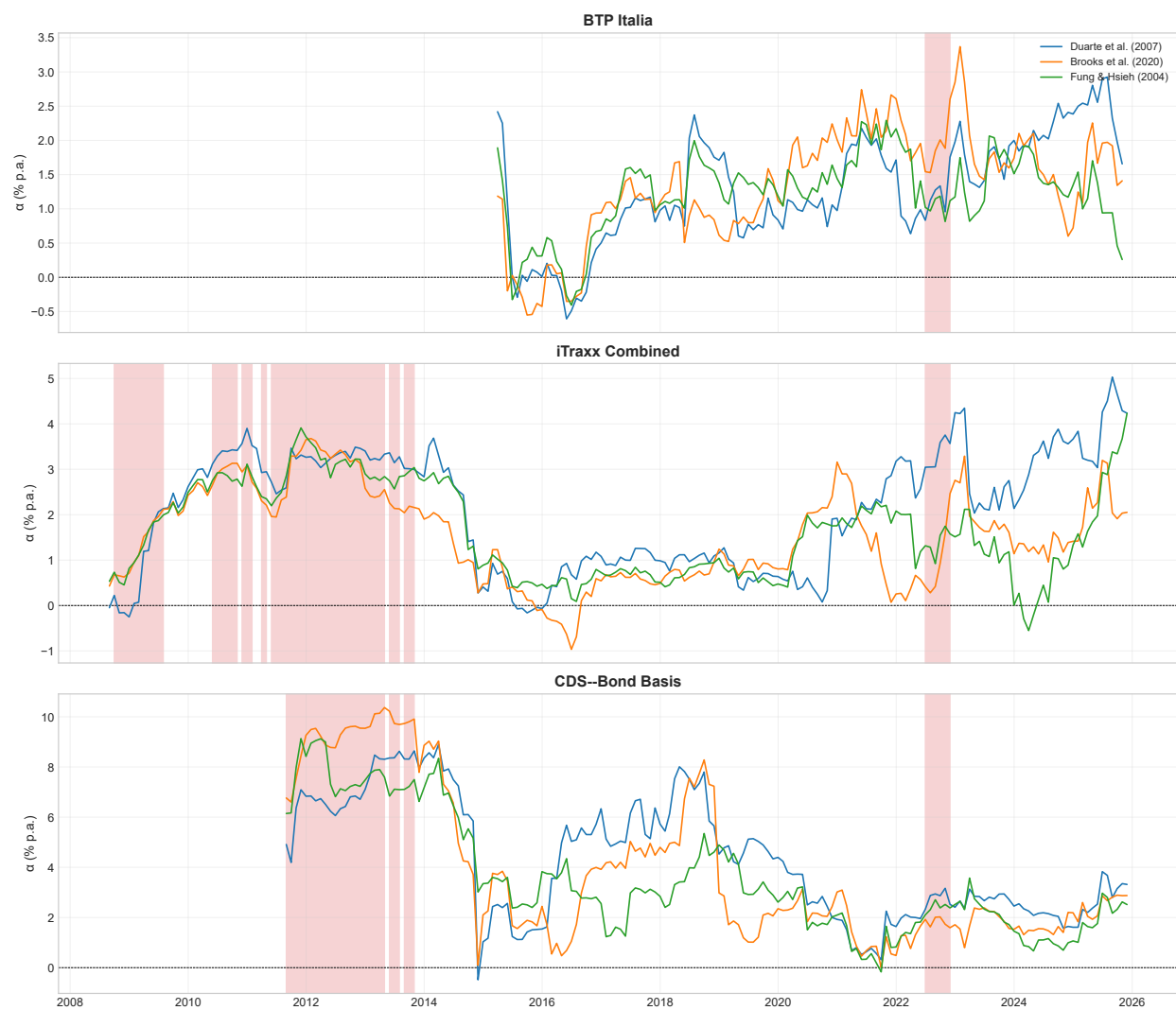
Market-Level Basis Time Series for Fixed Income Arbitrage Strategies. For BTP Italia, the basis is the cross-sectional mean of all outstanding issues with at least six months to maturity. For iTraxx Combined, it is the mean of the four on-the-run sub-index skews. For CDS-Bond Basis, it is the cross-sectional median of the single-name bases in the iTraxx Main universe. Dashed lines indicate entry thresholds.





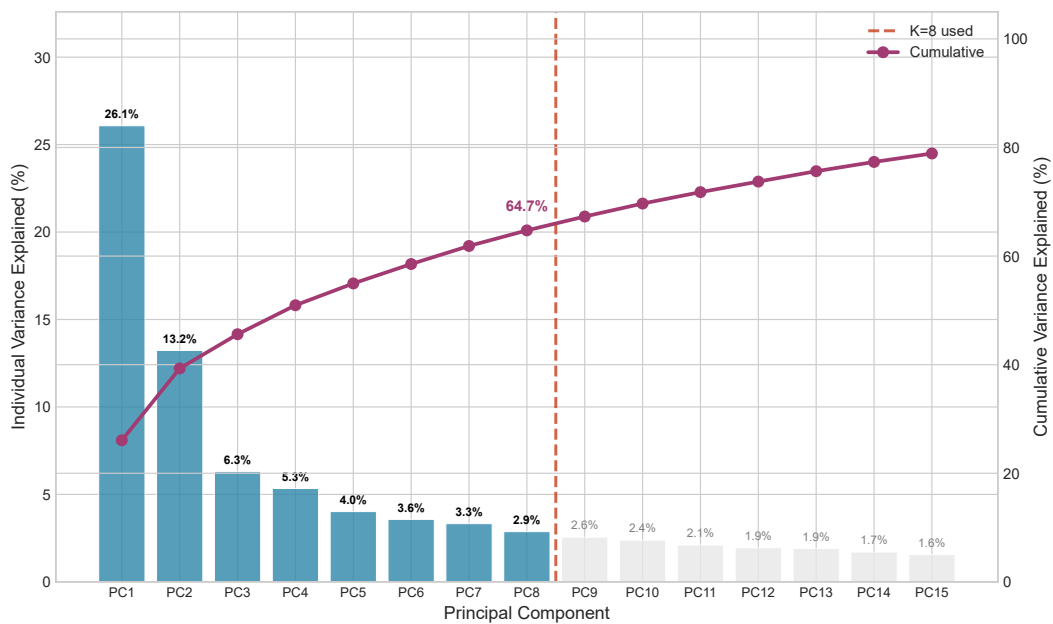
**Figure 3**

Cumulative Return Indices and Drawdowns of Fixed Income Arbitrage Strategies (Signal-Weighted). Each row shows one strategy: the upper sub-panel reports the cumulative return index from inception, and the lower sub-panel reports the running drawdown (peak-to-trough). Sample periods differ across strategies: BTP Italia from April 2012, iTraxx Combined from September 2005, CDS-Bond Basis from September 2008. All series end in May 2025 and are net of transaction costs.



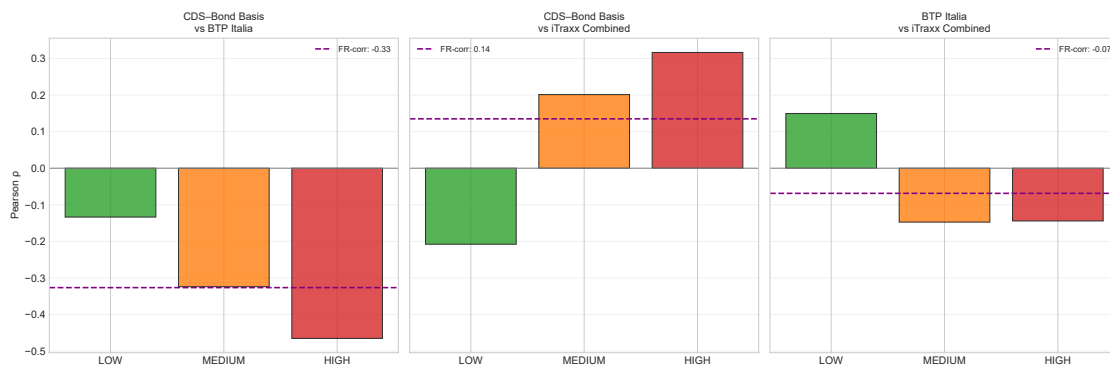
**Figure 4**

Rolling 36-Month Alpha Across Benchmark Factor Models. Each panel shows one strategy. Lines represent rolling alpha under Duarte et al. (2007, blue), Brooks et al. (2020, orange), and Fung & Hsieh (2004, green). Red shading: HIGH stress regime (iTraxx Main 5Y  $\geq$  100 bps).



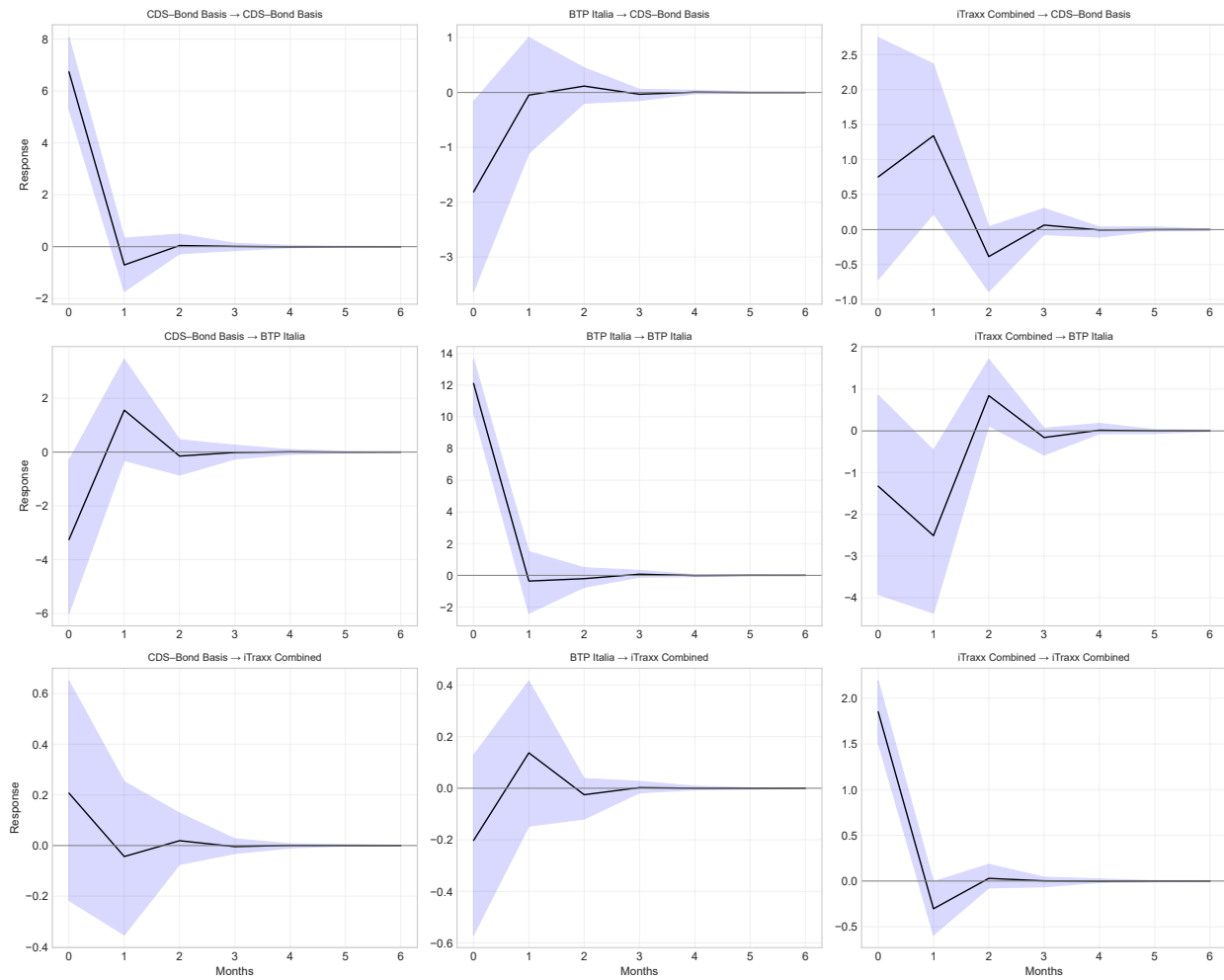
**Figure 5**

Scree Plot: Cumulative Variance Explained by Principal Components. Bars show the variance explained by each component; the line shows the cumulative share. The vertical dashed line marks the baseline  $K = 8$ , selected by the Bai–Ng  $IC_{p2}$  criterion; the first eight components capture 64.7% of total variance.



**Figure 6**

Pairwise Correlations by Stress Regime. Bars show Pearson correlations of  $\Delta m$  in LOW, MEDIUM, and HIGH stress regimes defined by iTraxx Main 5Y thresholds at 60 and 100 bps, as in Table XIV.



**Figure 7**  
Generalized Impulse Response Functions. Responses of  $\Delta m$  to one-standard-deviation shocks. Shaded areas: 95% bootstrap confidence intervals (1000 replications).